

Chemical Kinetics

Question1

Pre-exponential factors of two different reactions of same order are identical. Let activation energy of first reaction exceeds the activation energy of second reaction by 20 kJ mol^{-1} . If k_1 and k_2 are the rate constants of first and second reaction respectively at 300 K , then $\ln \frac{k_2}{k_1}$ will be _____. (nearest integer) $[R = 8.3 \text{ J K}^{-1} \text{ mol}^{-1}]$

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Answer: 8

Solution:

Arrhenius equation:

$$k = Ae^{-E_a/RT}$$

Given pre-exponential factors are identical, so $A_1 = A_2$.

Also, activation energy of first reaction exceeds that of second by 20 kJ mol^{-1} :

$$E_{a1} = E_{a2} + 20,000 \text{ J mol}^{-1}$$

Now,

$$\frac{k_2}{k_1} = \frac{Ae^{-E_{a2}/RT}}{Ae^{-E_{a1}/RT}} = e^{-(E_{a2}-E_{a1})/RT} = e^{(E_{a1}-E_{a2})/RT}$$

So,

$$\ln \left(\frac{k_2}{k_1} \right) = \frac{E_{a1}-E_{a2}}{RT} = \frac{20,000}{8.3 \times 300}$$

Compute:

$$8.3 \times 300 = 2490$$

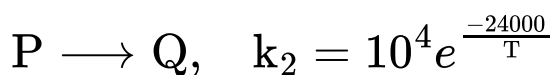
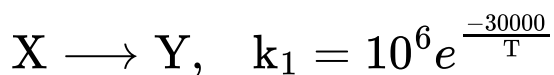
$$\ln\left(\frac{k_2}{k_1}\right) = \frac{20,000}{2490} \approx 8.03$$

Nearest integer:

8

Question2

The temperature at which the rate constants of the given below two gaseous reactions become equal is ____ K. (Nearest integer)



Given : $\ln 10 = 2.303$

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Answer: 1303

Solution:

Given,

$$k_1 = 10^6 e^{-30000/T}, \quad k_2 = 10^4 e^{-24000/T}$$

At the temperature T when $k_1 = k_2$:

$$10^6 e^{-30000/T} = 10^4 e^{-24000/T}$$

Divide both sides by $10^4 e^{-30000/T}$:

$$10^2 = e^{(-24000/T) - (-30000/T)} = e^{6000/T}$$

Take natural log on both sides:

$$\ln(10^2) = \frac{6000}{T}$$

$$2 \ln 10 = \frac{6000}{T}$$

Given $\ln 10 = 2.303$:

$$2(2.303) = \frac{6000}{T}$$

$$4.606 = \frac{6000}{T}$$

$$T = \frac{6000}{4.606} \approx 1302.65 \text{ K}$$

Nearest integer:

1303 K

Question3

Consider $A \xrightarrow{k_1} B$ and $C \xrightarrow{k_2} D$ are two reactions. If the rate constant (k_1) of the $A \longrightarrow B$ reaction can be expressed by the following equation $\log_{10} k = 14.34 - \frac{1.5 \times 10^4}{T/K}$ and activation energy of $C \longrightarrow D$ reaction (E_{a2}) is $\frac{1}{5}$ th of the $A \longrightarrow B$ reaction (E_{a1}), then the value of (E_{a2}) is

_____ kJmol^{-1} . (Nearest Integer)

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Answer: 57

Solution:

Given: $\log_{10} k = 14.34 - \frac{1.5 \times 10^4}{T/K}$

From the Arrhenius equation in base-10 form, we write:

$$\log_{10} k = \log_{10} A - \frac{E_a}{2.303RT}$$

On comparing this with $\log_{10} k = 14.34 - \frac{1.5 \times 10^4}{T/K}$, the coefficient of $\frac{1}{T}$ gives:

$$\frac{E_{a1}}{2.303R} = 1.5 \times 10^4$$

Now substitute $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$:

$$E_{a1} = 1.5 \times 10^4 \times 2.303 \times 8.314$$

$$E_{a_1} = 28.7207 \times 10^4 \text{ J}$$

Convert joules to kilojoules ($1 \text{ kJ} = 10^3 \text{ J}$):

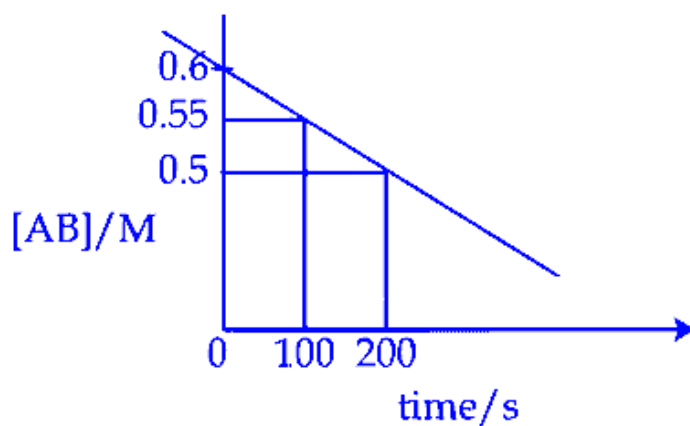
$$E_{a_1} = 287.207 \text{ kJ}$$

Given $E_{a_2} = \frac{1}{5}E_{a_1}$, so:

$$E_{a_2} = \frac{E_{a_1}}{5} = \frac{287.207}{5} = 57.44 \text{ kJ}$$

Question4

For the thermal decomposition of reactant AB(g) , the following plot is constructed.



The half life of the reaction is ' x ' min.

$x = \underline{\hspace{2cm}}$ min. (Nearest integer)

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Answer: 10

Solution:

The graph between concentration $[\text{AB}]$ and time t is a straight line that decreases with time. This type of graph shows that the reaction is **zero order** in $[\text{AB}]$.

For a zero-order reaction, the integrated rate law is:

$$[AB]_0 - [AB]_t = kt$$

$$0.60 - 0.55 = k(100)$$

$$k = 5 \times 10^{-4}$$

$$\text{Half life } (t_{1/2}) = \frac{[AB]_0}{2k}$$

$$= \frac{0.60}{2 \times 5 \times 10^{-4}}$$

$$= 600 \text{ sec}$$

$$= 10 \text{ min}$$

From the graph, the initial concentration is $[AB]_0 = 0.60$ and after 100 s the concentration is 0.55.

Substituting these values in the zero-order equation gives the rate constant $k = 5 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$.

For a zero-order reaction, the half-life is given by $t_{1/2} = \frac{[AB]_0}{2k}$. Putting the values, we get

$t_{1/2} = 600 \text{ s} = 10 \text{ min}$. Therefore, $x = 10 \text{ min}$ (nearest integer).

Question5

The half-life of ^{65}Zn is 245 days. After x days, 75% of original activity remained. The value of x in days is _____. (Nearest integer)

(Given: $\log 3 = 0.4771$ and $\log 2 = 0.3010$)

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Answer: 102

Solution:

$$t_{1/2} = \frac{\ln 2}{K}$$

Here, $t_{1/2} = 245$ days, so we find K as:

$$K = \frac{\ln 2}{245}$$

Now we use the radioactive decay relation for activity:

$$t = \frac{1}{K} \ln \frac{a_0}{a_t}$$

Given that 75% of the original activity remains, so $a_t = 0.75 a_0 = \frac{3}{4} a_0$.

So,

$$\frac{a_0}{a_t} = \frac{a_0}{\frac{3}{4} a_0} = \frac{4}{3}$$

Therefore,

$$t_{25\%} = \frac{1}{K} \ln \frac{4}{3}$$

Substitute $K = \frac{\ln 2}{245}$:

$$t_{25\%} = \frac{1}{\frac{\ln 2}{245}} \ln \frac{4}{3}$$

Simplifying:

$$t_{25\%} = 245 \frac{\ln \frac{4}{3}}{\ln 2} = 245 \left[\frac{2 \log 2 - \log 3}{\log 2} \right]$$

Now use the given values $\log 3 = 0.4771$ and $\log 2 = 0.3010$:

$$= 245 \left[\frac{2 \times 0.3010 - 0.4771}{0.3010} \right] = 101.66 \text{ day.}$$

Question 6

A $\rightarrow$ B (first reaction)

C $\rightarrow$ D (second reaction)

Consider the above two first-order reactions. The rate constant for first reaction at 500 K is double of the same at 300 K. At 500 K, 50% of the reaction becomes complete in 2 hour. The activation energy of the second reaction is half of that of first reaction. If the rate constant at 500 K of the second reaction becomes double of the rate constant of first reaction at the same temperature; then rate constant for the second reaction at 300 K is _____ $\times 10^{-1} \text{ hour}^{-1}$ (nearest integer).

Answer: 5

Solution:

For $A \xrightarrow{K_1} B$

$$\ln(2) = \frac{E_{a_1}}{R} \left[\frac{1}{300} - \frac{1}{500} \right]$$

$$E_{a_1} = \frac{\ln 2 \times R \times 1500}{2}$$

$$E_{a_2} = \frac{E_{a_1}}{2} = \frac{\ln 2 \times R \times 1500}{4}$$

$$(K_1)_{\text{at } 500 \text{ K}} = \frac{\ln 2}{2}$$

$$(K_2)_{\text{at } 500 \text{ K}} = \ln 2$$

Now for $C \xrightarrow{K_2} D$

$$\ln \left[\frac{(K_2)_{\text{at } 500 \text{ K}}}{(K_2)_{\text{at } 300 \text{ K}}} \right] = \left(\frac{\ln 2 \times R \times 1500}{4} \right) \times \frac{1}{R} \times \left[\frac{1}{300} - \frac{1}{500} \right]$$

$$(K_2)_{\text{at } 300 \text{ K}} = \frac{\ln 2}{\sqrt{2}} = 0.49$$

$$(K_2)_{\text{at } 300 \text{ K}} = 4.9 \times 10^{-1}$$

Ans is 5 .

Question 7

For reaction $A \rightarrow P$, rate constant $k = 1.5 \times 10^3 \text{ s}^{-1}$ at 27°C

If activation energy for the above reaction is 60 kJ mol^{-1} , then the temperature (in $^\circ \text{C}$) at which rate constant, $k = 4.5 \times 10^3 \text{ s}^{-1}$ is _____. (Nearest integer)

Given : $\log 2 = 0.30$, $\log 3 = 0.48$, $R = 8.3 \text{ J K}^{-1} \text{ mol}^{-1}$, $\ln 10 = 2.3$

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Answer: 41

Solution:

Use the Arrhenius equation in its two-temperature form:

$$\ln \left(\frac{k_2}{k_1} \right) = \frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

Given:

$$k_1 = 1.5 \times 10^3 \text{ s}^{-1}, \quad T_1 = 27^\circ\text{C} = 300 \text{ K}$$

$$k_2 = 4.5 \times 10^3 \text{ s}^{-1}$$

$$E_a = 60 \text{ kJ mol}^{-1} = 60000 \text{ J mol}^{-1}$$

$$R = 8.3 \text{ J K}^{-1} \text{ mol}^{-1}$$

First, find the ratio:

$$\frac{k_2}{k_1} = \frac{4.5 \times 10^3}{1.5 \times 10^3} = 3$$

So,

$$\ln 3 = \frac{60000}{8.3} \left(\frac{1}{300} - \frac{1}{T_2} \right)$$

Now convert $\ln 3$ using the given values:

$$\ln 3 = 2.3 \log 3 = 2.3 \times 0.48 = 1.104$$

Also,

$$\frac{60000}{8.3} \approx 7229$$

Hence,

$$1.104 = 7229 \left(\frac{1}{300} - \frac{1}{T_2} \right)$$

Therefore,

$$\frac{1}{300} - \frac{1}{T_2} = \frac{1.104}{7229}$$

$$\frac{1.104}{7229} \approx 1.53 \times 10^{-4}$$

So,

$$\frac{1}{T_2} = \frac{1}{300} - 1.53 \times 10^{-4}$$

$$\frac{1}{300} = 3.33 \times 10^{-3}$$

Thus,

$$\frac{1}{T_2} = 3.33 \times 10^{-3} - 0.153 \times 10^{-3} = 3.18 \times 10^{-3}$$

$$T_2 \approx \frac{1}{3.18 \times 10^{-3}} \approx 314 \text{ K}$$

Now convert into Celsius:

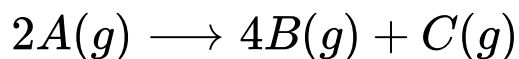
$$t = 314 - 273 = 41^\circ\text{C}$$

Hence, the required temperature is

$$\boxed{41^\circ\text{C}}$$

Question 8

Consider the following gas phase reaction being carried out in a closed vessel at 25°C .



time (min)	total pressure of the system (mm Hg)
30	300
∞	600

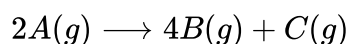
The pressure of $C(g)$ at 30 minutes time interval would be _____ mm Hg. (nearest integer)

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Answer: 20

Solution:

For the reaction



let the initial pressure of A be P_0 .

Since the vessel is closed and temperature is constant, pressure is proportional to number of moles.

At $t = \infty$, reaction is complete.

From the stoichiometry:

- 2 moles of A produce $4 + 1 = 5$ moles of products

So if initially only A is present, total moles increase by a factor

$$\frac{5}{2}$$

Hence final total pressure is

$$P_{\infty} = \frac{5}{2}P_0$$

Given:

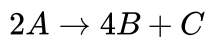
$$P_{\infty} = 600 \text{ mm Hg}$$

So,

$$\frac{5}{2}P_0 = 600$$

$$P_0 = 600 \times \frac{2}{5} = 240 \text{ mm Hg}$$

Now at 30 min, let the extent of reaction correspond to consumption of x pressure units of A according to:



Then pressure changes are:

- A : decreases by x
- B : increases by $2x$
- C : increases by $\frac{x}{2}$

This is because for $2A \rightarrow 4B + C$,

if 2 units of A react, 4 units of B and 1 unit of C are formed.

So total pressure at time t is

$$P_t = (240 - x) + 2x + \frac{x}{2}$$

$$P_t = 240 + \frac{3x}{2}$$

Given at 30 min,

$$240 + \frac{3x}{2} = 300$$

$$\frac{3x}{2} = 60$$

$$x = 40$$

Therefore pressure of C at 30 min is

$$P_C = \frac{x}{2} = 20 \text{ mm Hg}$$

So, the required pressure is

Question9

For a first order reaction $A \rightarrow B$

t/ min	[A]/M
0	0.6500
x	0.0650
20	0.00065

$x = \underline{\hspace{1cm}}$ min. (Nearest integer)

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Answer: 7

Solution:

For a first order reaction, the integrated rate law is

$$k = \frac{2.303}{t} \log \frac{[A]_0}{[A]_t}$$

We use the data at $t = 20$ min:

- Initial concentration, $[A]_0 = 0.6500$ M
- Concentration at 20 min, $[A]_{20} = 0.00065$ M

So,

$$k = \frac{2.303}{20} \log \frac{0.6500}{0.00065}$$

Now,

$$\frac{0.6500}{0.00065} = 1000$$

and

$$\log 1000 = 3$$

Therefore,

$$k = \frac{2.303}{20} \times 3$$

$$k = \frac{6.909}{20}$$

$$k = 0.34545 \text{ min}^{-1}$$

Now we use the concentration at time x :

$$\bullet [A]_x = 0.0650 \text{ M}$$

Again,

$$k = \frac{2.303}{x} \log \frac{[A]_0}{[A]_x}$$

$$0.34545 = \frac{2.303}{x} \log \frac{0.6500}{0.0650}$$

Now,

$$\frac{0.6500}{0.0650} = 10$$

and

$$\log 10 = 1$$

So,

$$0.34545 = \frac{2.303}{x}$$

Hence,

$$x = \frac{2.303}{0.34545}$$

$$x \approx 6.67 \text{ min}$$

Nearest integer:

$$\boxed{7 \text{ min}}$$

Question 10

If the half life of a first order reaction is 6.93 minutes then the time required for completion of 99% of the reaction will be _____ minutes.

(Given : $\log 2 = 0.3010$)

Answer: 46

Solution:

For a first order reaction,

$$t_{1/2} = \frac{0.693}{k}$$

Given:

$$t_{1/2} = 6.93 \text{ min}$$

So,

$$k = \frac{0.693}{6.93} = 0.1 \text{ min}^{-1}$$

Now, 99% completion means only 1% of the reactant is left.

So,

$$\frac{[A]_0}{[A]} = \frac{100}{1} = 100$$

For a first order reaction,

$$t = \frac{2.303}{k} \log \frac{[A]_0}{[A]}$$

Substituting the values:

$$t = \frac{2.303}{0.1} \log 100$$

Since,

$$\log 100 = 2$$

therefore,

$$t = \frac{2.303}{0.1} \times 2$$

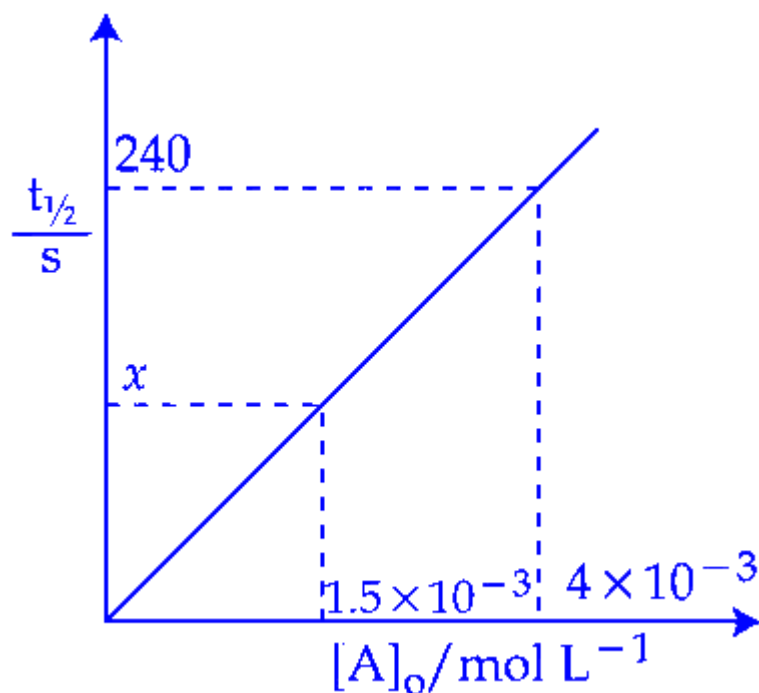
$$t = 23.03 \times 2 = 46.06 \text{ min}$$

Hence, the time required for 99% completion of the reaction is

46.06 minutes

Question11

For a reaction $A \rightarrow P$ at T K , the half life ($t_{1/2}$) is plotted as a function of initial concentration $[A]_0$ of A as given below.



The value of x in the given figure is ____ s (Nearest integer)

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Answer: 90

Solution:

From the graph, it is clear that the half-life $t_{1/2}$ increases directly with the initial concentration $[A]_0$.

We know that for a reaction of order N :

$$t_{1/2} \propto [A]_0^{1-N}$$

Here, since $t_{1/2} \propto [A]_0$, we can write:

$$1 - N = 1 \Rightarrow N = 0$$

This means the reaction is of **zero order**. For a zero-order reaction, the half-life is given by:

$$t_{1/2} = \frac{[A]_0}{2k}$$

From the data in the graph: For the first point, $t_{1/2} = 240$ s and $[A]_0 = 4 \times 10^{-3}$

So,

$$240 = \frac{4 \times 10^{-3}}{2k}$$

For the second point, $[A]_0 = 1.5 \times 10^{-3}$ and $t_{1/2} = x \text{ s}$

$$x = \frac{1.5 \times 10^{-3}}{2k}$$

Now, divide equation (1) by (2):

$$\frac{240}{x} = \frac{4}{1.5}$$

Therefore,

$$x = 90 \text{ s}$$

Question12

Sucrose hydrolyses in acidic medium into glucose and fructose by first order rate law with $t_{1/2} = 3$ hour. The percentage of sucrose remaining after 6 hours is

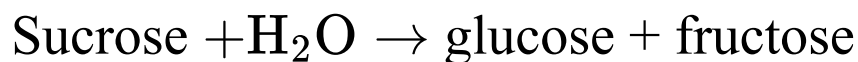
_____ . (Nearest integer)

(Given : $\log 2 = 0.3010$ and $\log 3 = 0.4771$)

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Answer: 25

Solution:



- The reaction is of the **first order**.
- The half-life of the reaction, $t_{1/2}$, is **3 hours**.
- We need to find the percentage of sucrose remaining after a total time, t , of **6 hours**.

Let's begin.

Step 1: Understand the concept of half-life ($t_{1/2}$)

For a chemical reaction, the half-life is the time it takes for the concentration of a reactant to reduce to exactly half of its initial value.

In this case, it means that every 3 hours, the amount of sucrose will become half of what it was at the start of that 3-hour period.

Step 2: Determine the number of half-lives passed

We are given a total time of 6 hours and a half-life of 3 hours. We can find the number of half-lives that have occurred, which we can call 'n'.

$$n = \frac{\text{Total time}}{\text{Half-life}} = \frac{t}{t_{1/2}}$$

Substituting the given values:

$$n = \frac{6 \text{ hours}}{3 \text{ hours}} = 2$$

So, a total of 2 half-lives have passed.

Step 3: Calculate the amount of sucrose remaining

Let's assume the initial percentage of sucrose is 100%.

- **After the first half-life (at t = 3 hours):**

The amount of sucrose remaining will be half of the initial amount.

$$\text{Sucrose remaining} = \frac{1}{2} \times 100\% = 50\%$$

- **After the second half-life (at t = 6 hours):**

The amount of sucrose will be half of the amount that was present at the end of the first half-life (which was 50%).

$$\text{Sucrose remaining} = \frac{1}{2} \times 50\% = 25\%$$

We can also use a general formula for the fraction of reactant remaining after 'n' half-lives:

$$\text{Fraction remaining} = \left(\frac{1}{2}\right)^n$$

In our case, $n = 2$, so:

$$\text{Fraction remaining} = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

To find the percentage remaining, we multiply this fraction by 100:

$$\text{Percentage remaining} = \frac{1}{4} \times 100 = 25\%$$

The question asks for the answer as the nearest integer.

Therefore, the percentage of sucrose remaining after 6 hours is **25**.

Question13

Decomposition of a hydrocarbon follows the equation

$k = \left(5.5 \times 10^{11} \text{ s}^{-1}\right) e^{\frac{-28000 \text{ K}}{T}}$. The activation energy of reaction is _____ kJ mol^{-1} . (Nearest Integer)

Given : $R = 8.3 \text{ J K}^{-1} \text{ mol}^{-1}$

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Answer: 232

Solution:

We are given the Arrhenius equation in the exponential form:

$$k = A e^{-\frac{E_a}{RT}}$$

Comparing this with the given equation:

$$k = (5.5 \times 10^{11} \text{ s}^{-1}) e^{-\frac{28000}{T}}$$

we can see that:

$$\frac{E_a}{R} = 28000$$

Now, substituting the given value of the gas constant $R = 8.3 \text{ J mol}^{-1} \text{ K}^{-1}$:

$$E_a = 28000 \times 8.3 \text{ J mol}^{-1}$$

$$E_a = 232400 \text{ J mol}^{-1}$$

Converting joules to kilojoules:

$$E_a = \frac{232400}{1000} = 232.4 \text{ kJ mol}^{-1}$$

Rounding to the nearest integer:

$E_a = 232 \text{ kJ mol}^{-1}$

Question14

Decomposition of A is a first order reaction at T(K) and is given by $A(g) \rightarrow B(g) + C(g)$.

In a closed 1 L vessel, 1 bar A(g) is allowed to decompose at T(K). After 100 minutes, the total pressure was 1.5 bar. What is the rate constant (in min^{-1}) of the reaction? ($\log 2 = 0.3$)

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Options:

A.

$$6.9 \times 10^{-4}$$

B.

$$6.9 \times 10^{-3}$$

C.

$$6.9 \times 10^{-1}$$

D.

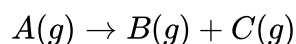
$$6.9 \times 10^{-2}$$

Answer: B

Solution:

Let initial pressure of A be $P_0 = 1$ bar in a 1 L closed vessel at constant T .

For the reaction:



Suppose x bar (in pressure-equivalent terms) of A decomposes.

Then after time t :

- Pressure of A left = $(1 - x)$ bar
- Pressure of B formed = x bar
- Pressure of C formed = x bar

So total pressure:

$$P_{\text{total}} = (1 - x) + x + x = 1 + x$$

Given after 100 min, $P_{\text{total}} = 1.5$ bar:

$$1 + x = 1.5 \Rightarrow x = 0.5$$

So partial pressure of A after 100 min:

$$P_A = 1 - x = 1 - 0.5 = 0.5 \text{ bar}$$

For a first order reaction:

$$k = \frac{2.303}{t} \log \left(\frac{[A]_0}{[A]_t} \right)$$

Here, pressure is proportional to concentration, so:

$$k = \frac{2.303}{100} \log \left(\frac{1}{0.5} \right)$$

$$= \frac{2.303}{100} \log 2$$

Given $\log 2 = 0.3$:

$$k = \frac{2.303}{100} \times 0.3 = 6.9 \times 10^{-3} \text{ min}^{-1}$$

Correct option: B) $6.9 \times 10^{-3} \text{ min}^{-1}$

Question 15

A $\rightarrow$ product (First order reaction).

Three sets of experiment were performed for a reaction under similar experimental conditions:

Run 1 $\Rightarrow$ 100 mL of 10 M solution of reactant A

Run 2 $\Rightarrow$ 200 mL of 10 M solution of reactant A

Run 3 $\Rightarrow$ 100 mL of 10 M solution of reactant A + 100 mL of H_2O added.

The correct variation of rate of reaction is

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Options:

A.

$$\text{Run 1} = \text{Run 2} = \text{Run 3}$$

B.

$$\text{Run 3} < \text{Run 1} < \text{Run 2}$$

C.

$$\text{Run 1} < \text{Run 2} < \text{Run 3}$$

D.

$$\text{Run 3} < \text{Run 1} = \text{Run 2}$$

Answer: D

Solution:

For a first order reaction, the rate law is:

$$\text{Rate} = k[\text{A}]$$

This means the rate depends only on the **concentration** of A, not on the total volume taken.

Run 1: 100 mL of 10 M A is used, so $[\text{A}] = 10 \text{ M}$.

Run 2: 200 mL of 10 M A is used, but the concentration is still $[\text{A}] = 10 \text{ M}$ (only volume increased, not concentration).

Run 3: 100 mL of 10 M A is mixed with 100 mL of water, so the solution gets diluted to double the volume.

So the new concentration becomes:

$$[\text{A}] = \frac{10 \times 100}{100 + 100} = 5 \text{ M}$$

Since $\text{Rate} = k[\text{A}]$, the rates are:

$$\text{Run 1: Rate} = k(10)$$

$$\text{Run 2: Rate} = k(10)$$

$$\text{Run 3: Rate} = k(5)$$

So, the correct variation is:

$$\text{Rate}_1 = \text{Rate}_2 > \text{Rate}_3$$

Question16

Correct statements regarding Arrhenius equation among the following are :

A. Factor $e^{-E_a/RT}$ corresponds to fraction of molecules having kinetic energy less than E_a .

B. At a given temperature, lower the E_a , faster is the reaction.

C. Increase in temperature by about 10°C doubles the rate of reaction.

D. Plot of $\log k$ vs $\frac{1}{T}$ gives a straight line with slope $= -\frac{E_a}{R}$.

Choose the correct answer from the options given below :

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Options:

A.

A and B Only

B.

B and D Only

C.

B and C Only

D.

A and C Only

Answer: C

Solution:

Arrhenius equation (NCERT) is

$$k = Ae^{-E_a/RT}$$

and in base-10 logarithm form,

$$\log k = \log A - \frac{E_a}{2.303 R} \left(\frac{1}{T} \right)$$

Now check each statement:

A. $e^{-E_a/RT}$ corresponds to fraction of molecules having kinetic energy **less than** E_a .

Incorrect. This factor represents the fraction of molecules having energy **equal to or greater than** E_a (able to cross the energy barrier).

B. At a given temperature, lower the E_a , faster is the reaction.

Correct. From $k = Ae^{-E_a/RT}$, smaller $E_a \Rightarrow$ larger $e^{-E_a/RT} \Rightarrow$ larger k .

C. Increase in temperature by about 10°C doubles the rate of reaction.

Correct as a common empirical rule mentioned in NCERT: for many reactions, rate approximately doubles for a 10°C rise.

D. Plot of $\log k$ vs $\frac{1}{T}$ gives a straight line with slope $= -\frac{E_a}{R}$.

Incorrect. For $\log$ (base 10), slope is

$$-\frac{E_a}{2.303 R}$$

The slope $-\frac{E_a}{R}$ is for plotting $\ln k$ vs $\frac{1}{T}$.

So, correct statements are **B and C only**.

Correct option: Option C.

Question17

Observe the following reactions at $T(\text{K})$.

I. $A \rightarrow \text{products}$.



Both the reactions are started at 10.00 am . The rates of these reactions at 10.10 am are same. The value of $-\frac{\Delta[\text{Br}^-]}{\Delta t}$ at 10.10 am is $2 \times 10^{-4} \text{ mol L}^{-1} \text{ min}^{-1}$. The concentration of A at 10.10 am is $10^{-2} \text{ mol L}^{-1}$. What is the first order rate constant (in min^{-1}) of reaction I ?

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Options:

A.

$$4 \times 10^{-3}$$

B.

$$2 \times 10^{-3}$$

C.

$$10^{-3}$$

D.

$$10^{-2}$$

Answer: A

Solution:

At $t = 10$ minutes (i.e., at 10:10 am)

For reaction II, 5 Br^- are consumed when the reaction proceeds once. So, the rate of reaction is related to the rate of decrease of $[\text{Br}^-]$ as:

$$\begin{aligned}\text{Rate of reaction} &= -\frac{1}{5} \frac{\Delta [\text{Br}^-]}{\Delta t} = \frac{1}{5} \times (2 \times 10^{-4}) \\ &= 4 \times 10^{-5}\end{aligned}$$

Now consider reaction I: $\text{A} \rightarrow \text{products}$.

At $t = 10$ minutes, both reactions have the same rate. So, the rate of reaction I at this time is also $4 \times 10^{-5} \text{ mol L}^{-1} \text{ min}^{-1}$.

Since reaction I is first order, its rate law is:

$$\text{Rate of reaction} = k[\text{A}]$$

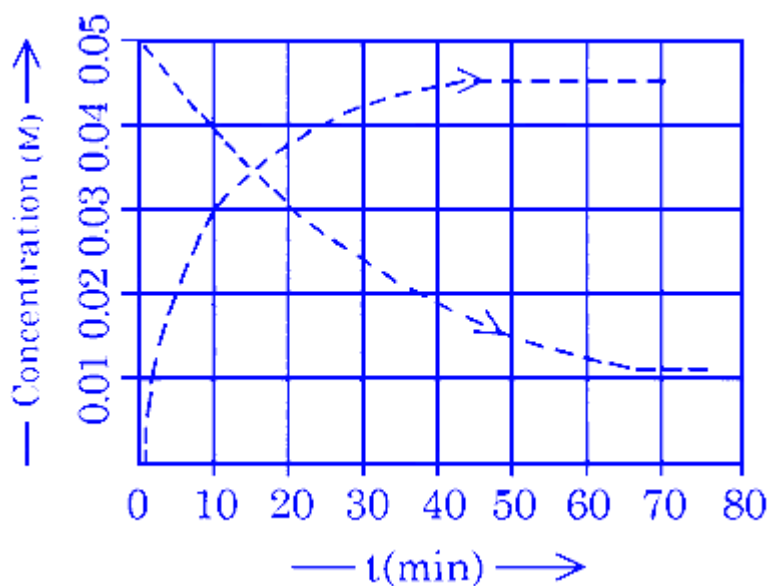
Given $[\text{A}] = 10^{-2} \text{ mol L}^{-1}$ at 10:10 am, we write:

$$\text{Rate of reaction} = 4 \times 10^{-5} = k[\text{A}]$$

So,

$$k = \frac{4 \times 10^{-5}}{10^{-2}} = 4 \times 10^{-3} \text{ min}^{-1}.$$

Question18



Given above is the concentration vs time plot for a dissociation reaction : $A \rightarrow nB$.

Based on the data of the initial phase of the reaction (initial 10 min), the value of n is _____ .

JEE Main 2026 (Online) 23rd January Evening Shift

Options:

A.

2

B.

5

C.

4

D.

3

Answer: D

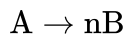
Solution:

From the graph, in the first 10 minutes the concentration of A decreases from 0.05 to 0.04.

So, amount of A dissociated in 10 minutes = $0.05 - 0.04 = 0.01$.

For the reaction $A \rightarrow nB$, when 0.01 of A reacts, B formed = $0.01 \times n$.

From the graph, in the same time interval, concentration of B formed becomes 0.03. So we equate $0.01 \times n$ to 0.03 and find n .



$$0.05 \quad 0$$

$$0.04 \quad 0.01 \times n$$

$$0.01 \times n = 0.03$$

$$n = 3$$

Question19

At 27°C in presence of a catalyst, activation energy of a reaction is lowered by 10 kJ mol^{-1} . The logarithm of ratio of $\frac{k(\text{catalysed})}{k(\text{uncatalysed})}$ is....

(Consider that the frequency factor for both the reactions is same)

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Options:

A.

$$1.741$$

B.

$$0.1741$$

C.

$$17.41$$

D.

$$3.482$$

Answer: A

Solution:

Arrhenius equation (NCERT):

$$k = A e^{-\frac{E_a}{RT}}$$

Given: catalyst lowers activation energy by $10 \text{ kJ mol}^{-1} = 10000 \text{ J mol}^{-1}$ and frequency factor A is same.

So,

$$\frac{k_{\text{cat}}}{k_{\text{uncat}}} = \frac{A e^{-\frac{E_{a,\text{cat}}}{RT}}}{A e^{-\frac{E_{a,\text{uncat}}}{RT}}} = e^{\frac{E_{a,\text{uncat}}}{RT} - \frac{E_{a,\text{cat}}}{RT}} = e^{\frac{10000}{RT}}$$

At 27°C , $T = 300 \text{ K}$ and $R = 8.314 \text{ J mol}^{-1}\text{K}^{-1}$:

$$\frac{10000}{RT} = \frac{10000}{8.314 \times 300} \approx 4.009$$

So,

$$\log_{10} \left(\frac{k_{\text{cat}}}{k_{\text{uncat}}} \right) = \log_{10} (e^{4.009}) = \frac{4.009}{2.303} \approx 1.741$$

Correct option: A (1.741)

Question20

$A \rightarrow D$ is an endothermic reaction occurring in three steps (elementary).

(i) $A \rightarrow B \Delta H_i = +\text{ve}$

(ii) $B \rightarrow C \Delta H_{ii} = -\text{ve}$

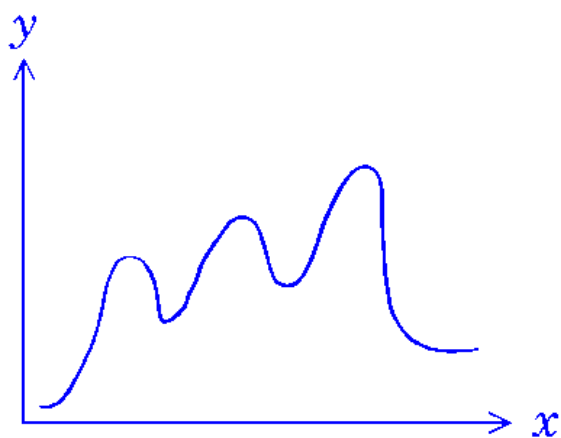
(iii) $C \rightarrow D \Delta H_{iii} = -\text{ve}$

Which of the following graphs between potential energy (y -axis) vs reaction coordinate (x -axis) correctly represents the reaction profile of $A \rightarrow D$?

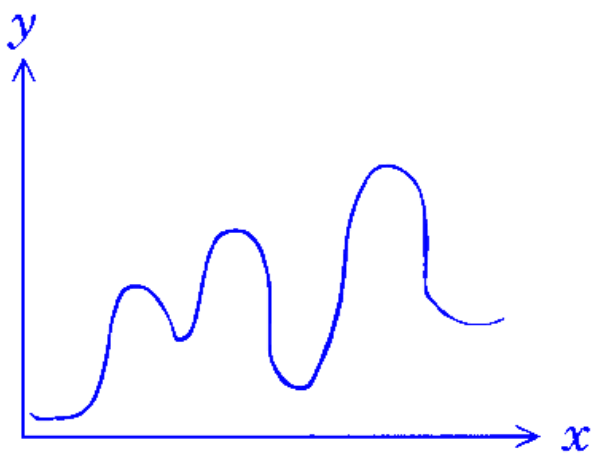
JEE Main 2026 (Online) 24th January Morning Shift

Options:

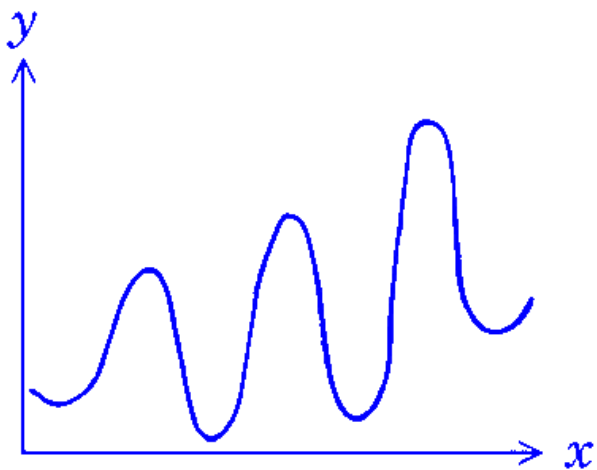
A.

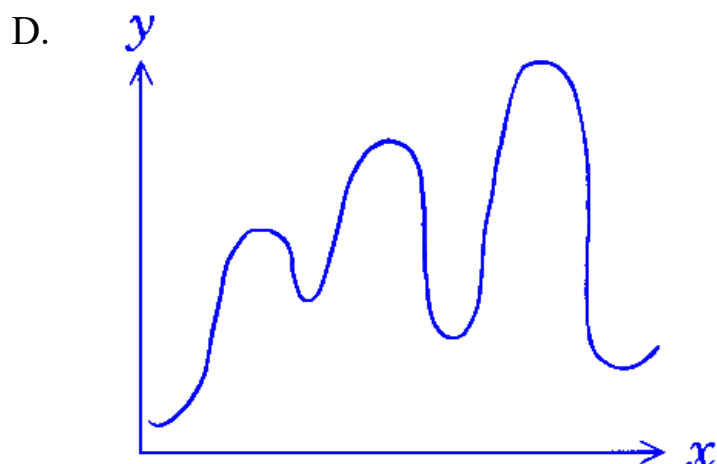


B.



C.

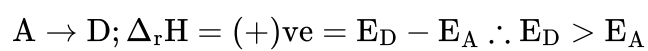




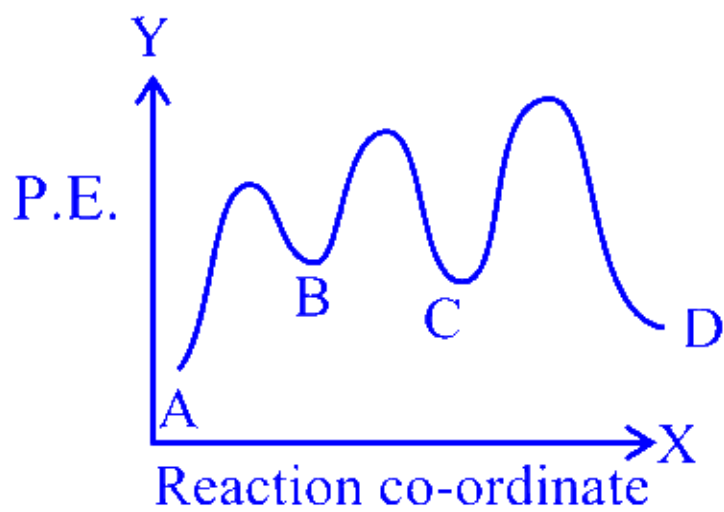
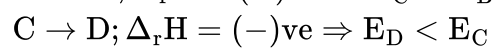
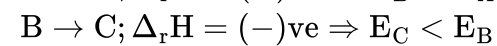
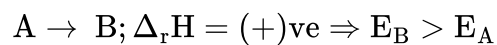
Answer: D

Solution:

Given :



Mechanism



Question21

An organic compound undergoes first order decomposition. The time taken for decomposition to $\left(\frac{1}{8}\right)^{\text{th}}$ and $\left(\frac{1}{10}\right)^{\text{th}}$ of its initial concentration are $t_{1/8}$ and $t_{1/10}$ respectively.

What is the value of $\frac{t_{1/8}}{t_{1/10}} \times 10$?

($\log 2 = 0.3$)

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Options:

A.

30

B.

9

C.

3

D.

0.9

Answer: B

Solution:

For a first order reaction, the integrated rate law is

$$t = \frac{2.303}{k} \log \left(\frac{[A]_0}{[A]_t} \right)$$

1) Time to reach $\left(\frac{1}{8}\right)^{\text{th}}$ of initial concentration

$$\text{Here, } [A]_t = \frac{[A]_0}{8}$$

$$t_{1/8} = \frac{2.303}{k} \log \left(\frac{[A]_0}{[A]_0/8} \right) = \frac{2.303}{k} \log(8)$$

$$\text{Now, } \log(8) = \log(2^3) = 3 \log 2 = 3 \times 0.3 = 0.9$$

So,

$$t_{1/8} = \frac{2.303}{k} \times 0.9$$

2) Time to reach $\left(\frac{1}{10}\right)^{\text{th}}$ of initial concentration

$$\text{Here, } [A]_t = \frac{[A]_0}{10}$$

$$t_{1/10} = \frac{2.303}{k} \log \left(\frac{[A]_0}{[A]_0/10} \right) = \frac{2.303}{k} \log(10)$$

$$\text{And } \log(10) = 1, \text{ so}$$

$$t_{1/10} = \frac{2.303}{k} \times 1$$

3) Required ratio

$$\frac{t_{1/8}}{t_{1/10}} = \frac{\frac{2.303}{k} \times 0.9}{\frac{2.303}{k} \times 1} = 0.9$$

So,

$$\frac{t_{1/8}}{t_{1/10}} \times 10 = 0.9 \times 10 = 9$$

Correct option: B (9)

Question22

$t_{100\%}$ is the time required for the 100% completion of the reaction while $t_{1/2}$ is the time required for 50% of the reaction to be

completed. Which of the following option correctly represents the relation between $t_{100\%}$ and $t_{1/2}$ for zero and first order reactions respectively?

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Options:

A.

$$t_{100\%} = (t_{1/2})^2 \text{ and } t_{100\%} = (t_{1/2})^{-\infty}$$

B.

$$t_{100\%} = 2t_{1/2} \text{ and } t_{100\%} = (t_{1/2})^{\infty}$$

C.

$$t_{100\%} = 2t_{1/2} \text{ and } t_{100\%} = (2t_{1/2})^2$$

D.

$$t_{100\%} = (t_{1/2})^{\infty} \text{ and } t_{100\%} = 2t_{1/2}$$

Answer: B

Solution:

For a **zero order reaction**:

The integrated rate law is

$$[A]_t = [A]_0 - kt$$

For **100% completion**, $[A]_t = 0$.

So,

$$0 = [A]_0 - kt_{100\%}$$

$$t_{100\%} = \frac{[A]_0}{k}$$

For **half-life**, $[A]_t = \frac{[A]_0}{2}$.

So,

$$\frac{[A]_0}{2} = [A]_0 - kt_{1/2}$$

$$kt_{1/2} = \frac{[A]_0}{2}$$

$$t_{1/2} = \frac{[A]_0}{2k}$$

Hence,

$$t_{100\%} = \frac{[A]_0}{k} = 2 \left(\frac{[A]_0}{2k} \right) = 2t_{1/2}$$

So for **zero order**,

$$t_{100\%} = 2t_{1/2}$$

For a **first order reaction**:

The integrated rate law is

$$[A]_t = [A]_0 e^{-kt}$$

For **100% completion**, we need $[A]_t = 0$.

That means

$$0 = [A]_0 e^{-kt}$$

But e^{-kt} can never become exactly zero for any finite value of t .

So, the reaction takes **infinite time** for complete consumption.

Thus,

$$t_{100\%} = \infty$$

Also, for first order reaction,

$$t_{1/2} = \frac{0.693}{k}$$

Hence $t_{100\%}$ is not a finite multiple of $t_{1/2}$, rather it is effectively represented as infinite.

So the correct relation is:

- **Zero order:** $t_{100\%} = 2t_{1/2}$
- **First order:** $t_{100\%} = (t_{1/2})^\infty$ in the given options, meaning complete reaction requires infinite time.

Therefore, the correct option is:

Option B

Question23

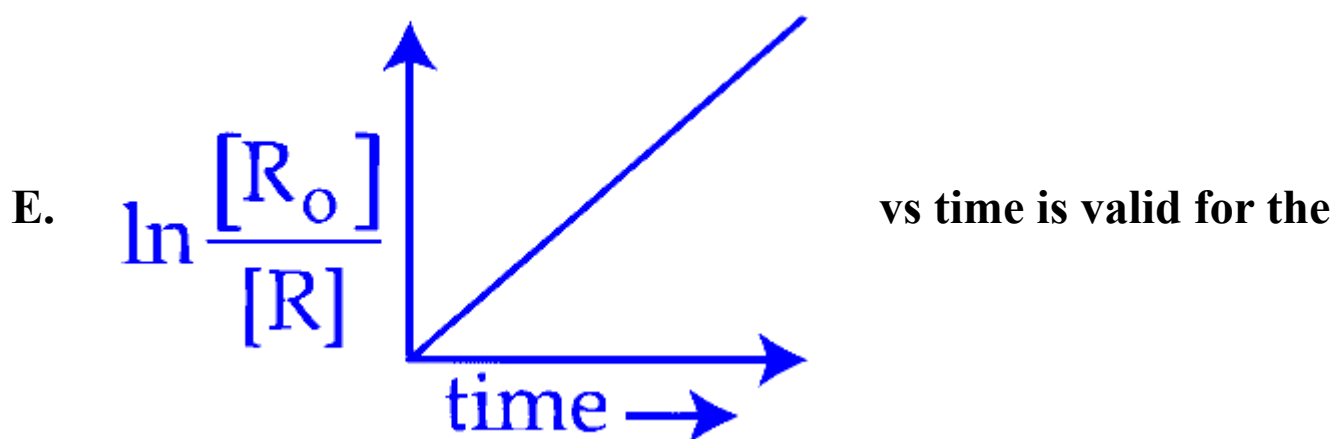
Consider the reaction $aX \rightarrow bY$, for which the rate constant at 30°C is $1 \times 10^{-3} \text{ mol}^{-1} \text{ L s}^{-1}$. Which of the following statements are true?

A. When concentration of 'X' is increased to four times, the rate of reaction becomes 16 times.

B. The reaction is a second order reaction.

C. The half-life period is independent of the concentration of X.

D. Decomposition of N_2O_5 is an example of the above reaction.



above reaction.

Choose the correct answer from the options given below:

JEE Main 2026 (Online) 2nd April Evening Shift

Options:

A.

A and B Only

B.

A, B and C Only

C.

A, B, D and E Only

D.

C and D Only

Answer: A

Solution:

The given rate constant is $k = 10^{-3} \text{ M}^{-1} \text{ s}^{-1}$.

Step 1: Determining the order of reaction

The general unit of the rate constant is $\text{M}^{1-n} \text{ s}^{-1}$, where n is the order of reaction. Here, the unit of k is $\text{M}^{-1} \text{ s}^{-1}$.

So, $1 - n = -1 \Rightarrow n = 2$. Hence, this is a **second order reaction**.

Step 2: Expression for the rate

The rate law for a second order reaction is: $\text{Rate} = k[A]^2$

Step 3: Effect of changing concentration

If the concentration of A is made 4 times, that is, $[A]' = 4[A]$, then New rate $= k(4[A])^2 = 16k[A]^2$. So, the rate increases by 16 times. Therefore, statement (A) is correct.

Step 4: Order of reaction

As we found $n = 2$, the reaction is second order. So, statement (B) is correct.

Step 5: Half-life relation

For a second order reaction, the half-life is inversely proportional to the initial concentration: $t_{1/2} = \frac{1}{k[A]_0}$. Hence, the half-life depends on concentration. So, statement (C) is incorrect because it is true only for a first order reaction.

Step 6: Example of reaction

The decomposition of N_2O_5 is a first order reaction. Thus, statement (D) is incorrect.

Step 7: Graph type

A straight-line graph of $\log \frac{[R]_0}{[R]_t}$ versus t is valid only for first order reactions, not second order. Therefore, statement (E) is incorrect.

Final Answer: The correct statements are (A) and (B).

Question24

Consider the first order reaction $\text{R} \rightarrow \text{P}$.

The fraction of molecules decomposed in the given first order reaction can be expressed as

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Options:

A.

$$1 - e^{k_1 t}$$

B.

$$1 + e^{k_1 t}$$

C.

$$1 + e^{-k_1 t}$$

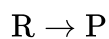
D.

$$1 - e^{-k_1 t}$$

Answer: D

Solution:

For a first order reaction



the integrated rate law is

$$[R]_t = [R]_0 e^{-k_1 t}$$

where:

- $[R]_0$ = initial concentration of reactant
- $[R]_t$ = concentration after time t

Now, the fraction of molecules left undecomposed after time t is

$$\frac{[R]_t}{[R]_0} = e^{-k_1 t}$$

So, the fraction of molecules decomposed is

$$1 - \frac{[R]_t}{[R]_0} = 1 - e^{-k_1 t}$$

Hence, the correct answer is

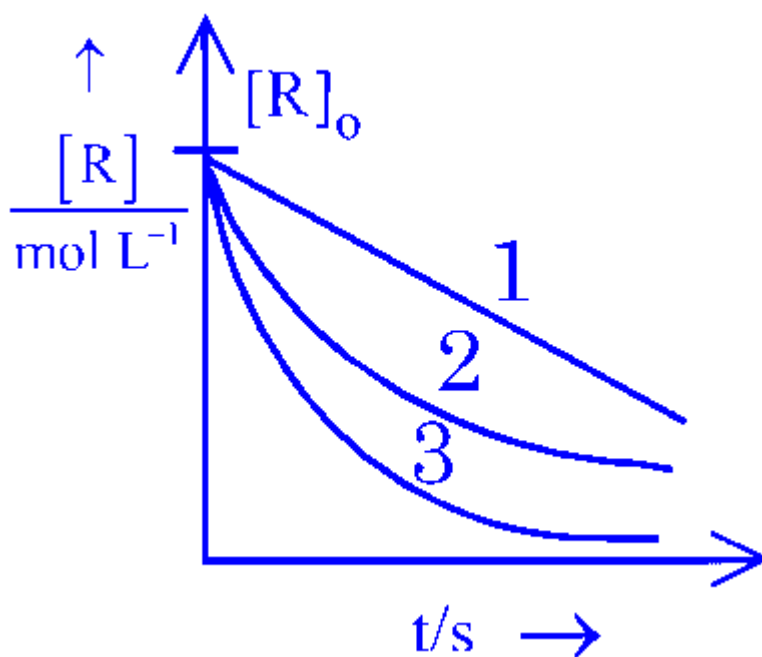
$$\boxed{1 - e^{-k_1 t}}$$

So, Option D is correct.

Question 25

Consider the given graph showing variation of reactant concentration with time.

Three different reactions were started with identical initial concentration of reactants. Which of the following statement is correct?



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Options:

A.

The order of all the three reactions is same.

B.

The rate constant of reaction 3 is larger than the rate constant of reaction 2 if the order of reaction is same for both.

C.

The SI unit of rate constant of reaction 1 is s^{-1} .

D.

Thermal decomposition of HI on gold surface is an example of reaction 2 .

Answer: B

Solution:

(A) The order of the reactions cannot be the same because curve (1) is a straight line, but curves (2) and (3) decrease exponentially with time.

A straight-line decrease of reactant concentration with time is the characteristic of a **zero order** reaction. Exponentially decaying curves are usually seen in **first order** (and sometimes other) reactions.

Rate expression for $n = 0$:

$$\begin{aligned}Kt &= [A_0] - [A] = [R_0] - [R] \\[R] &= [R_0] - kt \quad (\because y = c + mx) \\ \text{slope } (m) &= -k\end{aligned}$$

So, graph (1) matches the straight-line form $[R] = [R_0] - kt$, hence reaction (1) is zero order.

(B) Now assume that reactions (2) and (3) are first order reactions (their curves look exponential).

$$\begin{aligned}[R] &= [R_0]e^{-kt} \\ \text{At } t &= \text{constant} \\ [R_3] &< [R_2]\end{aligned}$$

At the same time t , curve (3) shows a smaller concentration than curve (2), i.e. $[R_3] < [R_2]$. This means reaction (3) is consuming reactant faster, so it has a larger rate constant.

And since Rate of reaction:

$$\begin{aligned}R &= K[A] \text{ or } K[R] \\ K &\propto \frac{1}{[R]}\end{aligned}$$

Hence, $K_3 > K_2$.

(c) SI unit for rate constant (k) = $\text{mol}^{1-n} \text{L}^{n-1} \text{s}^{-1}$. For 1st reaction order (n) = 0.

$$\begin{aligned}\text{Then si unit} &= \text{mol}^{1-0} \text{L}^{0-1} \text{s}^{-1} \\ &= \text{mol L}^{-1} \text{s}^{-1}.\end{aligned}$$

(D) Thermal decomposition of HI on Gold is an example of zero-order reaction because its rate expression is written as:

$$R = K[HI]^0$$

Question26

First order gas phase reaction



p_t = initial pressure of gas A, p_t = total pressure of the reaction mixture at time t

Expression of rate constant (k) is

JEE Main 2026 (Online) 5th April Morning Shift

Options:

A.

$$\frac{1}{t} \ln \frac{p_i}{2p_i - p_t}$$

B.

$$\frac{1}{t} \ln \frac{2p_i}{p_i - p_t}$$

C.

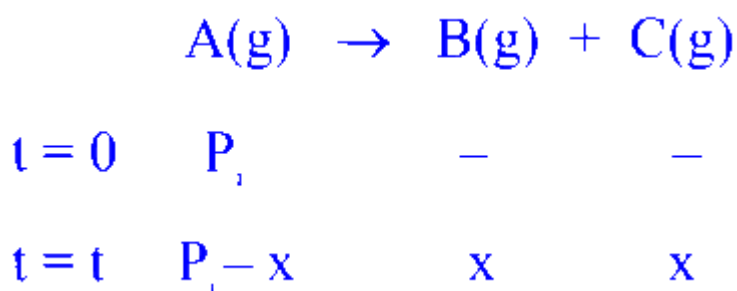
$$\frac{1}{t} \ln \frac{p_i}{3p_i - 2p_t}$$

D.

$$\frac{1}{t} \ln \frac{3p_i}{4p_i - p_t}$$

Answer: A

Solution:



So, total pressure at time t is

$$p_t = (p_i - x) + x + x = p_i + x$$

Hence,

$$x = p_t - p_i$$

Now pressure of A at time t is

$$p_A = p_i - x$$

Substituting $x = p_t - p_i$:

$$p_A = p_i - (p_t - p_i) = 2p_i - p_t$$

For a first-order reaction,

$$k = \frac{1}{t} \ln \frac{[A]_0}{[A]_t}$$

Since pressure is proportional to concentration for gases,

$$k = \frac{1}{t} \ln \frac{p_i}{2p_i - p_t}$$

Therefore, the correct expression is

$$k = \frac{1}{t} \ln \frac{p_i}{2p_i - p_t}$$

So, the correct option is:

Option A

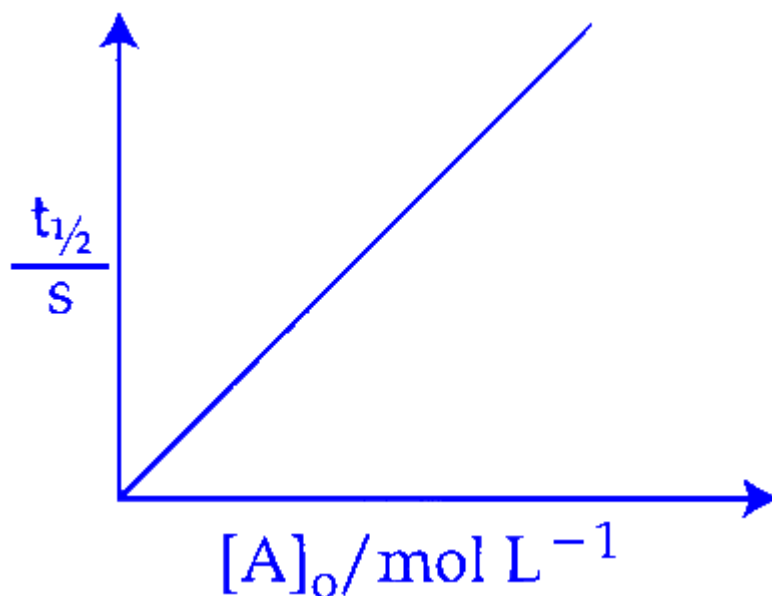
Question27

Given below are two statements :

$$R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1} \text{ and } 1\text{cal} = 4.2 \text{ J}$$

Statement I : When $E_a = 12.6\text{kcal/mol}$, the room temperature rate constant is doubled by a 10°C increase in temperature (298 K to 308 K)

Statement II : For a first order reactions $A \rightarrow B$,



Here $[A]_0$ is the initial concentration of A and $t_{1/2}$ is half life of reaction.

In the light of the above statements, choose the correct answer from the options given below :

JEE Main 2026 (Online) 8th April Evening Shift

Options:

A.

Both Statement I and Statement II are true

B.

Both Statement I and Statement II are false

C.

Statement I is true but Statement II is false

D.

Statement I is false but Statement II is true

Answer: C

Solution:

$$\text{S-I : } \ln \left(\frac{K_2}{K_1} \right) = \frac{12.6 \times 10^3}{\left(\frac{8.314}{4.2} \right)} \left[\frac{10}{298 \times 308} \right]$$

$$\Rightarrow \ln \left(\frac{K_2}{K_1} \right) = 0.693$$

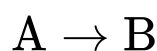
$$\Rightarrow \frac{K_2}{K_1} = 2$$

S-II : For 1st Order

$$t_{1/2} = \frac{\ln 2}{K}; t_{1/2} \propto [A]^0 = \text{constant.}$$

Statement-I is true but statement-II is false.

Question28



The molecule A changes into its isomeric form B by following a first order kinetics at a temperature of 1000 K . If the energy barrier with respect to reactant energy for such isomeric transformation is $191.48 \text{ kJ mol}^{-1}$ and the frequency factor is 10^{20} , the time required for 50% molecules of A to become B is _____ picoseconds (nearest integer). $\left[R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1} \right]$

JEE Main 2025 (Online) 22nd January Morning Shift

Answer: 69

Solution:

To determine the time required for 50% of molecule A to change into its isomeric form B, follow these steps:

Half-life Formula for First Order Kinetics:

The half-life ($t_{1/2}$) for a first-order reaction is given by:

$$t_{1/2} = \frac{0.693}{K}$$

Calculate the Rate Constant (K):

The rate constant K can be calculated using the Arrhenius equation:

$$K = A \cdot e^{-\frac{E_a}{RT}}$$

Given:

$$A \text{ (frequency factor)} = 10^{20}$$

$$E_a \text{ (activation energy)} = 191.48 \text{ kJ/mol} = 191.48 \times 10^3 \text{ J/mol}$$

$$R \text{ (universal gas constant)} = 8.314 \text{ J/mol} \cdot \text{K}$$

$$T = 1000 \text{ K}$$

Substitute the values into the Arrhenius equation:

$$K = 10^{20} \times e^{-\frac{191.48 \times 10^3}{8.314 \times 1000}}$$

Simplify this calculation:

$$K = 10^{20} \times e^{-23.031}$$

Simplifying further by recognizing that $e^{-23.031}$ is a very small number, gives:

$$K \approx \frac{10^{20}}{10^{10}} = 10^{10} \text{ sec}^{-1}$$

Calculate the Half-life:

Using the calculated value of K :

$$t_{1/2} = \frac{0.693}{10^{10}} = 6.93 \times 10^{-11} \text{ seconds}$$

Convert to Picoseconds:

Since 1 second = 10^{12} picoseconds:

$$t_{1/2} = 6.93 \times 10^{-11} \times 10^{12} \text{ picoseconds} = 69.3 \text{ picoseconds}$$

Therefore, the time required for 50% of the molecules of A to become B is approximately 69 picoseconds (nearest integer).

Question29

For the thermal decomposition of $\text{N}_2\text{O}_5(\text{g})$ at constant volume, the following table can be formed, for the reaction mentioned below.



Sr. No.	Time/s	Total pressure/(atm)
1	0	0.6
2	100	'x'

$$x = \underline{\hspace{2cm}} \times 10^{-3} \text{ atm [nearest integer]}$$

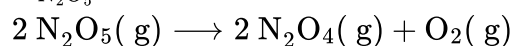
Given : Rate constant for the reaction is $4.606 \times 10^{-2} \text{ s}^{-1}$.

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Answer: 897

Solution:

$$K_{\text{N}_2\text{O}_5} = 2 \times 4.606 \times 10^{-2} \text{ S}^{-1}$$



$$P_i \quad 0.6 \quad 0 \quad 0$$

$$P_f \quad 0.6 - P \quad P \quad \frac{P}{2}$$

$$2 \times 4.606 \times 10^{-2} = \frac{2.303}{100} \log \frac{0.6}{0.6 - P}$$

$$4 \log_{10} \frac{0.6}{0.6 - P}$$

$$10^4 = \frac{0.6}{0.6 - P}$$

$$\Rightarrow 0.6 \times 10^4 - 10^4 P = 0.6$$

$$\Rightarrow 10^4 P = 0.6 (10^4 - 1)$$

$$P = (6000 - 0.6) \times 10^{-4}$$

$$= 5999. \times 10^{-4}$$

$$= 0.59994$$

$$P_{\text{Total}} = 0.6 + \frac{P}{2}$$

$$= 0.6 + 0.29997$$

$$= 0.89997$$

$$= 899.97 \times 10^{-3}$$

Ans. 900



$$t = 0 \quad 0.6 \quad 0 \quad 0$$

$$t = 100\text{s} \quad 0.6 - x \quad x \quad x/2$$

$$P_{\text{Total}} = 0.6 + \frac{x}{2}$$

As given in equation

$$K_r = 4.606 \times 10^{-2} \text{sec}^{-1}$$

(Here language conflict in question)

$$(K_r = \frac{KA}{2} \text{ not considered})$$

$$K_r t = \ln \frac{0.6}{0.6 - x}$$

$$4.606 \times 10^{-2} \times 100 = 2.303 \log \frac{0.6}{0.6 - x}$$

$$\begin{aligned} P_{\text{Total}} &= 0.6 + \frac{0.594}{2} = 0.897 \text{ atm} \\ &= 897 \times 10^{-3} \text{ atm} \end{aligned}$$

Question30

Consider a complex reaction taking place in three steps with rate constants k_1 , k_2 and k_3 respectively. The overall rate constant k is given by the expression $k = \sqrt{\frac{k_1 k_3}{k_2}}$. If the activation energies of the three steps are 60, 30 and 10 kJ mol⁻¹ respectively, then the overall energy of activation in kJmol⁻¹ is _____. (Nearest integer)

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Answer: 20

Solution:

To determine the overall energy of activation for the given complex reaction with rate constants k_1 , k_2 , and k_3 , we start with the expression for the overall rate constant k :

$$k = \sqrt{\frac{k_1 k_3}{k_2}}$$

The rate constant can also be expressed in terms of the Arrhenius equation:

$$A \cdot e^{-E_a/RT} = \sqrt{\frac{A_1 e^{-E_{a1}/RT} \cdot A_3 e^{-E_{a3}/RT}}{A_2 e^{-E_{a2}/RT}}}$$

By comparing the exponential terms from both sides, we have:

$$\frac{E_a}{RT} = \frac{1}{2} \left(\frac{E_{a1}}{RT} + \frac{E_{a3}}{RT} - \frac{E_{a2}}{RT} \right)$$

This simplifies to:

$$E_a = \frac{E_{a_1} + E_{a_3} - E_{a_2}}{2}$$

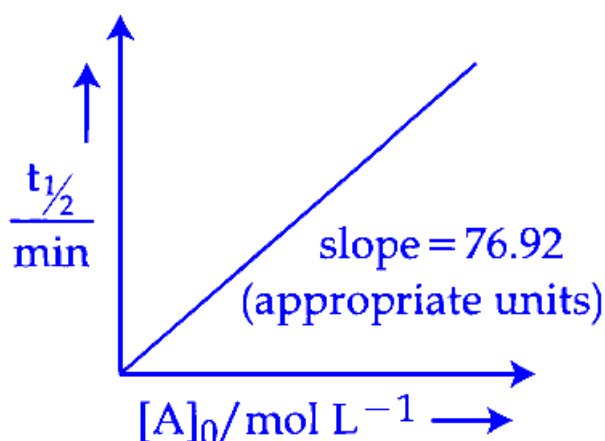
Substituting the given activation energies— $E_{a_1} = 60 \text{ kJ mol}^{-1}$, $E_{a_2} = 30 \text{ kJ mol}^{-1}$, and $E_{a_3} = 10 \text{ kJ mol}^{-1}$:

$$E_a = \frac{60+10-30}{2} = \frac{40}{2} = 20 \text{ kJ mol}^{-1}$$

Therefore, the overall energy of activation is 20 kJ/mol.

Question31

For the reaction $A \rightarrow \text{products}$.



The concentration of A at 10 minutes is _____ $\times 10^{-3} \text{ mol L}^{-1}$ (nearest integer). The reaction was started with 2.5 mol L^{-1} of A .

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Answer: 2435

Solution:

Order of Reaction:

Since $t_{1/2} \propto [A]_0$, the reaction follows a zero-order kinetics.

Half-life Equation for Zero-order Reaction:

The half-life ($t_{1/2}$) is calculated as:

$$t_{1/2} = \frac{[A]_0}{2K}$$

Given the slope from the graph is 76.92, which equals $\frac{1}{2K}$. This implies:

$$K = \frac{1}{2 \times 76.92}$$

Concentration of A at 10 Minutes:

Apply the zero-order kinetics equation:

$$[A]_{10} = -Kt + [A]_0$$

Substituting the values:

$$[A]_{10} = -\left(\frac{1}{2 \times 76.92}\right) \times 10 + 2.5 = 2.435 \text{ mol L}^{-1}$$

Final Concentration:

Convert to scientific notation:

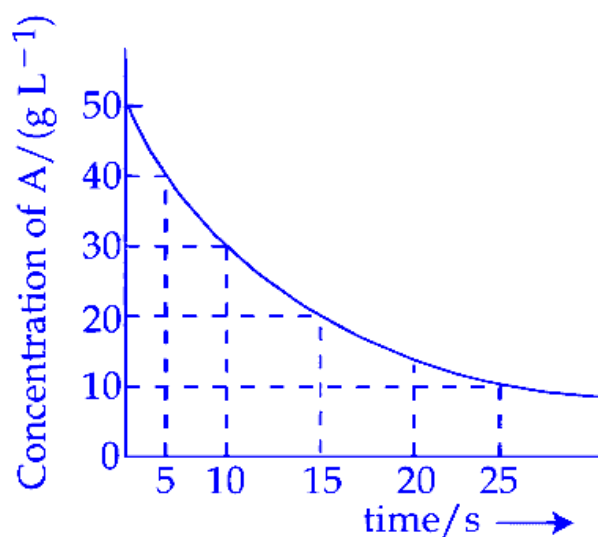
$$[A]_{10} = 2435 \times 10^{-3} \text{ mol L}^{-1}$$

Thus, the concentration of A at 10 minutes is approximately $2435 \times 10^{-3} \text{ mol L}^{-1}$.

Question32

For the reaction $A \rightarrow B$ the following graph was obtained. The time required (in seconds) for the concentration of A to reduce to 2.5 g L^{-1} (if the initial concentration of A was 50 g L^{-1}) is .
(Nearest integer)

Given : $\log 2 = 0.3010$



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Answer: 43

Solution:

To determine the time required for the concentration of A to decrease from an initial value of 50 g L^{-1} to 2.5 g L^{-1} in the reaction $A \rightarrow B$, we assume first-order kinetics. Although the graph does not provide a clear indication of the reaction order over the intervals $0 - 5$, $5 - 10$, and $10 - 15$ seconds, where the order appears to be zero, we'll proceed with the assumption of first-order kinetics, since the graph is not a straight line.

The rate constant K for first-order reactions can be calculated using the formula:

$$K = \frac{1}{t} \ln \frac{A_0}{A_t}$$

For the interval where $A_0 = 40 \text{ g/L}$ and $A_t = 20 \text{ g/L}$ after 10 seconds:

$$K = \frac{1}{10} \ln \frac{40}{20}$$

Now, to find the time t required for the concentration to reduce to 2.5 g/L :

$$K = \frac{1}{t} \ln \frac{50}{2.5}$$

Equating the two expressions for K :

$$\frac{1}{10} \ln 2 = \frac{1}{t} \ln 20$$

Solving for t :

$$t = \frac{1.3010 \times 10}{0.3010} = 43.3 \text{ sec}$$

Therefore, the time required for the concentration to decrease to 2.5 g/L is approximately 43 seconds.

Question33

Which of the following statement is not true for radioactive decay?

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Options:

- A. Half life is $\ln 2$ times of $\frac{1}{\text{rate constant}}$.
- B. Decay constant increases with increase in temperature.
- C. Decay constant does not depend upon temperature.
- D. Amount of radioactive substance remained after three half lives is $\frac{1}{8}$ th of original amount.

Answer: B

Solution:

Answer: Option B is not true.

Explanation:

Option A states that the half-life $t_{1/2}$ is $\ln 2$ times $\frac{1}{k}$, where k is the decay (rate) constant.

$$t_{1/2} = \frac{\ln 2}{k} = \ln 2 \times \frac{1}{k}.$$

This is correct.

Option B states that the decay constant k increases with an increase in temperature. However, for radioactive (nuclear) decay processes, the decay constant is essentially **independent of external factors** such as temperature and pressure. Hence, saying that it increases with temperature is **not true**.

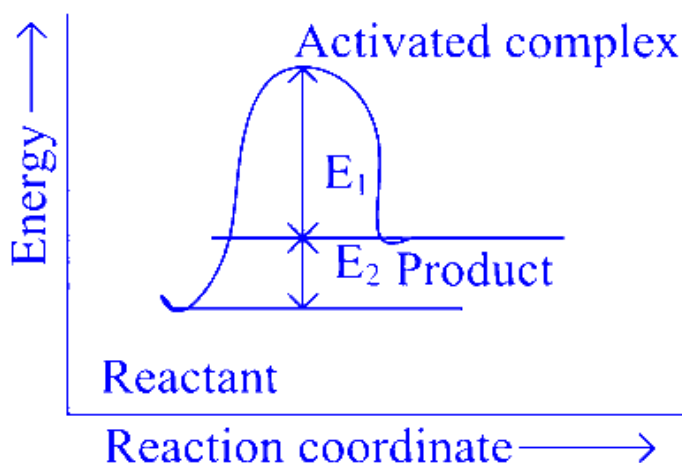
Option C states that the decay constant does not depend upon temperature, which **is** true for radioactive decay.

Option D states that the amount of radioactive substance left after three half-lives is $\frac{1}{8}$ of the original. Since one half-life leaves $\frac{1}{2}$ of the sample, three half-lives leave $\left(\frac{1}{2}\right)^3 = \frac{1}{8}$. This is correct.

Therefore, the statement that is **not** true is **Option B**.

Question34

Consider the given figure and choose the correct option :



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Options:

- A. Activation energy of both forward and backward reaction is $E_1 + E_2$ and reactant is more stable than product.
- B. Activation energy of forward reaction is $E_1 + E_2$ and product is more stable than reactant.
- C. Activation energy of backward reaction is E_1 and product is more stable than reactant.
- D. Activation energy of forward reaction is $E_1 + E_2$ and product is less stable than reactant.

Answer: D

Solution:

E_1 represents the activation energy for the backward reaction.

$E_1 + E_2$ is the activation energy for the forward reaction.

The product has more energy than the reactant, indicating that the product is less stable compared to the reactant.

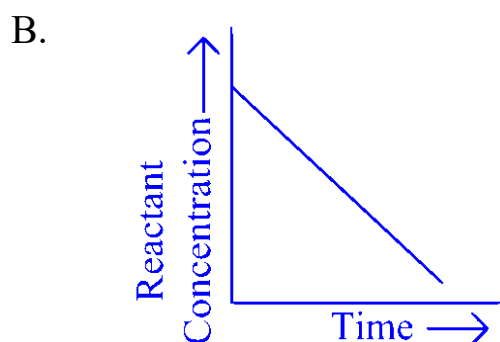
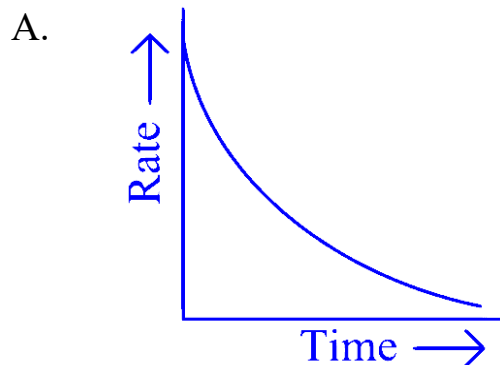
This information is crucial in determining the relative stability of reactants and products and the required energy for the reactions to proceed in either direction.

Question35

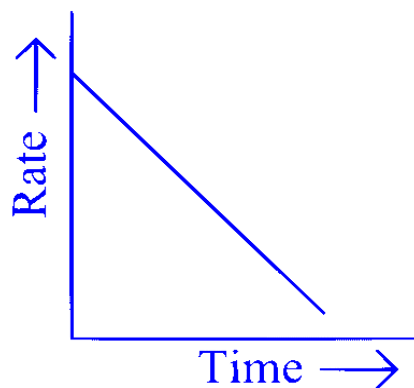
Which of the following graphs most appropriately represents a zero order reaction?

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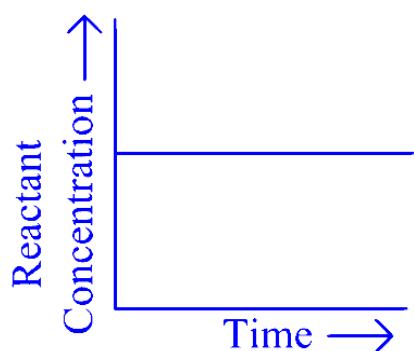
Options:



C.



D.



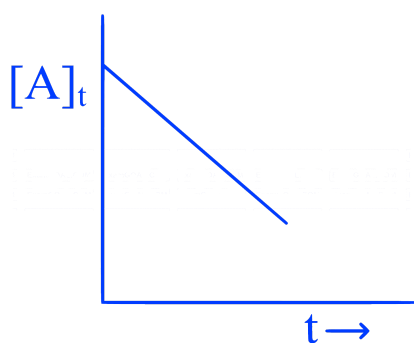
Answer: B

Solution:

For zero order reaction : $A \rightarrow P$

$$\text{Rate} = k$$

$$[A]_t = a_0 - kt$$



Question36

For a reaction, $\text{N}_2\text{O}_{5(\text{g})} \rightarrow 2\text{NO}_{2(\text{g})} + \frac{1}{2}\text{O}_{2(\text{g})}$ in a constant volume container, no products were present initially. The final pressure of the system when 50% of reaction gets completed is

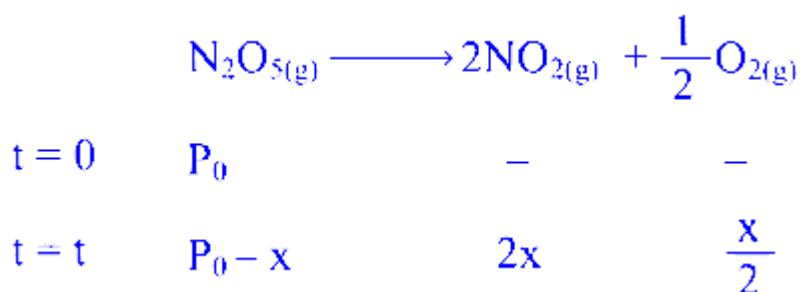
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Options:

- A. $5/2$ times of initial pressure
- B. $7/2$ times of initial pressure
- C. $7/4$ times of initial pressure
- D. 5 times of initial pressure

Answer: C

Solution:



$$x = \frac{P_0}{2}$$

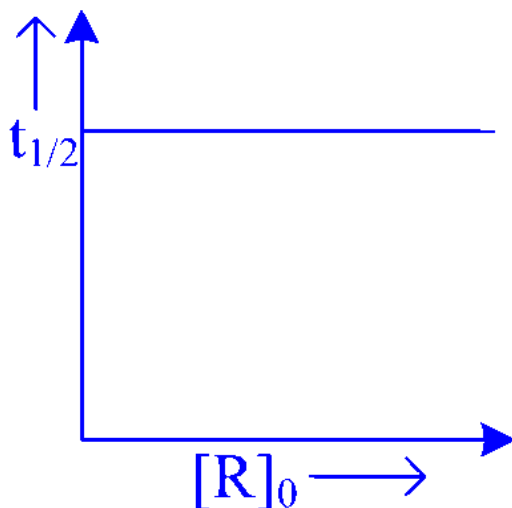
$$P_{\text{total}} = P_0 - \frac{P_0}{2} + P_0 + \frac{P_0}{4} = \frac{7}{4}P_0$$

Option (4)

Question37

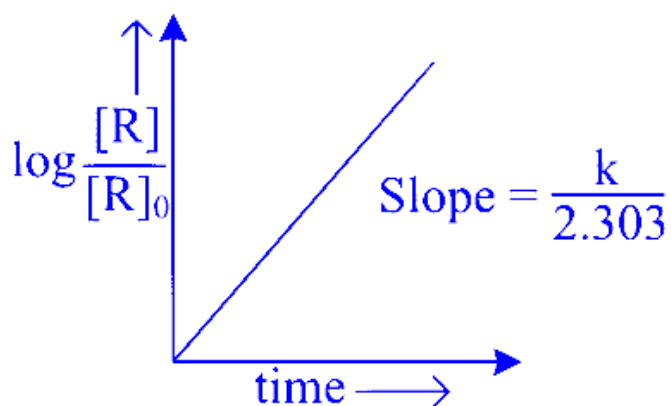
Given below are two statements :

Statement (I) :



is valid for first order reaction.

Statement (II) :



is valid for first order reaction.

In the light of the above statements, choose the correct answer from the options given below :

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Options:

- A. Both Statement I and Statement II are false
- B. Statement I is true but Statement II is false
- C. Statement I is false but Statement II is true
- D. Both Statement I and Statement II are true

Answer: B

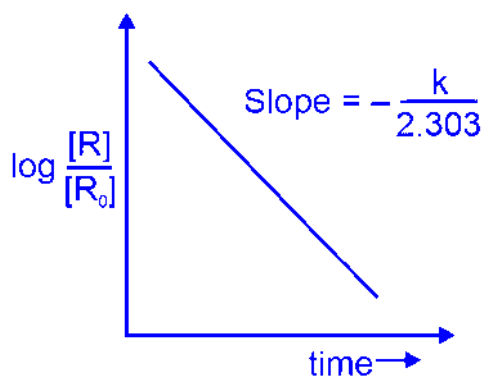
Solution:

Half life ($t_{1/2}$) of a first order reaction does not depend on initial concentration of the reactant $[R_0]$. Therefore Statement I is true. Rate constant (k) of a first order reaction is given by

$$k = \frac{2.303}{t} \log \frac{[R_0]}{[R]}$$
$$\log \frac{[R]}{[R_0]} = - \left(\frac{k}{2.303} \right) t$$

Plot of $\log \frac{[R]}{[R_0]}$ versus time is linear having

$$\text{slope} = - \frac{k}{2.303}$$



∴ Statement II is false.

Question38

$[A]_0 / \text{mol L}^{-1}$	$t_{1/2} / \text{min}$
0.100	200
0.025	100

For a given reaction $R \rightarrow P$, $t_{1/2}$ is related to $[A]_0$ as given in table.

Given: $\log 2 = 0.30$

Which of the following is true?

- A. The order of the reaction is $1/2$.
- B. If $[A]_0$ is 1 M, then $t_{1/2}$ is $200\sqrt{10}$ min
- C. The order of the reaction changes to 1 if the concentration of reactant changes from 0.100 M to 0.500 M.
- D. $t_{1/2}$ is 800 min for $[A]_0 = 1.6\text{M}$

Choose the correct answer from the options given below:

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Options:

- A. A and C Only
- B. A, B and D Only

C. A and B Only

D. C and D Only

Answer: B

Solution:

$$t_{1/2} \propto \frac{1}{A_0^{n-1}}$$

$$\frac{(t_{1/2})_1}{(t_{1/2})_2} = \frac{(A_0)_2^{n-1}}{(A_0)_1^{n-1}}$$

$$\frac{200}{100} = \left(\frac{0.025}{0.100}\right)^{n-1} \quad n - 1 = -\frac{1}{2}$$

$$2 = \left(\frac{1}{4}\right)^{n-1} \quad n = \frac{1}{2} \text{ (order)}$$

$$\Rightarrow t_{1/2} \propto \sqrt{A_0}$$

$$\frac{200}{t_{1/2}} = \frac{(0.1)^{1/2}}{(1)^{1/2}} \quad \text{when } A_0 = 1M$$

$$t_{1/2} = 200\sqrt{10} \text{ min}$$

*1st order kinetics have $t_{1/2}$ independent of their concentration. So upon changing the concentration $t_{1/2}$ should not change for first order reaction.

$$\frac{200}{t_{1/2}} = \frac{(0.1)^{1/2}}{(1.6)^{1/2}} \quad \text{when } A_0 = 1.6M$$

$$t_{1/2} = 800 \text{ min}$$

Question39

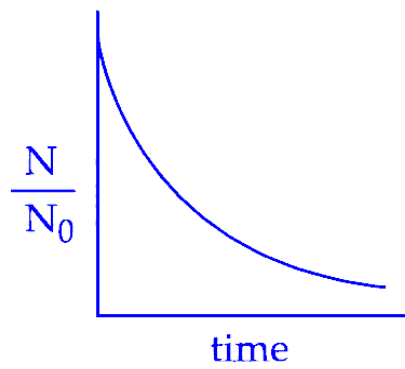
For bacterial growth in a cell culture, growth law is very similar to the law of radioactive decay. Which of the following graphs is most suitable to represent bacterial colony growth ?

Where N - Number of Bacteria at any time, N_0 - Initial number of Bacteria.

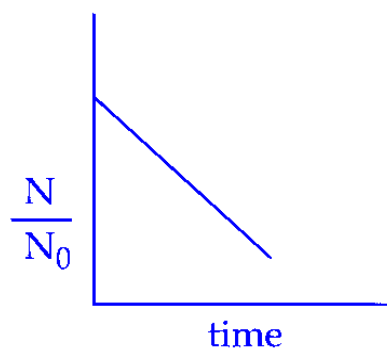
JEE Main 2025 (Online) 28th January Evening Shift

Options:

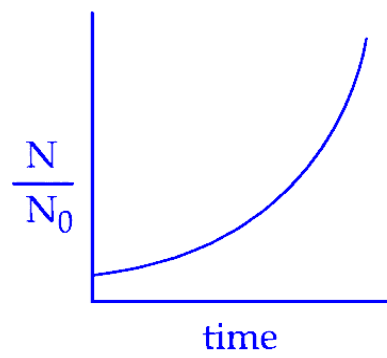
A.



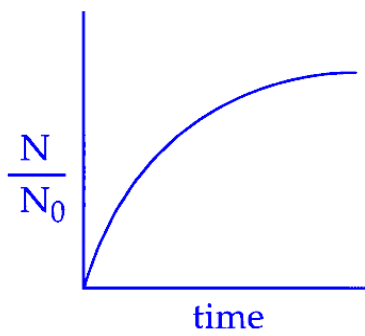
B.



C.



D.



Answer: C

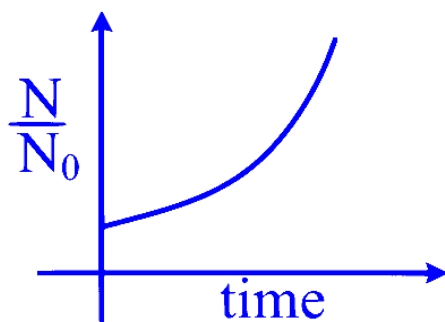
Solution:

Because no. of bacteria initial = N_0 and No. of bacteria at any time $t = N$

Since bacterial growth is given as

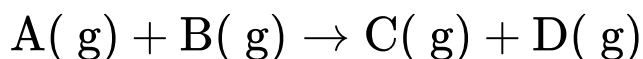
$$N = N_0 e^{Kt}$$

Where K = growth constant for bacterial growth



Question40

Consider an elementary reaction



If the volume of reaction mixture is suddenly reduced to $\frac{1}{3}$ of its initial volume, the reaction rate will become ' x ' times of the original reaction rate. The value of x is :

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Options:

A.

B.

$$9$$

C.

$$\frac{1}{3}$$

D.

$$\frac{1}{9}$$

Answer: B

Solution:

$$R_1 = K[A]^1[B]^1$$

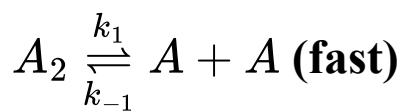
$$R_1 = K \left[\frac{n_A}{V} \right]^1 \left[\frac{n_B}{V} \right]^1$$

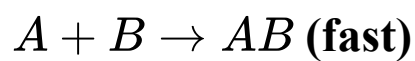
$$R_2 = K \left[\frac{3n_A}{V} \right]^1 \left[\frac{3n_B}{V} \right]^1$$

$$R_2 = 9R_1$$

Question41

The reaction $A_2 + B_2 \rightarrow 2AB$ follows the mechanism:





The overall order of the reaction is:

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Options:

A.

3

B.

2.5

C.

1.5

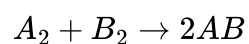
D.

2

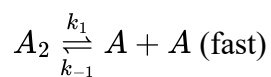
Answer: C

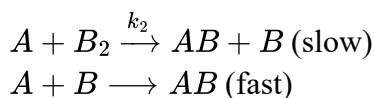
Solution:

Reaction given:



Mechanism:

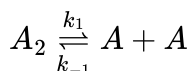




The rate determining step in a mechanism is the slow step.

$$\text{So, rate} = k_2[A][B_2]$$

A is the intermediate; to determine the concentration of A, the equilibrium is considered.



$$\frac{k_1}{k_{-1}} = \frac{[A][A]}{[A_2]}$$

$$= \frac{[A]^2}{[A_2]}$$

$$S_0, [A]^2 = \frac{k_1}{k_{-1}}[A_2]$$

$$[A] = \left\{ \frac{k_1}{k_{-1}}[A_2] \right\}^{1/2}$$

Substitute [A] in the rate,

$$\text{rate} = k_2[A][B_2]$$

$$= k_2 \left(\frac{k_1}{k_{-1}} \right)^{1/2} [A_2]^{1/2} [B_2]$$

$$= k_2 \left(\frac{k_1}{k_{-1}} \right)^{1/2} [A_2]^{1/2} [B_2]^1$$

$$\text{Overall rate} = \frac{1}{2} + 1$$

$$= \frac{3}{2}$$

$$= 1.5$$

Question 42

Drug X becomes ineffective after 50% decomposition. The original concentration of drug in a bottle was 16mg/mL which becomes

4mg/mL in 12 months. The expiry time of the drug in months is _____.

Assume that the decomposition of the drug follows first order kinetics.

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Options:

A.

12

B.

3

C.

6

D.

2

Answer: C

Solution:

Original concentration, $a = 16$ mg/mL

Concentration at time

$t = 12$ months, $a - x = 4$ mg/mL

For first order kinetics, the equation for late constant

$$k = \frac{2.303}{t} \log \frac{a}{a-x}$$

$a \rightarrow$ Initial concentration

$a - x \rightarrow$ Concentration at time, t

$$= \frac{2.303}{12 \text{ months}} \log \frac{16 \text{ mg/mL}}{4 \text{ mg/mL}}$$

$$= \frac{2.303}{12} \log 4$$

$$= 0.1155 \text{ months}^{-1}$$

Drug is ineffective after 50% decomposition. So, the expiry time is after 50% decomposition. The time at which 50% decomposition occurs in a reaction is known as half life ($t_{1/2}$). So, $t_{1/2}$ is the expiry time of the drug.

$t_{1/2}$ formula is

$$t_{1/2} = \frac{0.693}{k}$$

Substitute $k = 0.1155 \text{ months}^{-1}$,

$$t_{1/2} = \frac{0.693}{0.1155} \text{ months}^{-1}$$

$$= 6 \text{ months}$$

Expiry time can also be calculate as,

$$t = \frac{2.303}{k} \log \frac{a}{a-x}$$

Initial is taken as 100%

The percent at expiry time is 50%

$$= \frac{2.303}{0.1155 \text{ months}^{-1}} \times \log \frac{100\%}{50\%}$$

$$= \frac{2.303}{0.1155} \times \log \frac{100}{50}$$

$$= \frac{2.303}{0.1155} \times 0.301$$

$$= 6.002$$

$$= 6 \text{ months}$$

Correct answer is option (3) 6

Question43

Reactant A converts to product D through the given mechanism (with the net evolution of heat):

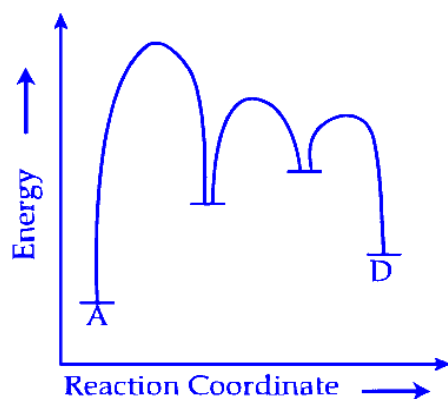


Which of the following represents the above reaction mechanism?

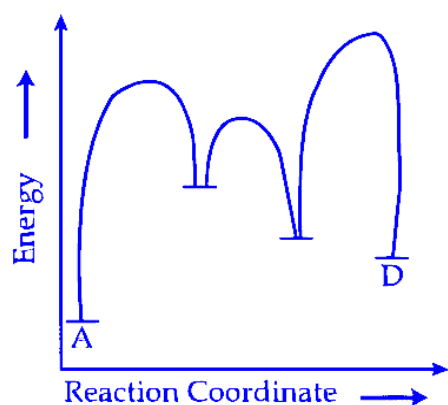
JEE Main 2025 (Online) 2nd April Evening Shift

Options:

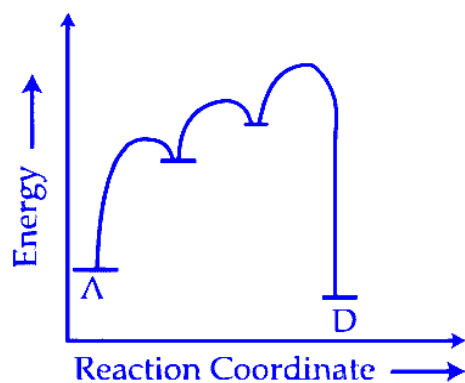
A.



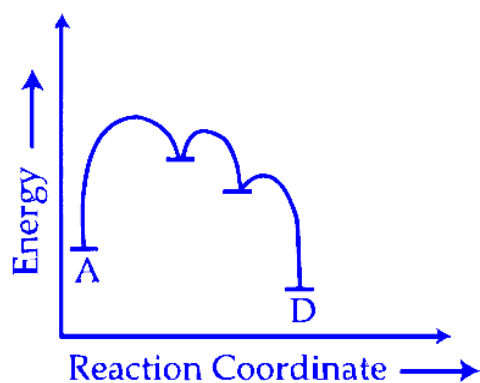
B.



C.

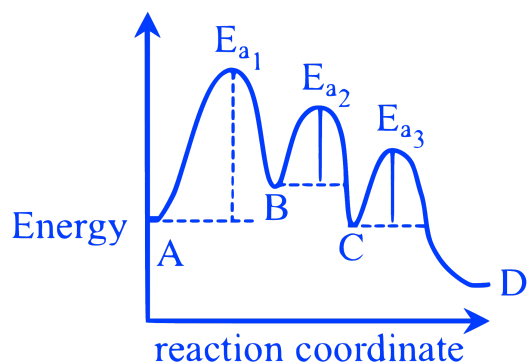


D.



Answer: D

Solution:



A $\rightarrow$ B Slow

$\Delta H = +ve$ $E_{a1} \rightarrow$ High

$B \rightarrow C$ Fast

$\Delta H = -ve$ $E_{a_2} \rightarrow \text{Low}$

$C \rightarrow D$

$\Delta H = -ve$ $E_{a_3} \rightarrow \text{Low}$

Question44

In a reaction $A + B \rightarrow C$, initial concentrations of A and B are related as $[A]_0 = 8[B]_0$. The half lives of A and B are 10 min and 40 min, respectively. If they start to disappear at the same time, both following first order kinetics, after how much time will the concentration of both the reactants be same?

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Options:

A. 20 min

B. 40 min

C. 80 min

D. 60 min

Answer: B

Solution:

The initial conditions are:

$$[A]_0 = 8[B]_0$$

Half-life of A, $[t_{1/2}]_A = 10 \text{ min}$

Half-life of B, $[t_{1/2}]_B = 40 \text{ min}$

Both reactants follow first-order kinetics, and we want to find the time, t , when $[A]_t = [B]_t$.

Step-by-step Derivation:

Write the First-order Rate Equation:

For first-order reactions, the concentration $[X]_t$ at time t is given by:

$$[X]_t = [X]_0 e^{-k_X t}$$

where k_X is the rate constant for substance X .

Express Half-life through Rate Constant:

For first-order reactions, the rate constant k is related to the half-life $t_{1/2}$ by:

$$k = \frac{\ln 2}{t_{1/2}}$$

Substituting the known half-lives:

$$k_A = \frac{\ln 2}{10}$$

$$k_B = \frac{\ln 2}{40}$$

Set Up the Equality from the Condition $[A]_t = [B]_t$:

$$[A]_0 e^{-k_A t} = [B]_0 e^{-k_B t}$$

By inserting the initial condition $[A]_0 = 8[B]_0$, it becomes:

$$8[B]_0 e^{-k_A t} = [B]_0 e^{-k_B t}$$

Simplify and Solve for t :

Divide both sides by $[B]_0$:

$$8e^{-k_A t} = e^{-k_B t}$$

Taking the natural logarithm of both sides:

$$\ln 8 = -k_A t + k_B t$$

Replace k_A and k_B with their expressions:

$$\ln 8 = \left(\frac{\ln 2}{10} - \frac{\ln 2}{40} \right) t$$

Simplify Further:

$$\ln 8 = \ln 2 \left(\frac{1}{10} - \frac{1}{40} \right) t$$

Simplify $\left(\frac{1}{10} - \frac{1}{40} \right)$ to get:

$$\frac{1}{10} - \frac{1}{40} = \frac{4-1}{40} = \frac{3}{40}$$

So,

$$\ln 8 = \ln 2 \times \frac{3}{40} \times t$$

Final Expression for t :

$$t = \frac{\ln 8}{\ln 2 \times \frac{3}{40}}$$

Calculate the value of t using $\ln 8 = 3 \ln 2$:

$$t = \frac{3 \ln 2}{\ln 2 \times \frac{3}{40}} = \frac{40}{1} = 40 \text{ min}$$

Therefore, after 40 minutes, the concentration of both reactants A and B will be the same.

Question 45

Consider the following statements related to temperature dependence of rate constants.

Identify the correct statements.

A. The Arrhenius equation holds true only for an elementary homogenous reaction.

B. The unit of A is same as that of k in Arrhenius equation.

C. At a given temperature, a low activation energy means a fast reaction.

D. A and E_a as used in Arrhenius equation depend on temperature.

E. When $E_a \gg RT$, A and E_a become interdependent.

Choose the correct answer from the options given below:

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Options:

A. A , C and D Only

B. B , D and E Only

C. B and C Only

D. A and B Only

Answer: C

Solution:

Let's examine each statement in the light of

$$k = A e^{-\frac{E_a}{RT}}.$$

A. "Arrhenius holds true only for an elementary homogeneous reaction."

- False – it's an empirical law and is routinely used (and found valid) for complex, composite or even heterogeneous processes.

B. "The unit of A is the same as that of k ."

- True – since $e^{-E_a/RT}$ is dimensionless, A must carry whatever units k has.

C. "At a given temperature, a low activation energy means a fast reaction."

- True – smaller E_a makes $\exp(-E_a/RT)$ larger, hence a larger k .

D. " A and E_a as used in Arrhenius depend on temperature."

- False – in the simple Arrhenius model both are treated as constants (over modest T -ranges).

E. “When $E_a \gg RT$, A and E_a become interdependent.”

• False – although in some data sets you observe a correlation (compensation effect), the basic Arrhenius form still treats them as independent parameters.

Therefore the only correct statements are B and C.

Answer: B and C only.

Question 46

Rate law for a reaction between A and B is given by

$$r = k[A]^n[B]^m$$

If concentration of A is doubled and concentration of B is halved from their initial value, the ratio of new rate of reaction to the initial rate of reaction $\left(\frac{r_2}{r_1}\right)$ is

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Options:

A. $(n - m)$

B. $2^{(n-m)}$

C. $\frac{1}{2^{m+n}}$

D. $(m + n)$

Answer: B

Solution:

$$r_1 = k[A]^n[B]^m$$

Now A is doubled & B is halved in concentration

$$\Rightarrow r_2 = k2^n[A]^n \cdot \frac{[B]^m}{2^m}$$

$$\text{Now } \frac{r_2}{r_1} = 2^{(n-m)}$$

Question47

For $A_2 + B_2 \rightleftharpoons 2AB$

E_a for forward and backward reaction are 180 and 200 kJ mol^{-1} respectively

If catalyst lowers E_a for both reaction by 100 kJ mol^{-1} .

Which of the following statement is correct?

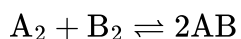
JEE Main 2025 (Online) 4th April Morning Shift

Options:

- A. Catalyst does not alter the Gibbs energy change of a reaction.
- B. The enthalpy change for the reaction is $+20 \text{ kJ mol}^{-1}$.
- C. Catalyst can cause non-spontaneous reactions to occur.
- D. The enthalpy change for the catalysed reaction is different from that of uncatalysed reaction.

Answer: A

Solution:



$$E_f = 180 \text{ kJ mol}^{-1}$$

$$E_b = 200 \text{ kJ mol}^{-1}$$

$$\Delta H = E_f - E_b = -20 \text{ kJ mol}^{-1}$$

In presence of catalyst :

$$E_f = 180 - 100 = 80 \text{ kJ mol}^{-1}$$

$$E_b = 200 - 100 = 100 \text{ kJ mol}^{-1}$$

Question48

Half life of zero order reaction $A \rightarrow \text{product}$ is 1 hour, when initial concentration of reactant is 2.0 mol L^{-1} . The time required to decrease concentration of A from 0.50 to 0.25 mol L^{-1} is :

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Options:

A. 0.5 hour

B. 15 min

C. 60 min

D. 4 hour

Answer: B

Solution:

For zero order reaction

$$\text{Half life} = \frac{A_o}{2k}$$

$$60 \text{ min} = \frac{2}{2k}$$

$$k = \frac{1}{60} \text{ M/min}$$

Now

$$A_t = A_o - kt$$

$$t = \frac{A_o - A_t}{k}$$

$$= \frac{0.5 - 0.25}{1/60}$$

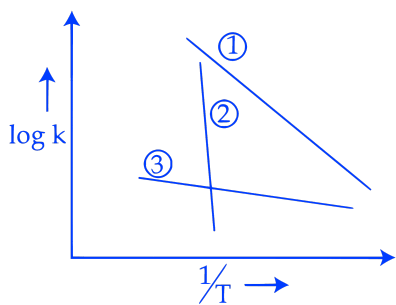
$$0.25 \times 60$$

$$t = 15 \text{ min}$$

Question49

Consider the following plots of log of rate constant $k(\log k)$ vs $\frac{1}{T}$ for three different reactions. The correct order of activation energies

of these reactions is :



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Options:

A. $E_{a2} > E_{a1} > E_{a3}$

B. $E_{a1} > E_{a3} > E_{a2}$

C. $E_{a3} > E_{a2} > E_{a1}$

D. $E_{a1} > E_{a2} > E_{a3}$

Answer: A

Solution:

$$K = Ae^{-E_a/RT}$$

$$\log k = \log A - \frac{E_a}{2.303RT}$$

For graph between $\log k$ with $\frac{1}{T}$

$$\left| \text{Slope of curve} \right| = \frac{E_a}{2.303R}$$

From given graph

$$\text{Magnitude of slope} \Rightarrow (2) > (1) > (3)$$

$$\text{Hence } E_{a2} > E_{a1} > E_{a3}$$

Question 50

Reaction $A(g) \rightarrow 2 B(g) + C(g)$ is a first order reaction. It was started with pure A

t/min	Pressure of system at time t/mm Hg
10	160
∞	240

Which of the following option is incorrect?

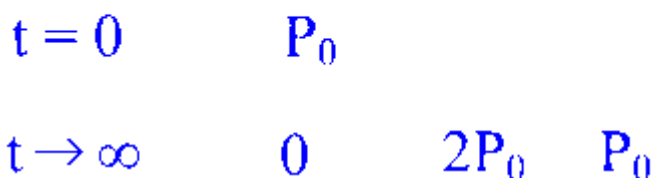
JEE Main 2025 (Online) 7th April Morning Shift

Options:

- A. Initial pressure of A is 80 mm Hg
- B. The reaction never goes to completion
- C. Partial pressure of A after 10 minute is 40 mm Hg
- D. Rate constant of the reaction is 1.693 min^{-1}

Answer: D

Solution:



$$P_{\infty} = 3P_0 = 240$$

$$P_0 = 80 \text{ mm of Hg}$$

$$Kt = \ln \left(\frac{P_{\infty} - P_0}{P_{\infty} - P_t} \right)$$

$$K \times 10 = \ln \left(\frac{240 - 80}{240 - 160} \right)$$

$$K = \frac{\ln 2}{10} = 0.0693 \text{ min}^{-1}$$

Option (3) is incorrect

Question 51

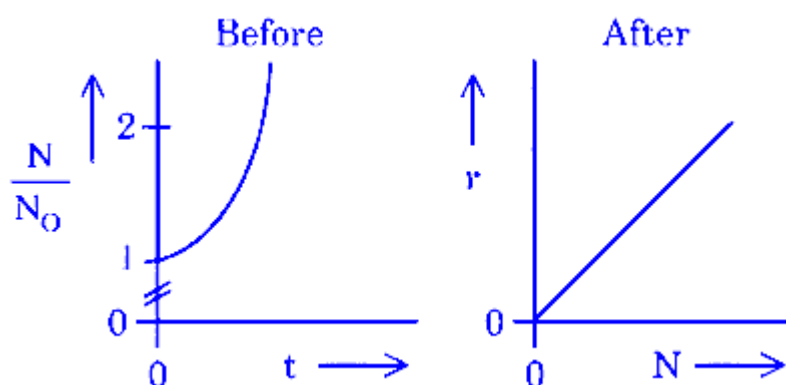
A person's wound was exposed to some bacteria and then bacterial growth started to happen at the same place. The wound was later treated with some antibacterial medicine and the rate of bacterial decay (r) was found to be proportional with the square of the existing number of bacteria at any instance. Which of the following set of graphs correctly represents the 'before' and 'after' situation of the application of the medicine?

[Given: N = No. of bacteria, t = time, bacterial growth follows 1st order kinetics.]

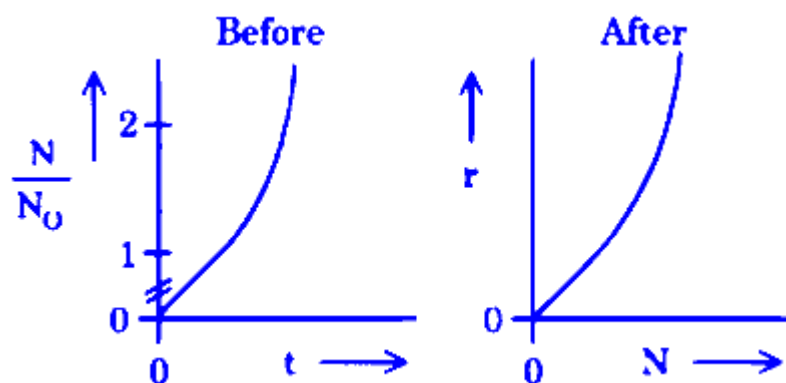
JEE Main 2025 (Online) 7th April Morning Shift

Options:

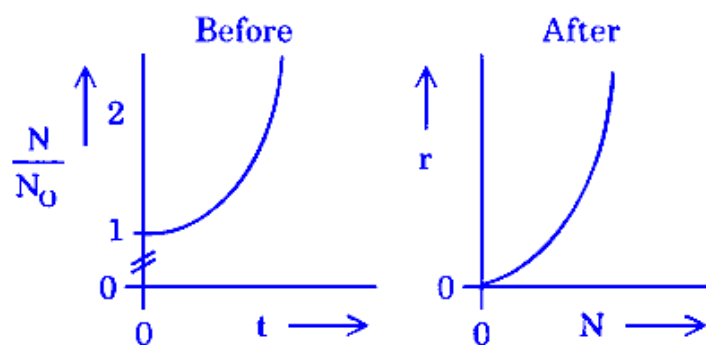
A.



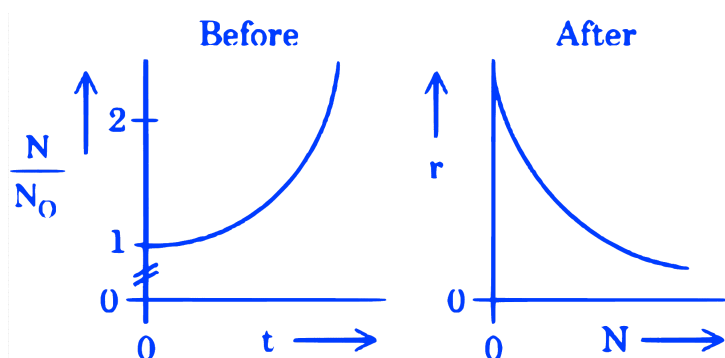
B.



C.



D.



Answer: C

Solution:

*Before applying medicine

$$\frac{dA}{dt} = K[A] \text{ (First order growth) (Rate law)}$$

$$\frac{A}{A_0} = \frac{N}{N_0} = e^{Kt}$$

*After applying medicine

Active Bacteria $\longrightarrow$ Inactive Bacteria

(A)

(I)

$$r = -\frac{dA}{dt} = K[A]^2 \quad \text{(Rate law)}$$

$y = K_x^2$ Parabola

Question52

$A(g) \longrightarrow B(g) + C(g)$ is a first order reaction.

Time	t	∞
P_{system}	P_t	P_{∞}

The reaction was started with reactant A only. Which of the following expressions is correct for rate constant k?

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Options:

A. $k = \frac{1}{t} \ln \frac{P_{\infty}}{P_t}$

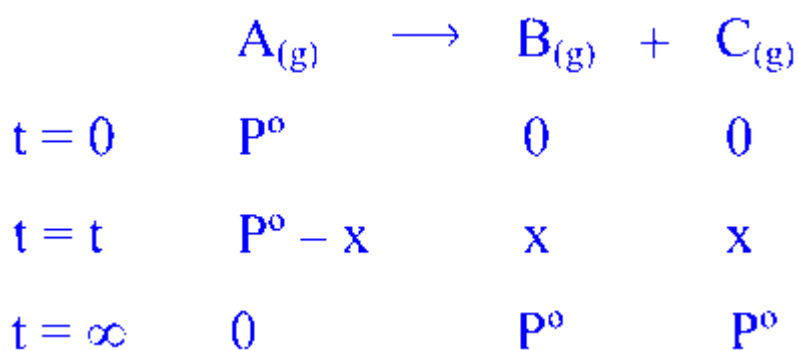
B. $k = \frac{1}{t} \ln \frac{P_{\infty}}{2(P_{\infty} - P_t)}$

C. $k = \frac{1}{t} \ln \frac{2(P_{\infty} - P_t)}{P_t}$

D. $k = \frac{1}{t} \ln \frac{P_{\infty}}{(P_{\infty} - P_t)}$

Answer: B

Solution:



$$P_t = P^{\circ} + x \Rightarrow x = P_t - P^{\circ} = P_t - \frac{P_{\infty}}{2}$$

$$P_{\infty} = 2P^{\circ} \Rightarrow P^{\circ} = \frac{P_{\infty}}{2}$$

$$k = \frac{1}{t} \ln \frac{P^{\circ}}{P^{\circ} - x}$$

$$k = \frac{1}{t} \ln \frac{P_{\infty}}{2(P_{\infty} - P_t)}$$

Question53

In a first order decomposition reaction, the time taken for the decomposition of reactant to one fourth and one eighth of its initial concentration are t_1 and t_2 (s), respectively. The ratio t_1/t_2 will be:

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Options:

A.

$$\frac{4}{3}$$

B.

$$\frac{3}{2}$$

C.

$$\frac{3}{4}$$

D.

$$\frac{2}{3}$$

Answer: D

Solution:

For a first-order reaction, the integrated rate law is given by:

$$\ln \frac{[A]}{[A]_0} = -kt$$

where:

$[A]_0$ is the initial concentration,

$[A]$ is the concentration at time t ,

k is the rate constant.

Let's find the times t_1 and t_2 corresponding to when the concentration becomes $\frac{1}{4}[A]_0$ and $\frac{1}{8}[A]_0$ respectively.

For concentration $\frac{1}{4}[A]_0$:

Substitute into the integrated rate law:

$$\ln\left(\frac{1}{4}\right) = -kt_1$$

Recognizing that $\ln\left(\frac{1}{4}\right) = -\ln 4$, we have:

$$-\ln 4 = -kt_1 \implies t_1 = \frac{\ln 4}{k}$$

For concentration $\frac{1}{8}[A]_0$:

Substitute into the integrated rate law:

$$\ln\left(\frac{1}{8}\right) = -kt_2$$

Recognizing that $\ln\left(\frac{1}{8}\right) = -\ln 8$, we have:

$$-\ln 8 = -kt_2 \implies t_2 = \frac{\ln 8}{k}$$

Now, to find the ratio $\frac{t_1}{t_2}$:

$$\frac{t_1}{t_2} = \frac{\frac{\ln 4}{k}}{\frac{\ln 8}{k}} = \frac{\ln 4}{\ln 8}$$

We can simplify further by expressing the logarithms in terms of $\ln 2$:

$$\ln 4 = \ln(2^2) = 2 \ln 2$$

$$\ln 8 = \ln(2^3) = 3 \ln 2$$

Thus,

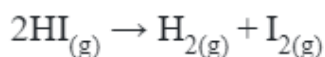
$$\frac{t_1}{t_2} = \frac{2 \ln 2}{3 \ln 2} = \frac{2}{3}$$

So, the ratio $\frac{t_1}{t_2}$ is $\frac{2}{3}$.

This corresponds to Option D.

Question 54

Consider the following data for the given reaction



	1	2	3
HI(mol L ⁻¹)	0.005	0.01	0.02
Rate (mol L ⁻¹ s ⁻¹)	7.5×10^{-4}	3.0×10^{-3}	1.2×10^{-2}

[27-Jan-2024 Shift 1]

Answer: 2

Solution:

Let, $R = k[\text{HI}]^n$ using any two of given data,

$$\frac{3 \times 10^{-3}}{7.5 \times 10^{-4}} = \left(\frac{0.01}{0.005} \right)^n$$

$$n = 2$$

Question55

For a reaction taking place in three steps at same temperature,

overall rate constant $K = \frac{K_1 K_2}{K_3}$. If E_{a1} , E_{a2} and E_{a3} are 40, 50 and 60 kJ/mol respectively, the overall E_a is kJ/mol.

[29-Jan-2024 Shift 1]

Answer: 30

Solution:

$$K = \frac{K_1 K_2}{K_3} = \frac{A_1 \cdot A_2}{A_3} \cdot e^{-\frac{(E_{a1} + E_{a2} - E_{a3})}{RT}}$$

$$A \cdot e^{-E_a/RT} = \frac{A_1 A_2}{A_3} \cdot e^{-\frac{(E_{a1} + E_{a2} - E_{a3})}{RT}}$$

$$E_a = E_{a1} + E_{a2} - E_{a3} = 40 + 50 - 60 = 30 \text{ kJ/mole.}$$

Question56

The half-life of radioisotopic bromine - 82 is 36 hours. The fraction which remains after one day is _____ $\times 10^{-2}$.

(Given $\text{antilog } 0.2006 = 1.587$)

[29-Jan-2024 Shift 2]

Answer: 63

Solution:

Half life of bromine - 82 = 36 hours

$$t_{1/2} = \frac{0.693}{K}$$

$$K = \frac{0.693}{36} = 0.01925 \text{ hr}^{-1}$$

1st order rxn kinetic equation

$$t = \frac{2.303}{K} \log \frac{a}{a-x}$$

$$\log \frac{a}{a-x} = \frac{t \times K}{2.303} \quad (t = 1 \text{ day} = 24 \text{ hr})$$

$$\log \frac{a}{a-x} = \frac{24 \text{ hr} \times 0.01925 \text{ hr}^{-1}}{2.303}$$

$$\log \frac{a}{a-x} = 0.2006$$

$$\frac{a}{a-x} = \text{anti log}(0.2006)$$

$$\frac{a}{a-x} = 1.587$$

$$\text{If } a = 1$$

$$\frac{1}{1-x} = 1.587 \Rightarrow 1-x = 0.6301 = \text{Fraction remain}$$

after one day

Question57

The rate of first order reaction is $0.04\text{molL}^{-1} \text{ s}^{-1}$ at 10 minutes and $0.03\text{molL}^{-1} \text{ s}^{-1}$ at 20 minutes after initiation. Half life of the reaction is _____ minutes. (Given $\log 2 = 0.3010$, $\log 3 = 0.4771$)

[30-Jan-2024 Shift 1]

Answer: 24

Solution:

$$0.04 = k[A]_0 e^{-k \times 10 \times 60} \dots\dots\dots(1)$$

$$0.03 = k[A]_0 e^{-k \times 20 \times 60} \dots\dots\dots(2)$$

$$(1)/(2)$$

$$\frac{4}{3} = e^{600k(2-1)}$$

$$\frac{4}{3} = e^{600k}$$

$$\ln \frac{4}{3} = 600k$$

$$\ln \frac{4}{3} = 600 \times \frac{\ln 2}{t_{1/2}}$$

$$t_{1/2} = 600 \frac{\ln 2}{\ln \frac{4}{3}} \text{ sec}$$

$$t_{1/2} = 600 \times \frac{\log 2}{\log 4 - \log 3} \text{ sec.} = 10 \times \frac{0.3010}{0.6020 - 0.477} \text{ min}$$

$$t_{1/2} = 24.08 \text{ min}$$

Ans. 24

Question58

NO_2 required for a reaction is produced by decomposition of N_2O_5 in CCl_4 as by equation $2\text{N}_2\text{O}_{5(\text{g})} \rightarrow 4\text{NO}_{2(\text{g})} + \text{O}_{2(\text{g})}$

The initial concentration of N_2O_5 is 3molL^{-1} and it is 2.75molL^{-1} after 30 minutes.

The rate of formation of NO_2 is $x \times 10^{-3}\text{molL}^{-1} \text{min}^{-1}$, value of x is _____

[30-Jan-2024 Shift 2]

Answer: 17

Solution:

Rate of reaction (ROR)

$$= -\frac{1}{2} \frac{\Delta[\text{N}_2\text{O}_5]}{\Delta t} = \frac{1}{4} \frac{\Delta[\text{NO}_2]}{\Delta t} = \frac{\Delta[\text{O}_2]}{\Delta t}$$

$$\text{ROR} = -\frac{1}{2} \frac{\Delta[\text{N}_2\text{O}_5]}{\Delta t} = -\frac{1}{2} \frac{(2.75 - 3)}{30} \text{molL}^{-1} \text{min}^{-1}$$

$$\text{ROR} = -\frac{1}{2} \frac{(-0.25)}{30} \text{molL}^{-1} \text{min}^{-1}$$

$$\text{ROR} = \frac{1}{240} \text{molL}^{-1} \text{min}^{-1}$$

$$\text{Rate of formation of } \text{NO}_2 = \frac{\Delta[\text{NO}_2]}{\Delta t} = 4 \times \text{ROR}$$

$$= \frac{4}{240} = 16.66 \times 10^{-3} \text{molL}^{-1} \text{min}^{-1} \simeq 17 \times 10^{-3}$$

Question59

Integrated rate law equation for a first order gas phase reaction is given by (where P_i is initial pressure and P_t is total pressure at time t)

[31-Jan-2024 Shift 1]

Options:

A.

$$k = \frac{2.303}{t} \times \log \frac{P_i}{(2P_i - P_t)}$$

B.

$$k = \frac{2.303}{t} \times \log \frac{2P_i}{(2P_i - P_t)}$$

C.

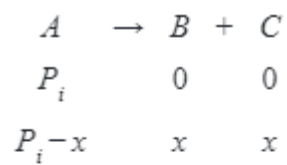
$$k = \frac{2.303}{t} \times \log \frac{(2P_i - P_t)}{P_i}$$

D.

$$k = \frac{2.303}{t} \times \frac{P_i}{(2P_i - P_t)}$$

Answer: A

Solution:



$$P_t = P_i + x$$

$$P_i - x = P_i - P_t + P_i$$

$$= 2P_i - P_t$$

$$K = \frac{2.303}{t} \log \frac{P_i}{2P_i - P_t}$$

Question60

$r = k[A]$ for a reaction, 50% of A is decomposed in 120 minutes. The time taken for 90% decomposition of A is _____ minutes.

[31-Jan-2024 Shift 2]

Answer: 399

Solution:

$$r = k[A]$$

So, order of reaction = 1

$$t_{1/2} = 120 \text{ min}$$

For 90% completion of reaction

$$\Rightarrow k = \frac{2.303}{t} \log \left(\frac{a}{a-x} \right)$$

$$\Rightarrow \frac{0.693}{t_{1/2}} = \frac{2.303}{t} \log \frac{100}{10}$$

$$\therefore t = 399 \text{ min.}$$

Question61

The ratio of $^{14}\text{C}/^{12}\text{C}$ in a piece of wood is $1/8$ part that of atmosphere. If half life of ^{14}C is 5730 years, the age of wood sample is years.

[1-Feb-2024 Shift 1]

Answer: 17190

Solution:

$$\lambda t = \ln \frac{(^{14}\text{C}/^{12}\text{C})_{\text{atmosphere}}}{(^{14}\text{C}/^{12}\text{C})_{\text{wood sample}}}$$

As per the question,

$$\frac{(^{14}\text{C}/^{12}\text{C})_{\text{wood}}}{(^{14}\text{C}/^{12}\text{C})_{\text{atmosphere}}} = \frac{1}{8}$$

$$\text{So, } \lambda t = \ln 8$$

$$\frac{\ln 2}{t_{1/2}} t = \ln 8$$

$$t = 3 \times t_{1/2} = 17190 \text{ years}$$

Question62

The following data were obtained during the first order thermal decomposition of a gas A at constant volume:



S.No	Time/s	Total pressure/(atm)
1.	0	0.1
2.	115	0.28

The rate constant of the reaction is ____ $\times 10^{-2} \text{ s}^{-1}$ (nearest integer)

[1-Feb-2024 Shift 2]

Answer: 2

Solution:



$$t = 0 \quad 0.1$$

$$t = 115 \text{ sec.} \quad 0.1 - x \quad 2x \quad x$$

$$0.1 + 2x = 0.28$$

$$2x = 0.18$$

$$x = 0.09$$

$$K = \frac{1}{115} \ln \frac{0.1}{0.1 - 0.09}$$

$$= 0.0200 \text{ sec}^{-1}$$

$$= 2 \times 10^{-2} \text{ sec}^{-1}$$

Question63

The number of correct statement/s from the following is ____

- A. Larger the activation energy, smaller is the value of the rate constant.
- B. The higher is the activation energy, higher is the value of the temperature coefficient.
- C. At lower temperatures, increase in temperature causes more change in the value of k than at higher temperature.
- D. A plot of $\ln k$ vs $\frac{1}{T}$ is a straight line with slope equal to $-\frac{E_a}{R}$

[24-Jan-2023 Shift 1]

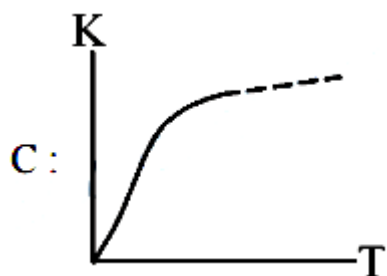
Answer: 3

Solution:

$$A : k = A e^{-\frac{E_a}{RT}}$$

As E_a increases k decreases

$$B : \text{Temperature coefficient} = \frac{k_{T+10}}{k_T}$$



Option (C) is wrong. Δk may be greater or lesser depending on temperature.

$$D : \ln k = \ln A - \frac{E_a}{RT}$$

Question 64

A student has studied the decomposition of a gas AB_3 at 25°C . He obtained the following data.

p(mm Hg)	50	100	200	400
Relative $t_{1/2}$ (s)	4	2	1	0.5

The order of the reaction is
[24-Jan-2023 Shift 2]

Options:

A. 0.5

B. 2

C. 1

D. 0 (zero)

Answer: B

Solution:

$$\begin{aligned}
 t_{1/2} &\propto (P_0)^{1-n} \\
 \frac{(t_{1/2})_1}{(t_{1/2})_2} &= \frac{(P_0)_1^{1-n}}{(P_0)_2^{1-n}} \\
 \Rightarrow \left(\frac{4}{2}\right) &= \left(\frac{50}{100}\right)^{1-n}
 \end{aligned}$$

$$\Rightarrow 2 = \left(\frac{1}{2}\right)^{1-n}$$

$$\Rightarrow 2 = (2)^{n-1}$$

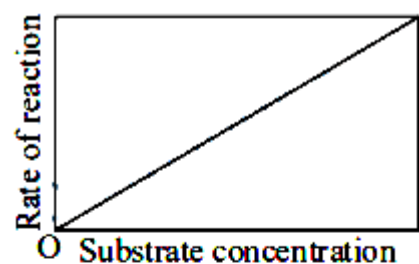
$$\Rightarrow n-1 = 1$$

$$\Rightarrow n = 2$$

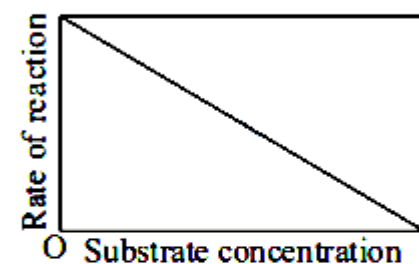
Question 65

The variation of the rate of an enzyme catalyzed reaction with substrate concentration is correctly represented by graph

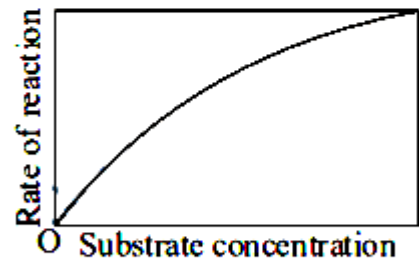
(A)



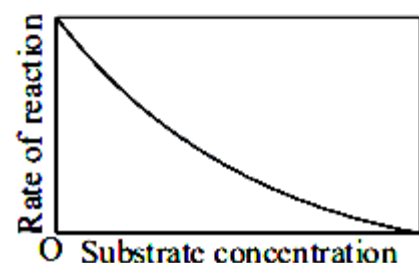
(B)



(C)



(D)



[25-Jan-2023 Shift 1]

Options:

A. b

B. c

C. d

D. a

Answer: B

Solution:

Fact base.

Question66

For the first order reaction $A \rightarrow B$ the half life is 30 min. The time taken for 75% completion of the reaction is _____ min. (Nearest integer)

Given : $\log 2 = 0.3010$

$\log 3 = 0.4771$

$\log 5 = 0.6989$

[25-Jan-2023 Shift 1]

Answer: 60

Solution:

$$t_{1/2} = T_{50} = 30 \text{ min}$$

$$T_{75} = 2t_{1/2} = 30 \times 2 = 60 \text{ min}$$

Question67

A first order reaction has the rate constant, $k = 4.6 \times 10^{-3} \text{ s}^{-1}$. The number of correct statement/s from the following is/are _____.

Given : $\log 3 = 0.48$

- A. Reaction completes in 1000 s.**
B. The reaction has a half-life of 500 s.
C. The time required for 10% completion is 25 times the time required for 90% completion.
D. The degree of dissociation is equal to $(1 - e^{-kt})$.
E. The rate and the rate constant have the same unit.
[25-Jan-2023 Shift 2]

Answer: 1

Solution:

$$t_{10\%} = \frac{1}{K} \ln \left(\frac{a}{a-x} \right) = \frac{1}{K} \ln \left(\frac{100}{90} \right)$$

$$t_{10\%} = \frac{2.303}{K} (\log 10 - \log 9)$$

$$t_{10\%} = \frac{2.093}{K} \times (0.04)$$

Similarly

$$t_{90\%} = \frac{1}{K} \ln \left(\frac{100}{10} \right)$$

$$t_{90\%} = \frac{2.303}{K}$$

$$\frac{t_{90\%}}{t_{10\%}} = \frac{1}{0.04} = 25$$

$$e^{kt} = \frac{a}{a-x}$$

$$\frac{a-x}{a} = e^{-kt}$$

$$1 - \frac{x}{a} = e^{-kt}$$

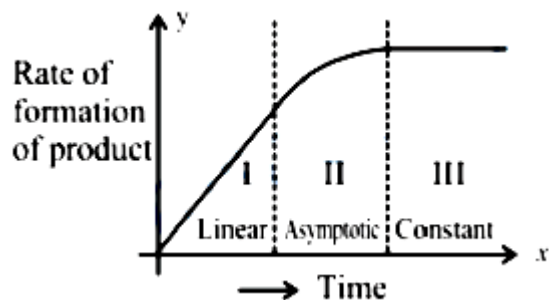
$$x = a(1 - e^{-kt})$$

$$\alpha = \frac{x}{a} = (1 - e^{-kt})$$

Question68

For certain chemical reaction $X \rightarrow Y$, the rate of formation of product is plotted against the time as shown in the figure. The

number of Correct statement/s from the following is _____



- (A) Over all order of this reaction is one
 - (B) Order of this reaction can't be determined
 - (C) In region-I and III, the reaction is of first and zero order respectively
 - (D) In region-II, the reaction is of first order
 - (E) In region-II, the order of reaction is in the range of 0.1 to 0.9.
- [29-Jan-2023 Shift 1]

Answer: 2

Solution:

Only option (B) is correct as order cannot be determined

Question69

For conversion of compound $A \rightarrow B$, the rate constant of the reaction was found to be $4.6 \times 10^{-5} \text{ Lmol}^{-1} \text{ s}^{-1}$. The order of the reaction is _____.

[29-Jan-2023 Shift 2]

Answer: 2

Solution:

As unit of rate constant is $(\text{conc.})^{1-n} \text{ time}^{-1}$

$$\Rightarrow (\text{L mol}^{-1}) \Rightarrow 1 - n = -1$$

$$n = 2$$

Question70

If compound A reacts with B following first order kinetics with rate constant $2.011 \times 10^{-3} \text{ s}^{-1}$. The time taken by A (in seconds) to reduce from 7g to 2g will be _____. (Nearest Integer)

[$\log 5 = 0.698$, $\log 7 = 0.845$, $\log 2 = 0.301$]

[30-Jan-2023 Shift 1]

Answer: 623

Solution:



$$t = t = t_{2g}$$

at constant volume

$$\begin{aligned} t &= \frac{2.303}{K} \log \frac{[A]_0}{[A]_t} \\ &= \frac{2 \cdot 303}{2 \cdot 011 \times 10^{-3}} \log \frac{7}{2} \\ &= \frac{2 \cdot 303 \times 0 \cdot 544}{2 \cdot 011 \times 10^{-3}} \\ &= 622.989 \\ &\approx 623 \end{aligned}$$

Question71

An organic compound undergoes first order decomposition. If the time taken for the 60% decomposition is 540 s, then the time required for 90% decomposition will be is _____ S. (Nearest integer).

Given : $\ln 10 = 2.3$; $\log 2 = 0.3$
[30-Jan-2023 Shift 2]

Answer: 1350

Solution:

$$\begin{aligned}\frac{t_1}{t_2} &= \frac{\frac{1}{K} \ln \frac{a_0}{0.4a_0}}{\frac{1}{K} \ln \frac{a_0}{0.1a_0}} \\ \frac{540}{t_2} &= \frac{\ln \frac{10}{4}}{\ln 10} \\ \frac{540}{t_2} &= \frac{\log 10 - \log 4}{\log 10} \\ \frac{540}{t_2} &= \frac{1 - 0.6}{1} \\ \Rightarrow \frac{540}{t_2} &= 0.4 \\ \Rightarrow t_2 &= \frac{540}{0.4} = 1350 \text{ sec}\end{aligned}$$

Question72

A $\rightarrow$ B

The rate constants of the above reaction at 200K and 300K are 0.03min^{-1} and 0.05min^{-1} respectively. The activation energy for the reaction is _____ J (Nearest integer)

(Given : $\ln 10 = 2.3$

$$\mathbf{R = 8.3JK^{-1}mol^{-1}}$$

$$\mathbf{\log 5 = 0.70}$$

$$\mathbf{\log 3 = 0.48}$$

$$\mathbf{\log 2 = 0.30}$$

[31-Jan-2023 Shift 1]

Answer: 2520

Solution:

$$\log \frac{K_{300}}{K_{200}} = \frac{E_a}{2.3 \times 8.314} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

$$\log \frac{0.05}{0.03} = \frac{E_a}{2.305 \times 8.314} \times \left[\frac{1}{200} - \frac{1}{300} \right]$$

$$E_a = 2519.88 \text{ J} \Rightarrow E_a = 2520 \text{ J}$$

Question 73

Match List-I with List-II

List-I	List-II
(A) Physisorption	I Singlelayer adsorption
(B) Chemisorption	II $20 - 40 \text{ kJ mol}^{-1}$
(C) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \xrightarrow{\text{Fe(s)}} 2\text{NH}_3(\text{g})$	III Chromatography
(D) Analytical Application or Adsorption	IV Heterogeneous catalysis

**Choose the correct answer from the options given below :
[31-Jan-2023 Shift 2]**

Options:

A. A - II, B - III, C - I, D - IV

B. A - III, B - IV, C - I, D - II

C. A - IV, B - II, C - III, D - I

D. A - II, B - I, C - IV, D - III

Answer: D

Solution:

- (A) Physisorption = 20 – 40 kJ/mol and Chemisorption = 80 – 240 kJ/mol
(B) Physisorption is multi-layered and chemisorption is unimolecular layered.
(C) In heterogeneous catalysis, medium and catalyst are in different phases.
(D) Chromatography uses adsorption to purify/separate mixtures.
-

Question 74

The rate constant for a first order reaction is 20min^{-1} . The time required for the initial concentration of the reactant to reduce to its $\frac{1}{32}$ level is _____ $\times 10^{-2}$ min. (Nearest integer)

(Given :

$$\ln 10 = 2.303$$

$$\log 2 = 0.3010)$$

[31-Jan-2023 Shift 2]

Answer: 17

Solution:

$$C = \frac{C_o}{2^n} = \frac{C_o}{32}$$

$$n = 5$$

$$t = 5t_{1/2}$$

$$= \frac{5 \times 0.693}{20} = \frac{0.693}{4}$$

$$= 0.17325 \text{ min} = 17.325 \times 10^{-2} \text{ min.}$$

Question 75

A and B are two substances undergoing radioactive decay in a container. The half life of A is 15 min and that of B is 5 min. If the initial concentration of B is 4 times that of A and they both start decaying at the same time, how much time will it take for the

concentration of both of them to be same? _____ min.
[1-Feb-2023 Shift 1]

Answer: 15

Solution:

$$[A]_t = [A]_0 e^{-kt}$$

For A : Let $[A]_t$ be y and $[A]_0$ be x; $k = \frac{\ln 2}{t_{1/2}} =$

$$\frac{\ln 2}{15 \text{ min}}$$

$$y = x e^{-kt}$$

$$= x e^{-\left(\frac{\ln 2}{15}\right)t}$$

For B : $[B]_t = [B]_0 e^{-kt}$

Let $[B]_t = y$; $[B]_0 = 4x$; $k = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{5 \text{ min}}$

$$y = 4x e^{-\left(\frac{\ln 2}{5}\right)t}$$

$$\Rightarrow x e^{-\left(\frac{\ln 2}{15}\right)t} = 4x e^{-\left(\frac{\ln 2}{5}\right)t}$$

$$e^{t\left(\frac{\ln 2}{5} - \frac{\ln 2}{15}\right)} = 4$$

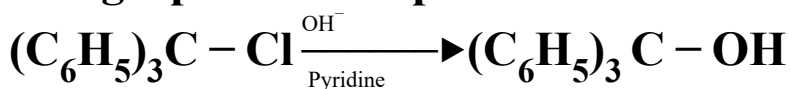
$$t \times \left[\frac{\ln 2}{5} - \frac{\ln 2}{15} \right] = \ln 4$$

$$t \times \ln 2 \left[\frac{1}{5} - \frac{1}{15} \right] = 2 \ln 2$$

$$t = 15 \text{ min}$$

Question 76

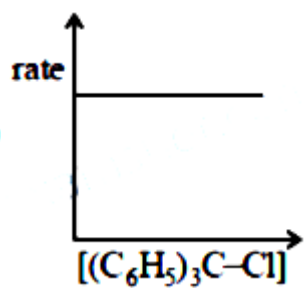
The graph which represents the following reaction is:



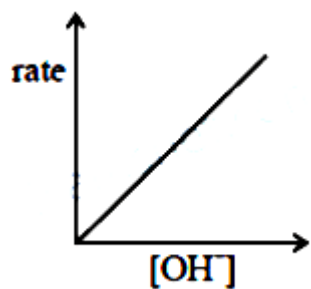
[1-Feb-2023 Shift 2]

Options:

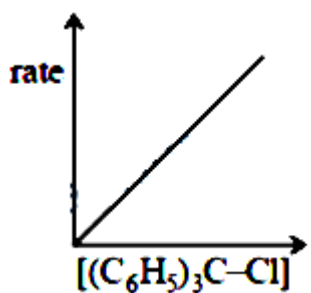
A.



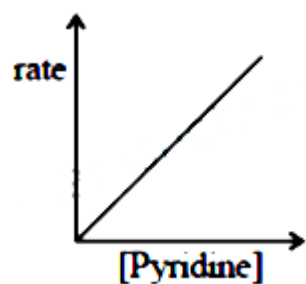
B.



C.



D.

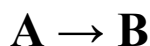


Answer: C

Solution:

(It is SN_1 reaction so rate of reaction depends on the concentration of alkyl halide only.)

Question77



The above reaction is of zero order. Half life of this reaction is 50 min. The time taken for the concentration of A to reduce to one-fourth of its initial value is _____ min.

(Nearest integer)

[1-Feb-2023 Shift 2]

Answer: 75

Solution:

Assume reaction starts with 1 mole A

$$\left(t_{1/2} = \frac{a}{2k}, K = \frac{1}{2 \times 50} \right)$$

For 75% completion

$$a - \frac{a}{4} = kt$$

$$t = \frac{3}{4} \frac{a}{k} = \frac{3}{4} \times \frac{100}{a} = 75$$

Question 78

For the adsorption of hydrogen on platinum, the activation energy is 30 kJ mol^{-1} and for the adsorption of hydrogen on nickel, the activation energy is 41.4 kJ mol^{-1} . The logarithm of the ratio of the rates of chemisorption on equal areas of the metals at 300K is

(Nearest integer)

Given: $\ln 10 = 2.3$

$R = 8.3 \text{ JK}^{-1} \text{ mol}^{-1}$

[6-Apr-2023 shift 1]

Answer: 2

Solution:

$$K = Ae^{-\frac{E_a}{RT}}$$

$$K_1 = Ae^{-\frac{(E_a)_1}{RT}}$$

$$K_2 = Ae^{-\frac{(E_a)_2}{RT}}$$

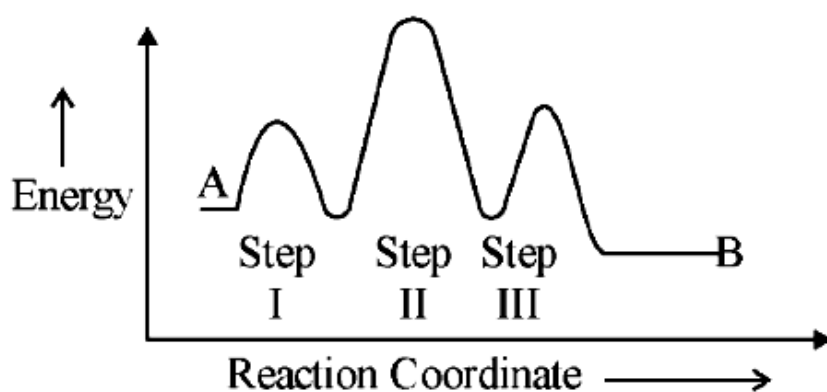
$$\frac{K_2}{K_1} = e^{\frac{(E_a)_1 - (E_a)_2}{RT}}$$

$$\log \frac{K_2}{K_1} = \frac{(E_a)_1 - (E_a)_2}{2.3 RT}$$

$$= \frac{(41.4 - 30) \times 1000}{2.3 \times 8.3 \times 300} = 1.99$$

Question 79

Consider the following reaction that goes from A to B in three steps as shown below:-



Choose the correct

	Number of intermediates	Number of Activated complexes	Rate determining step
(1)	3	2	II
(2)	2	3	II
(2)	2	3	I
(4)	2	3	III

[6-Apr-2023 shift 2]

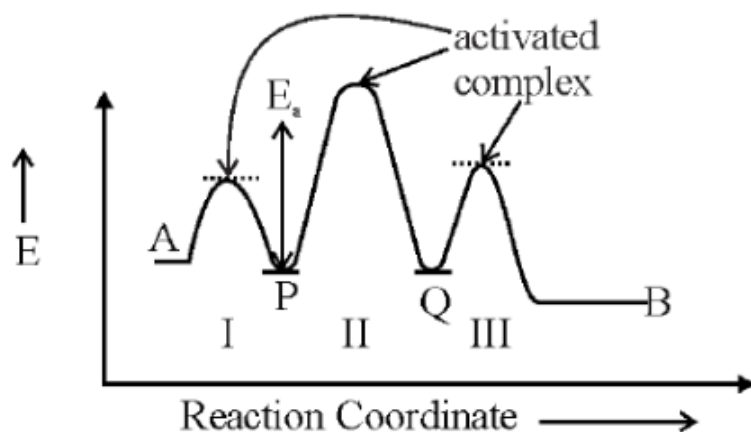
Options:

- A. (1)
- B. (2)
- C. (3)
- D. (4)

Answer: B

Solution:

Step with highest activation energy is RDS, so step II is RDS
No. of activated complex = 3



P and Q are intermediates
(Number of intermediates = 2)

Question80

The number of given statements/s which is/are correct is _____
[8-Apr-2023 shift 1]

Options:

- A. The stronger the temperature dependence of the rate constant, the higher is the activation energy.
- B. If a reaction has zero activation energy, its rate is independent of temperature.
- C. The stronger the temperature dependence of the rate constant, the smaller is the activation energy

D. If there is no correlation between the temperature and the rate constant then it means that the reaction has negative activation energy.

Answer: B

Solution:

Clearly, if $E_a = 0$, K is temperature independent

if $E_a > 0$, K increase with increase in temperature

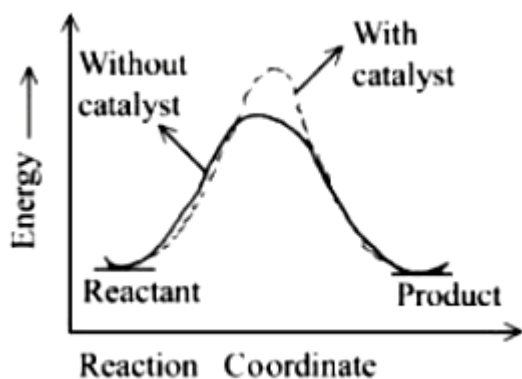
if $E_a < 0$, K decrease with increase in temperature

Question81

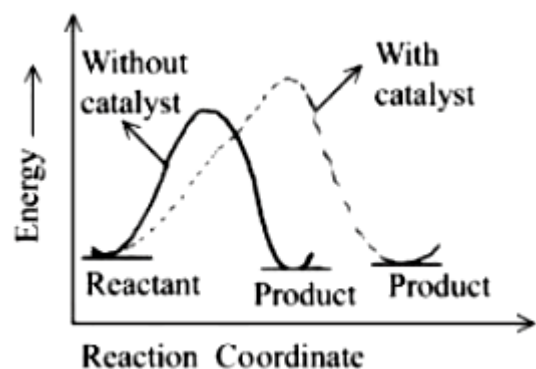
The correct reaction profile diagram for a positive catalyst reaction.
[8-Apr-2023 shift 2]

Options:

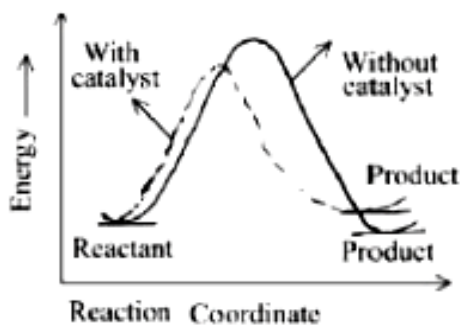
A.



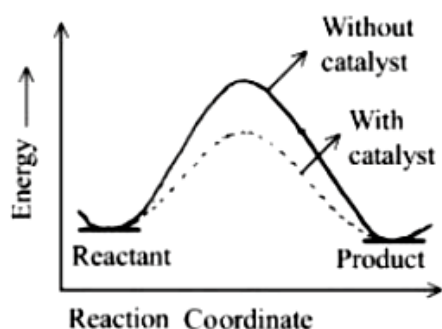
B.



C.



D.



Answer: D

Solution:

Catalysts decrease activation energy only.

Question82

A molecule undergoes two independent first order reactions whose respective half lives are 12 min and 3 min. If both the reactions are occurring then the time taken for the 50% consumption of the reactant is _____ min. (Nearest integer)

[10-Apr-2023 shift 1]

Answer: 2

Solution:

$$k_{\text{eff}} = k_1 + k_2$$

$$\frac{\ln^2}{t_{\text{eff}}} = \frac{\ln^2}{t_1} + \frac{\ln^2}{t_2}$$

$$\frac{1}{t_{\text{eff}}} = \frac{1}{12} + \frac{1}{3} = \frac{1+4}{12} = 5/12$$

$$t_{\text{eff}} = \frac{12}{5} = 2.4 = 2$$

Question83

The number of incorrect statement/s from the following is

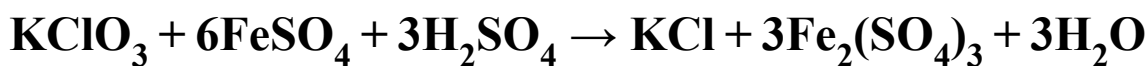
- A. The successive half lives of zero order reactions decreases with time.
 - B. A substance appearing as reactant in the chemical equation may not affect the rate of reaction
 - C. Order and molecularity of a chemical reaction can be a fractional number
 - D. The rate constant units of zero and second order reaction are $\text{mol L}^{-1} \text{s}^{-1}$ and $\text{mol}^{-1} \text{L s}^{-1}$ respectively.
- [10-Apr-2023 shift 2]

Answer: 1

Solution:

- (A) For zero order $t_{1/2} = \frac{[A]_0}{2K}$ as concentration decreases half life decreases (Correct statement)
 - (B) If order w.r.t. that reactant is zero then it will not affect rate of reaction. (Correct statement)
 - (C) Order can be fractional but molecularity can not be (Incorrect statement)
 - (D) For zero order reaction unit is $\text{mol L}^{-1} \text{s}^{-1}$ and for second order reaction unit is $\text{mol}^{-1} \text{L s}^{-1}$ (Correct statement)
-

Question84



The above reaction was studied at 300K by monitoring the concentration of FeSO_4 in which initial concentration was 10M and after half an hour became 8.8M. The rate of production of $\text{Fe}_2(\text{SO}_4)_3$ is _____ $\times 10^{-6} \text{ mol L}^{-1} \text{ s}^{-1}$
[11-Apr-2023 shift 1]

Answer: 333

Solution:

$$\frac{-\Delta \text{FeSO}_4}{\Delta t} = \frac{10 - 8.8}{30 \times 60} = \frac{1.2}{1800}$$

From given equation :

$$-\frac{1}{6} \frac{\Delta \text{FeSO}_4}{\Delta t} = \frac{1}{3} \times (\text{Rate of production of } \text{Fe}_2(\text{SO}_4)_3)$$

$$\text{Rate of production of } \text{Fe}_2(\text{SO}_4)_3 = \frac{3}{6} \times \frac{1.2}{1800}$$

$$= \frac{1}{3} \times 10^{-3}$$

$$= \frac{1000}{3} \times 10^{-6}$$

$$= 333.33 \times 10^{-6}$$

Question85

For a chemical reaction $\text{A} + \text{B} \rightarrow \text{Product}$, the order is 1 with respect to A and B.

Rate $\text{molL}^{-1}\text{s}^{-1}$	[A] molL^{-1}	[B] molL^{-1}
0.10	20	0.5
0.40	x	0.5
0.80	40	Y

What is the value of x and y ?
[11-Apr-2023 shift 2]

Options:

- A. 80 and 2
- B. 40 and 4
- C. 80 and 4
- D. 160 and 4

Answer: A

Solution:

$$r = K[A]^1[B]^1$$

$$0.1 = K(20)^1(0.5)^1 \dots (i)$$

$$0.40 = K(x)^1(0.5)^1 \dots (ii)$$

$$0.80 = K(40)^1(y)^1 \dots (iii)$$

From (i) and (ii)

$$x = 80$$

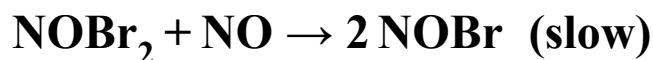
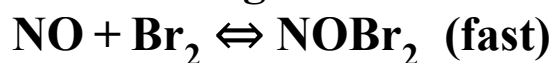
From (i) and (iii)

$$y = 2$$

Question86

The reaction $2 \text{NO} + \text{Br}_2 \rightarrow 2 \text{NOBr}$ takes places through the

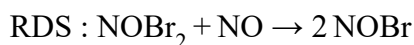
mechanism given below:



The overall order of the reaction is _____.
[12-Apr-2023 shift 1]

Answer: 3

Solution:



$$r = K[\text{NOBr}_2][\text{NO}] \dots\dots (i)$$

$$K_{eq} = \frac{[\text{NOBr}_2]}{[\text{NO}][\text{Br}_2]} \dots\dots (ii)$$

From (i) & (ii)

$$R = K \cdot K_{eq} \cdot [\text{NO}][\text{Br}_2][\text{NO}]$$

$$R = K'[\text{NO}]^2[\text{Br}_2]$$

$$\text{Overall order} = 3$$

Question87

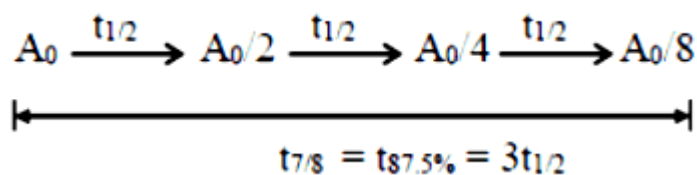
$t_{87.5}$ is the time required for the reaction to undergo 87.5% completion and t_{50} is the time required for the reaction to undergo 50% completion. The relation between $t_{87.5}$ and t_{50} for a first order reaction is _____ $t_{87.5} = x \times t_{50}$. The value of x is _____ .

(Nearest integer)

[13-Apr-2023 shift 1]

Answer: 3

Solution:



Question88

$A(g) \rightarrow 2B(g) + C(g)$ is first order reaction. The initial pressure of the system was found to be 800 mm Hg which increased to 1600 mm Hg after 10 min. The total pressure of the system after 30 min will be _____ mm Hg. (Nearest integer)
[13-Apr-2023 shift 2]

Answer: 2200

Solution:

$$t_{\frac{1}{2}} = 10 \text{ minutes}$$

$$(P_A)_{30 \text{ min}} = (P_A)_0 \left(\frac{1}{2} \right)^{30/10}$$

$$(P_A)_{30 \text{ min}} = 100 \text{ mm Hg}$$

	$A(g)$	$\longrightarrow$	$2B(g)$	$+$	$C(g)$
at $t = 0$	800 mm		0		0
at $t = 30$	100 mm		1400 mm		700 mm
Total pressure after 30 minutes = 2200 mm Hg					

Question89

For a reversible reaction $A \rightleftharpoons B$, the $\Delta H_{\text{forward reaction}} = 20 \text{ kJ mol}^{-1}$. The activation energy of the uncatalysed forward reaction is 300 kJ mol^{-1} . When the reaction is catalysed keeping the reactant

concentration same, the rate of the catalyzed forward reaction at 27°C is found to be same as that of the uncatalysed reaction at 327°C . The activation energy of the catalyzed backward reaction is _____ KJ mol^{-1} .

[15-Apr-2023 shift 1]

Answer: 130

Solution:

$$E_a = 300 \text{ kJ mol}^{-1}$$

$$\frac{E_a}{T} = \frac{E'_a}{T'}$$

(Since rate of catalyzed and uncatalysed reaction is same)

$$\frac{300}{600} = \frac{E_{a,f}'}{300}$$

$$E_{a,f}' = 150$$

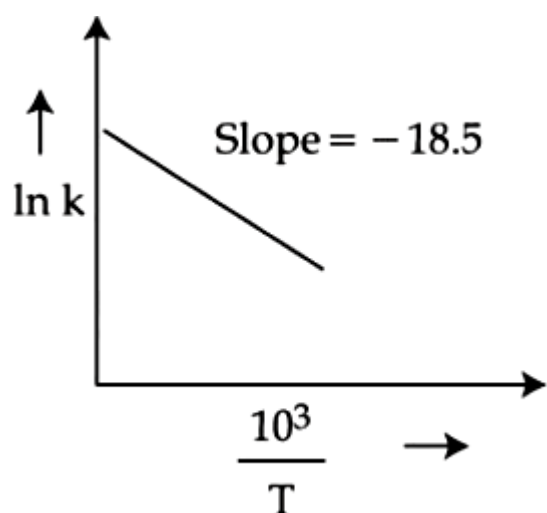
$$20 = 150$$

$$20 = 150 - E_{a,b}'$$

$$E_{a,b}' = 130$$

Question90

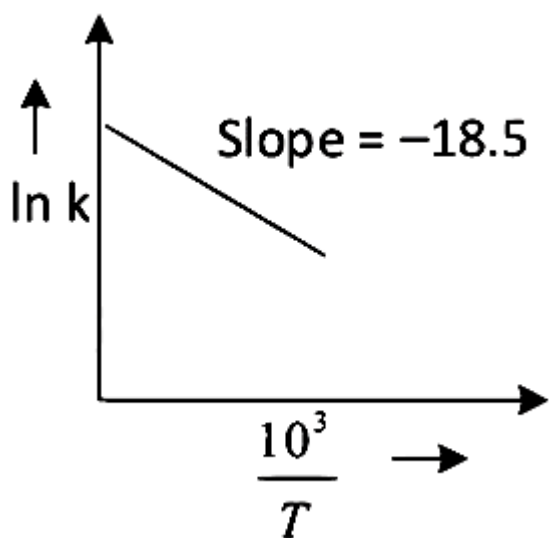
The rate constants for decomposition of acetaldehyde have been measured over the temperature range $700 - 1000\text{K}$. The data has been analysed by plotting $\ln k$ v s $\frac{10^3}{T}$ graph. The value of activation energy for the reaction is kJ mol^{-1} . _____ (Nearest integer)
(Given : $R = 8.31\text{JK}^{-1}\text{mol}^{-1}$)



[24-Jun-2022-Shift-1]

Answer: 154

Solution:



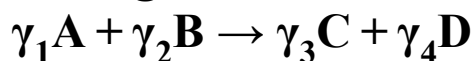
$$\ln k = \ln A - \frac{E_a}{RT}$$

$$\therefore \text{Slope of the graph} = - \frac{E_a}{R \times 10^3} = -18.5$$

$$\therefore E_a = 18.5 \times 8.31 \times 1000 \approx 154 \text{ kJ mol}^{-1}$$

Question91

For a given chemical reaction



Concentration of C changes from 10 mmol dm^{-3} to 20 mmol dm^{-3} in 10 seconds. Rate of appearance of D is 1.5 times the rate of disappearance of B which is twice the rate of disappearance A. The rate of appearance of D has been experimentally determined to be $9 \text{ mmol dm}^{-3} \text{ s}^{-1}$. Therefore, the rate of reaction is ____ $\text{mmol dm}^{-3} \text{ s}^{-1}$. (Nearest Integer)
[25-Jun-2022-Shift-1]

Answer: 1

Solution:

$$\text{Rate} = \frac{1}{r_1} \left(\frac{-d[A]}{dt} \right) = \frac{1}{r_2} \left(\frac{-d[B]}{dt} \right) = \frac{1}{r_3} \left(\frac{-d[C]}{dt} \right)$$

$$= \frac{1}{r_4} \left(\frac{d[D]}{dt} \right)$$

$$\frac{d[D]}{dt} = \frac{r_4}{r_2} \left(\frac{-d[B]}{dt} \right)$$

$$\frac{r_4}{r_2} = \frac{3}{2}$$

$$\frac{-d[B]}{dt} = \frac{r_2}{r_1} \left(\frac{-d[A]}{dt} \right) \Rightarrow \frac{r_2}{r_1} = 2$$

$$r_2 = 2r_1$$

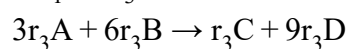
$$r_4 = 1.5r_2 = 3r_1$$

$$\frac{d[C]}{dt} = 1 \text{m} \cdot \text{mol dm}^{-3} \text{sec}^{-1}$$

$$\frac{d[D]}{dt} = \frac{r_4}{r_3} \cdot \frac{d[C]}{dt} \Rightarrow \frac{r_4}{r_3} = 9$$

$$r_4 = 9r_3 = 3r_1$$

$$\Rightarrow r_1 = 3r_3$$



$$\therefore \text{rate of reaction} = \frac{1}{9} \times 9 \text{m} \cdot \text{mol dm}^{-3} \text{sec}^{-1}$$

$$= 1 \text{m} \cdot \text{mol dm}^{-3} \text{sec}^{-1}$$

Question92

At 345K, the half life for the decomposition of a sample of a gaseous compound initially at 55.5 kPa was 340 s. When the pressure was 27.8 kPa, the half life was found to be 170s. The order of the reaction is _____. [integer answer]
[25-Jun-2022-Shift-2]

Answer: 0

Solution:

$$t_{1/2} \propto \frac{1}{[P_0]^{n-1}}$$
$$\frac{t_1}{t_2} = \frac{(P_2)^{n-1}}{(P_1)^{n-1}}$$
$$\frac{340}{170} = \left(\frac{27.8}{55.5} \right)^{n-1}$$
$$\Rightarrow 2 = \frac{1}{(2)^{n-1}}$$
$$n = 0$$

Question93

A flask is filled with equal moles of A and B. The half lives of A and B are 100 s and 50 s respectively and are independent of the initial concentration. The time required for the concentration of A to be four times that of B is _____ s.
(Given : $\ln 2 = 0.693$)
[26-Jun-2022-Shift-1]

Answer: 200

Solution:

$$k_A = \frac{\ln 2}{100}; k_B = \frac{\ln 2}{50}$$

$$A_t = A_0 \times e^{-k_A t}$$

$$A_t = A_0 \times e^{\left(\frac{-\ln 2}{100} \times t\right)}$$

$$B_t = B_0 \times e^{\left(\frac{-\ln 2}{50} \times t\right)}$$

$$A_0 = B_0$$

$$A_t = 4B_t$$

$$e^{-\frac{\ln 2}{100} \times t} = 4 \times e^{-\frac{\ln 2}{50} \times t}$$

$$e^{\frac{\ln 2}{100} \times t} = 4$$

$$e^{\frac{\ln 2}{100} \times t} = 4$$

$$\frac{\ln 2}{100} \times t = \ln 4 = 2 \ln 2$$

$$t = 200 \text{ sec}$$

Question94

Catalyst A reduces the activation energy for a reaction by 10 kJ mol^{-1} at 300K. The ratio of rate constants, $\frac{k_{T, \text{Catalysed}}}{k_{T, \text{Uncatalysed}}}$ is e^x .

The value of x is _____ [nearest integer]

[Assume that the pre-exponential factor is same in both the cases.

Given $R = 8.31 \text{ JK}^{-1} \text{ mol}^{-1}$]

[26-Jun-2022-Shift-2]

Answer: 4

Solution:

$$K = A e^{\frac{-E_a}{RT}}$$

$$K_{\text{cat}} = A e^{\frac{E_a}{RT}}, \quad K_{\text{uncat.}} = A e^{\frac{E_a}{RT}}$$

$$\frac{K_{\text{cat}}}{K_{\text{uncat.}}} = e^{\frac{E_a}{RT}} = e^{\frac{10 \times 1000}{8.3 \times 300}} = e^{4.009} = e^x$$

$$\therefore x = 4$$

Question 95

The rate constant for a first order reaction is given by the following equation:

$$\ln k = 33.24 - \frac{2.0 \times 10^4 \text{K}}{T}$$

The activation energy for the reaction is given by ____ kJ mol^{-1} . (In nearest integer) (Given : $R = 8.3 \text{ JK}^{-1} \text{ mol}^{-1}$)
[27-Jun-2022-Shift-1]

Answer: 166

Solution:

$$\ln k = \ln A - \frac{E_A}{RT}$$

$$\text{Given: } \ln k = 33.24 - \frac{2.0 \times 10^4}{T}$$

$$\therefore \text{ on comparing } \frac{E_A}{R} = 2.0 \times 10^4$$

$$\therefore E_A = 2.0 \times 10^4 \times R$$

$$\Rightarrow E_A = 2.0 \times 10^4 \times 8.3 \text{ J}$$

$$\Rightarrow E_A = 16.6 \times 10^4 \text{ J} = 166 \text{ kJ}$$

Question 96

It has been found that for a chemical reaction with rise in temperature by 9K the rate constant gets doubled. Assuming a reaction to be occurring at 300K, the value of activation energy is found to be _____ kJ mol⁻¹. [nearest integer]
(Given $\ln 10 = 2.3$, $R = 8.3 \text{ JK}^{-1} \text{ mol}^{-1}$, $\log 2 = 0.30$)
[27-Jun-2022-Shift-2]

Answer: 59

Solution:

$$T_1 = 300\text{K}$$

(Rate constant)

$$K_2 = 2K_1, \text{ on increase temperature by } 9\text{K}$$

$$T_2 = 309\text{K}$$

$$E_a = ?$$

$$\log \frac{K_2}{K_1} = \frac{E_a}{2.3R} \left[\frac{T_2 - T_1}{T_2 \cdot T_1} \right]$$

$$\log 2 = \frac{E_a}{2.3 \times 8.3} \left[\frac{9}{309 \times 300} \right]$$

$$E_a = \frac{0.3 \times 309 \times 300 \times 2.3 \times 8.3}{9}$$

$$= 58988.1 \text{ J/mole}$$

$$\approx 59 \text{ kJ/mole}$$

Question97

For a first order reaction $A \rightarrow B$, the rate constant, $k = 5.5 \times 10^{-14} \text{ s}^{-1}$. The time required for 67% completion of reaction is $x \times 10^{-1}$ times the half life of reaction. The value of x is _____
(Nearest integer)
(Given : $\log 3 = 0.4771$)
[28-Jun-2022-Shift-1]

Answer: 16

Solution:

$$t_{67\%} = \frac{1}{k} \ln \left(\frac{1}{1-0.67} \right) = \frac{t_{1/2}}{\ln 2} \times \ln \left(\frac{1}{1 - \frac{2}{3}} \right)$$

$$t_{67\%} = \frac{t_{1/2}}{\log 2} \times \log 3 = \frac{t_{1/2} \times 0.4771}{0.301}$$

$$\Rightarrow t_{67\%} = 1.585 \times t_{1/2}$$

$$X \times 10^{-1} = 1.585$$

$$\Rightarrow X = 15.85$$

Ans.16

Question98

A radioactive element has a half life of 200 days. The percentage of original activity remaining after 83 days is _____ (Nearest integer)
(Given : antilog 0.125 = 1.333, antilog 0.693 = 4.93)
[28-Jun-2022-Shift-2]

Answer: 75

Solution:

$$\lambda = \frac{2.303}{t} \log \frac{A_0}{A}$$

$$\frac{0.693}{200} = \frac{2.303}{83} \log \frac{A_0}{A}$$

$$\frac{A}{A_0} = 0.75$$

Hence, percentage of original activity remaining after 83 days is 75%

Question99

The activation energy of one of the reactions in a biochemical process is $532611 \text{ J mol}^{-1}$. When the temperature falls from 310 K to 300K, the change in rate constant observed is $k_{300} = x \times 10^{-3} k_{310}$. The value of x is _____

[Given : $\ln 10 = 2.3$, $R = 8.3 \text{ J K}^{-1} \text{ mol}^{-1}$]

[29-Jun-2022-Shift-1]

Answer: 1

Solution:

$$\ln \left(\frac{K_2}{K_1} \right) = \frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

$$\ln \left(\frac{K_2}{K_1} \right) = \frac{532611}{8.3} \times \left(\frac{10}{310 \times 300} \right)$$

where K_2 is at 310K & K_1 is at 300K

$$\ln \left(\frac{K_2}{K_1} \right) = 6.9$$

$$= 3 \times \ln 10$$

$$\ln \frac{K_2}{K_1} = \ln 10^3$$

$$K_2 = K_1 \times 10^3$$

$$K_1 = K_2 \times 10^{-3}$$

$$\text{So } K = 1$$

Question 100

The equation

$$k = (6.5 \times 10^{12} \text{ s}^{-1}) e^{-26000 \text{ K} / T}$$

is followed for the decomposition of compound A. The activation energy for the reaction is kJ mol^{-1} . [nearest integer]

(Given : $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$)

[29-Jun-2022-Shift-2]

Answer: 216

Solution:

$$k = Ae^{\frac{-E_a}{RT}}$$
$$\frac{E_a}{RT} = \frac{26000}{T}$$
$$E_a = 26000 \times 8.314$$
$$= 216164\text{J}$$
$$= 216\text{ kJ}$$

Question101

The half life for the decomposition of gaseous compound A is 240s when the gaseous pressure was 500 Torr initially. When the pressure was 250 Torr, the half life was found to be 4.0min. The order of the reaction is (Nearest integer)
[25-Jul-2022-Shift-1]

Answer: 1

Solution:

$$(t_{1/2})_A = 240\text{s when } P = 500 \text{ torr}$$

$$(t_{1/2})_A = 4 \text{ min} = 4 \times 60 = 240 \text{ sec when } P = 250 \text{ torr}$$

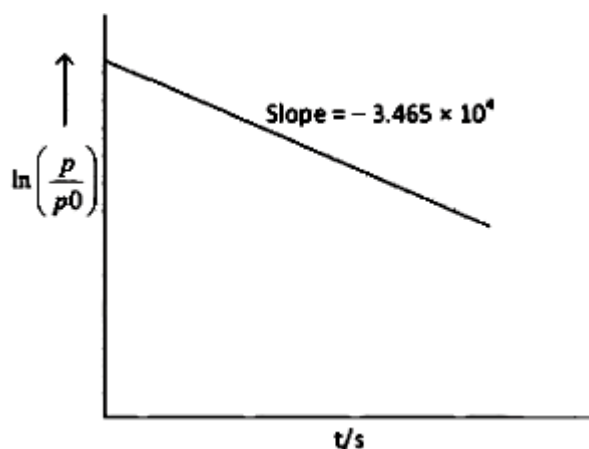
If means half-life is independent of concentration of reactant present.

$\therefore$ Order of reaction = 1

Question102

For the decomposition of azomethane.

$\text{CH}_3\text{N}_2\text{CH}_3(\text{g}) \rightarrow \text{CH}_3\text{CH}_3(\text{g}) + \text{N}_2(\text{g})$, a first order reaction, the variation in partial pressure with time at 600K is given as



The half life of the reaction is _____ $\times 10^{-5}$ s. [Nearest integer]
[25-Jul-2022-Shift-2]

Answer: 2

Solution:

For first order reaction,

$$\ln A = \ln A_0 - kt$$

$$\text{Hence Slope} = -k$$

$$-k = -3.465 \times 10^4$$

$$k = \frac{0.693}{t_{1/2}}$$

$$3.465 \times 10^4 = \frac{0.693}{t_{1/2}}$$

$$t_{1/2} = 2 \times 10^{-5} \text{ s}$$

Question103

For a reaction $\text{A} \rightarrow 2\text{B} + \text{C}$ the half lives are 100s and 50s when the concentration of reactant A is 0.5 and 1.0 mol L^{-1} respectively. The order of the reaction is _____. (Nearest Integer)
[26-Jul-2022-Shift-1]

Answer: 2

Solution:

$$t_{1/2} \propto \frac{1}{(a_0)^{n-1}}$$

$$t_{1/2} = 100 \text{ sec } a_0 = 0.5$$

$$t_{1/2} = 50 \text{ sec } a_0 = 1$$

$$\frac{100}{50} = \left(\frac{1}{0.5} \right)^{n-1}$$

$$(2) = (2)^{n-1}$$

$$n - 1 = 1$$

$$n = 2$$

Question104

At 30°C, the half life for the decomposition of AB₂ is 200s and is independent of the initial concentration of AB₂. The time required for 80% of the AB₂ to decompose is Given: log 2 = 0.30 log 3 = 0.48 [26-Jul-2022-Shift-2]

Options:

A. 200s

B. 323s

C. 467s

D. 532s

Answer: C

Solution:

Since, half-life is independent of the initial concentration of AB₂. Hence, the reaction is "First Order".

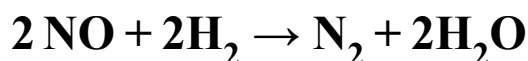
$$k = \frac{2.303 \log 2}{t_{1/2}}$$

$$\frac{2.303 \log 2}{t_{1/2}} = \frac{2.303}{t} \log \frac{100}{(100 - 80)}$$

$$\frac{2.303 \times 0.3}{200} = \frac{2.303}{t} \log 5$$

$$t = 467s$$

Question 105



The above reaction has been studied at 800°C. The related data are given in the table below

Reaction serial number	Initial Pressure of H_2 / kPa	Initial Pressure of NO / kPa	Initial rate $\left(\frac{-dp}{dt} \right) / (\text{kPa/s})$
1	65.6	40.0	0.135
2	65.6	20.1	0.033
3	38.6	65.6	0.214
4	19.2	65.6	0.106

The order of the reaction with respect to NO is
[27-Jul-2022-Shift-1]

Answer: 2

Solution:

Let the rate of reaction (r) is as

$$r = K[\text{NO}]^n[\text{H}_2]^m$$

From 1st data

$$0.135 = K[40]^n \cdot (65.6)^m \dots (1)$$

From 2nd data

$$0.033 = K(20.1)^n \cdot (65.6)^m \dots (2)$$

On dividing equation (1) by equation (2)

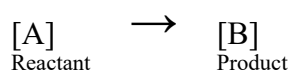
$$\frac{0.135}{0.033} = \left(\frac{40}{20.1} \right)^n$$

$$4 = (2)^n$$

$$\therefore n = 2$$

$\therefore$ Order of reaction w.r.t. NO is 2 .

Question106



If formation of compound [B] follows the first order of kinetics and after 70 minutes the concentration of [A] was found to be half of its initial concentration. Then the rate constant of the reaction is

$x \times 10^{-6} \text{ s}^{-1}$. The value of x is _____.(Nearest Integer)

[27-Jul-2022-Shift-2]

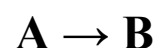
Answer: 165

Solution:

$$\begin{aligned} K &= \frac{0.693}{t_{1/2}} = \frac{0.693}{70 \times 60} \\ &= \frac{6930}{7 \times 6} \times 10^{-6} \\ &= 165 \times 10^{-6} \text{ s}^{-1} \end{aligned}$$

Question107

For the given first order reaction



the half life of the reaction is 0.3010 min. The ratio of the initial concentration of reactant to the concentration of reactant at time 2.0 min will be equal to _____. (Nearest integer)
[28-Jul-2022-Shift-1]

Answer: 100

Solution:

$$t_{1/2} = \frac{0.693}{K} \quad t_{1/2} \text{ given} = 0.3010$$

$$K = \frac{0.693}{0.3010}$$

$$K = 2.30$$

$$K = \frac{2.303}{t} \log \frac{(A_0)}{(A_t)}$$

$A_0 \rightarrow$ initial concentration of reactant

$A_t \rightarrow$ concentration of reactant at time t

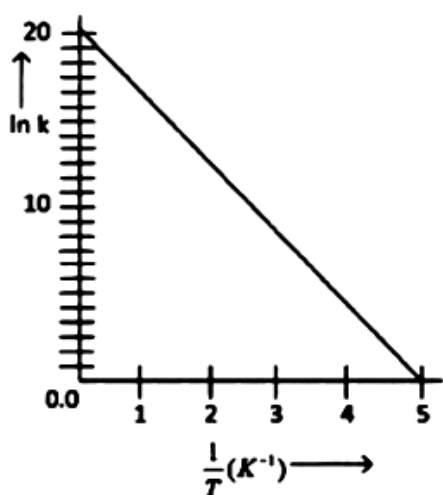
$$2.303 = \frac{2.303}{2} \log \frac{(A_0)}{(A_t)}$$

$$2 = \log \frac{(A_0)}{(A_t)}$$

$$100 = \frac{A_0}{A_t}$$

Question108

For a reaction, given below is the graph of $\ln k$ vs $\frac{1}{T}$. The activation energy for the reaction is equal to ____ (nearest integer)
(Given : $R = 2 \text{ cal K}^{-1} \text{ mol}^{-1}$)



[28-Jul-2022-Shift-2]

Answer: 8

Question109

The reaction between X and Y is first order with respect to X and zero order with respect to Y.

Experiment	$\frac{[X]}{\text{mol L}^{-1}}$	$\frac{[Y]}{\text{mol L}^{-1}}$	$\frac{\text{Initial rate}}{\text{mol L}^{-1}\text{min}^{-1}}$
I	0.1	0.1	2×10^{-3}
I	L	0.2	4×10^{-3}
III	0.4	0.4	$M \times 10^{-3}$
IV	0.1	0.2	2×10^{-3}

Examine the data of table and calculate ratio of numerical values of

M and L (Nearest Integer) [29-Jul-2022-Shift-1]

Answer: 40

Solution:

$$r = k[X][Y]^0 = k[X]$$

Using I & II

$$\frac{4 \times 10^{-3}}{2 \times 10^{-3}} = \left(\frac{L}{0.1} \right) \Rightarrow L = 0.2$$

Using I & III

$$\frac{M \times 10^{-3}}{2 \times 10^{-3}} = \frac{0.4}{0.1} \Rightarrow M = 8$$

$$\frac{M}{L} = \frac{8}{0.2} = 40$$

Question110

Assuming 1 μ g of trace radioactive element X with a half life of 30 years is absorbed by a growing tree. The amount of X remaining in the tree after 100 years is _____ $\times 10^{-1}$ μ g.

[Given : $\ln 10 = 2.303$; $\log 2 = 0.30$]

[29-Jul-2022-Shift-2]

Answer: 1

Solution:

$$t = \frac{1}{\lambda} \ln \left(\frac{a}{a-x} \right)$$
$$\Rightarrow 100 = \left(\frac{30}{\ln 2} \right) \left[\ln \left(\frac{1}{w} \right) \right]$$

$$\Rightarrow \left[\frac{100 \times \log 2}{30} \right] = \log \left(\frac{1}{w} \right)$$

$$\Rightarrow 1 = \log \left(\frac{1}{w} \right)$$

$$\Rightarrow \frac{1}{w} = 10$$

$$\text{So } w = 0.1 \mu\text{g}$$

Question111

Sucrose hydrolyses in acid solution into glucose and fructose following first order rate law with a half-life of 3.33h at 25°C. After 9h, the fraction of sucrose remaining is f. The value of $\log_{10} \left(\frac{1}{f} \right)$ is $\times 10^{-2}$

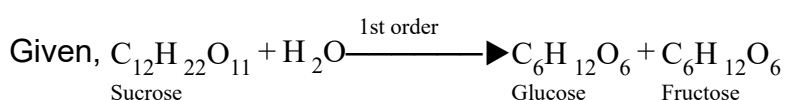
(Rounded off to the nearest integer).

[Assume, $\ln 10 = 2.303$, $\ln 2 = 0.693$]

[24 Feb 2021 Shift 2]

Answer: 81

Solution:



$$t_{1/2} = \frac{10}{3} \text{h}$$

$$t = 0, \quad a = [A]_0 \quad (\text{initial conc.})$$

$$\text{At } t = 9\text{h} \quad a - x = [A]_t \quad (\text{conc. at time } t)$$

For using 1st order equation,

$$K = \frac{2.303}{t} \log \frac{[A]_0}{[A]_t} \Rightarrow \frac{K \times t}{2.303} = \log \frac{[A]_0}{[A]_t}$$

$$\frac{\ln 2 \times 9}{10/3 \times 2303} = \log \left(\frac{1}{F} \right) \Rightarrow \log \left(\frac{1}{F} \right) = 0.8124 \quad \left(\because k = \frac{\ln 2}{t_{1/2}} \right)$$

$$\log \left(\frac{1}{F} \right) = 81.24 \times 10^{-2}$$

$$x = 81.24 \quad \text{or } x \approx 81$$

Question112

The rate constant of a reaction increases by five times on increase in temperature from 27°C to 52°C . The value of activation energy in kJ mol^{-1} is (Rounded off to the nearest integer)

$$[R = 8.314\text{J K}^{-1}\text{mol}^{-1}]$$

[25 Feb 2021 Shift 2]

Answer: 52

Solution:

$$T_1 = (273 + 27) = 300\text{K}, T_2 = (273 + 52) = 325\text{K}$$

Given, temperature coefficient of the reaction,

$$\alpha_T = \frac{K_{325}}{K_{300}} = 5$$

$$\log \frac{K_{T_2}}{K_{T_1}} = \frac{E_a}{2.303R} \times \left(\frac{T_2 - T_1}{T_1 T_2} \right)$$

$$\log \frac{K_{325}}{K_{300}} = \frac{E_a}{2.303 \times 8.314} \left(\frac{325 - 300}{300 \times 325} \right)$$

$$\log 5 = \frac{E_a}{2.303 \times 8319} \times \frac{25}{300 \times 325}$$

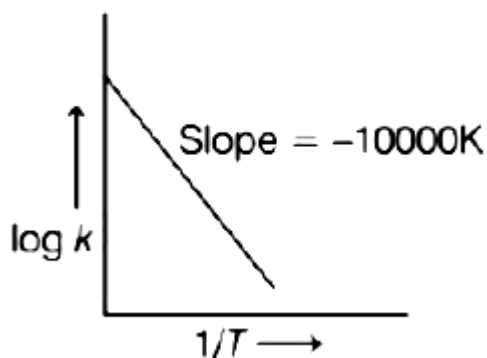
$$E_a = 52194.78\text{J mol}^{-1}$$

$$= 52.194\text{kJ mol}^{-1}$$

$$= 52\text{kJ mol}^{-1}$$

Question113

For the reaction, $aA + bB \rightarrow cC + dD$, the plot of $\log k$ v s $\frac{1}{T}$ is given below



The temperature at which the rate constant of the reaction is 10^{-4} s^{-1} is K
(Rounded off to the nearest integer).

[Given : The rate constant of the reaction is 10^{-5} s^{-1} at 500K]
[25 Feb 2021 Shift 1]

Answer: 526

Solution:

According to Arrhenius equation,

$$\log K = \log A - \frac{E_a}{2.303RT}$$

$$\text{Given, Slope} = \frac{E_a}{2.303R} = 10,000$$

$$K_1 = 10^{-5}, K_2 = 10^{-4}$$

$$T_1 = 500\text{K}$$

$$\log_{10} \left[\frac{K_1}{K_2} \right] = \frac{E_a}{2.303R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$$

$$\log \left[\frac{10^{-4}}{10^{-5}} \right] = 10000 \left[\frac{1}{500} - \frac{1}{T_2} \right]$$

$$\log 10 = 10000 \left[\frac{1}{500} - \frac{1}{T_2} \right]$$

$$T_2 = 526.31 \sim 526\text{K}$$

Question114

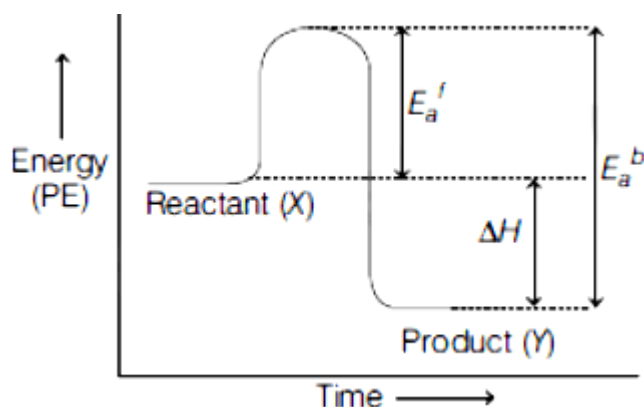
An exothermic reaction $X \rightarrow Y$ has an activation energy 30kJ mol^{-1} . If energy change ΔE during the reaction is -20kJ mol^{-1} then the activation energy for the reverse reaction in kJ is

[26 Feb 2021 Shift 1]

Answer: 50

Solution:

$X \rightarrow Y$, it is an exothermic reaction whose ΔE or $\Delta H = -20\text{kJ mol}^{-1}$



Given, activation energy of the forward reaction ($X \rightarrow Y$),

$$E_a^f = 30\text{kJ mol}^{-1}$$

Activation energy of the backward or reverse reaction ($Y \rightarrow X$), E_a^b can be calculated as,

$$\Delta H = E_a^f - E_a^b$$

$$\Rightarrow E_a^b = E_a^f - \Delta H = 30 - (-20) = 50\text{kJ mol}^{-1}$$

Question115

A reaction has a half-life of 1min. The time required for 99.9% completion of the reaction is min (Round off to the nearest integer).

[Use : $\ln 2 = 0.69$, $\ln 10 = 2.3$]

[18 Mar 2021 Shift 2]

Answer: 10

Solution:

Using formula,

$$k = \frac{1}{t} \ln \left(\frac{C_0}{C_t} \right)$$

$$\text{As } k = \frac{\ln 2}{t_{1/2}}$$

$$t_{1/2} = 1 \text{ min [Given]}$$

$$\frac{\ln 2}{1} = \frac{1}{t} \ln \left(\frac{100}{0.1} \right)$$

$$\Rightarrow t = \frac{\ln 1000}{\ln 2}$$

$$t = \frac{3 \times 2.3}{0.69} = 10$$

Question116

For a certain first order reaction 32% of the reactant is left after 570s. The rate constant of this reaction is $\times 10^{-3} \text{ s}^{-1}$. (Round off to the nearest integer).

[Given, $\log_{10} 2 = 0.301$, $\ln 10 = 2.303$]

[17 Mar 2021 Shift 1]

Answer: 2

Solution:

For first order reaction,

$$k \times t = \ln \frac{A_0}{A_t} \dots (i)$$

where, rate = rate constant

t = time

Initial conc. of reactant, $A_0 = 100$

Conc. of reactant after 't' time, $A_t = 32$

Put all values in equation (i)

$t = \text{time}$

$$A_t = 32$$

$$k \times 570 = \ln \frac{100}{32}$$

$$k \times 570 = 2.303(\log 10^2 - \log 2^5)$$

$$\Rightarrow k = \frac{2.303}{570}(2 - 5 \log 2)$$

$$= \frac{2.303}{570}[2 - (5 \times 0.3)]$$

$$= \frac{2.303}{570} \times 0.5$$

$$k = 2 \times 10^{-3} \text{ s}^{-1}$$

Question117

The decomposition of formic acid on gold surface follows first order kinetics. If the rate constant at 300K is $1.0 \times 10^{-3} \text{ s}^{-1}$ and the activation energy, $E_a = 11.488 \text{ kJ mol}^{-1}$, the rate constant at 200K is $\times 10^{-5} \text{ s}^{-1}$ (Round off to the nearest integer).

[Given, $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$]

[16 Mar 2021 Shift 1]

Answer: 10

Solution:

Given, $T_1 = 200 \text{ K}$, $T_2 = 300 \text{ K}$

$E_a = \text{Activation energy} = 11.488 \text{ kJ/mol}$

Rate constant (k_2) at 300K = 10^{-3} s^{-1}

To find = rate constant (k_1) at 200K

We know,

$$\log \frac{k_2}{k_1} = \frac{E}{2.303R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

$$\log \frac{10^{-3}}{k_1} = \frac{11.488 \times 10^3}{2.303 \times 8.314} \left(\frac{1}{200} - \frac{1}{300} \right)$$

$$\log \frac{10^{-3}}{k_1} = 1$$

$$\log \frac{10^{-3}}{k_1} = \log 10$$

$$\Rightarrow k_1 = 10^{-4}$$

$$= 10 \times 10^{-5} \text{ s}^{-1}$$

Question 118

A and B decompose via first order kinetics with half-lives 54.0 min and 18.0 min respectively. Starting from an equimolar non-reactive mixture of A and B, the time taken for the concentration of A to become 16 times that of B is min. (Round off to the nearest integer).

[16 Mar 2021 Shift 2]

Answer: 108

Solution:

Given, $(t_{1/2})_1 = 54 \text{ min}$

$(t_{1/2})_2 = 18 \text{ min}$

A B

$t = 0, 'x' \text{ M}$ $t = 0, 'x' \text{ M}$

To calculate, $[A_t] = 16 \times [B_t] \dots (i)$

time = ?

For 1st order reaction, $[A_t] = \frac{A_0}{(2)^n}$

n = number of half-lives

From Eq. (i), $[A_t] = 16 \times [B_t]$

$$\frac{x}{(2)^{n_1}} = \frac{x}{(2)^{n_2}} \times 16$$

$$(2)^{n_2} = (2)^{n_1} \times (2)^4$$

$$\Rightarrow n_2 = n_1 + 4$$

$$\frac{t}{(t_{1/2})_2} = \frac{t}{(t_{1/2})_1} + 4 \Rightarrow t \left(\frac{1}{18} - \frac{1}{54} \right) = 4$$

$$\Rightarrow t = \frac{4 \times 18 \times 54}{36} = 108 \text{ min}$$

Question119

The reaction $2A + B_2 \rightarrow 2AB$ is an elementary reaction. For a certain quantity of reactants, if the volume of the reaction vessel is reduced by a factor of 3, the rate of the reaction increases by a factor of (Round off to the nearest integer).
[17 Mar 2021 Shift 2]

Answer: 27

Solution:

Given, the reaction $2A + B_2 \rightarrow 2AB$ is an elementary reaction.

So, rate of reaction

$$= k[A]^2[B_2]$$

$$\text{Initial rate} = k(a/V)^2(b/V)^1$$

where, a and b are the number of moles of A and B and V is the volume.

On reducing the volume by a factor of 3, the concentrations of A and B_2 will become 3 times

$$\text{Final rate} = k(3a/V)^2(3b/V)^1 = 3^2 \times 3k\left(\frac{a}{V}\right)^2\left(\frac{b}{V}\right)^1 = 27 \times \text{initial rate}$$

Hence, the rate becomes $3^2 \times 3 = 27$ times of initial rate.

Question120

The inactivation rate of a viral preparation is proportional to the amount of virus. In the first minute after preparation, 10% of the virus is inactivated. The rate constant for viral inactivation is

$$\underline{\hspace{2cm}} \times 10^{-3} \text{ min}^{-1}. \text{ (Nearest integer)}$$

[Use : $\ln 10 = 2.303$; $\log_{10} 3 = 0.477$; property of logarithm

$$:\log x^y = y \log x]$$

[20 Jul 2021 Shift 1]

Answer: 106

Solution:

As the unit of rate constant is min^{-1} so it must be a first order reaction

$$K \times t = 2.303 \log A_0 / A_t$$

in 1 min 10% is in activated so taking

$$A_0 = 100 \quad A_t = 90 \text{ in 1min}$$

$$\text{So } K \times 1 = 2.303 \times \log \frac{100}{90}$$

$$= 2.303 \times (\log 10 - 2 \log 3)$$

$$= 2.303 \times (1 - 2 \times 0.477)$$

$$= 0.10593$$

$$= 105.93 \times 10^{-3}$$

$$\approx 106$$

Question121

For a chemical reaction $A \rightarrow B$, it was found that concentration of B is increased by 0.2 mol L^{-1} in 30 min.

The average rate of the reaction is _____ $\times 10^{-1} \text{ mol L}^{-1} \text{ h}^{-1}$. (in nearest integer)

[25 Jul 2021 Shift 2]

Answer: 4

Solution:



$$t = 0 \quad 0$$

$$t = 30 \text{ min} \quad 0.2 \text{ M}$$

$$\text{Av. rate of reaction} = -\frac{\Delta[A]}{\Delta t} = \frac{\Delta[B]}{\Delta t} = \frac{(0.2 - 0)}{\frac{1}{2}}$$

$$= 0.4 = 4 \times 10^{-1} \text{ mol / L} \times \text{hr}$$

Question122



In the above first order reaction the concentration of PCl_5 reduces from initial concentration 50 mol L^{-1} to 10 mol L^{-1} in 120 minutes at 300°K . The rate constant for the reaction at 300°K is $x \times 10^{-2} \text{ min}^{-1}$.

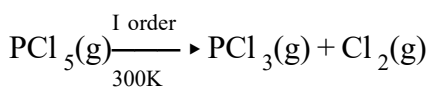
The value of x is _____.

[Given $\log 5 = 0.6989$]

[20 Jul 2021 Shift 2]

Answer: 1

Solution:



$$t = 0 \quad 50\text{M}$$

$$t = 120\text{min} \quad 10\text{M}$$

$$\Rightarrow K = \frac{2.303}{t} \log \frac{[A_0]}{[A_t]}$$

$$\Rightarrow K = \frac{2.303}{120} \log \frac{50}{10}$$

$$\Rightarrow K = \frac{2.303}{120} \times 0.6989 = 0.013413 \text{ min}^{-1}$$

$$= 1.3413 \times 10^{-2} \text{ min}^{-1}$$

$$1.34 \Rightarrow \text{Nearest integer} = 1$$

Question123

For the first order reaction $\text{A} \rightarrow 2\text{B}$, 1 mole of reactant A gives 0.2 moles of B after 100 minutes. The half life of the reaction is min. (Round off to the nearest integer).

[Use : $\ln 2 = 0.69$, $\ln 10 = 2.3$]

Properties of logarithms : $\ln x^y = y \ln x$;

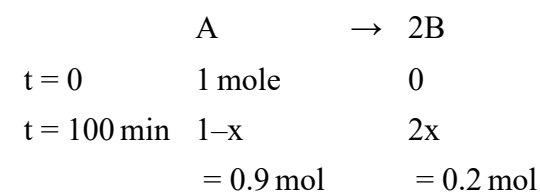
$$\ln\left(\frac{x}{y}\right) = \ln x - \ln y$$

(Round off to the nearest integer)

[27 Jul 2021 Shift 2]

Answer: 300

Solution:



$$\text{Now, } t = \frac{t_{1/2}}{\ln 2} \times \frac{[A_0]}{[A_t]}$$

$$100 = \frac{t_{1/2}}{\ln 2} \times \ln \frac{1}{0.9} \Rightarrow t_{1/2} = 690 \text{ min.}$$

(taking $\ln 3 = 1.11$)

Ans. 600 to 700

Question124

For a reaction of order n, the unit of the rate constant is :

[27 Jul 2021 Shift 1]

Options:

A. $\text{mol}^{1-n} \text{L}^{1-n} \text{s}$

B. $\text{mol}^{1-n} \text{L}^{2n-1} \text{s}^{-1}$

C. $\text{mol}^{1-n} \text{L}^{n-1} \text{s}^{-1}$

D. $\text{mol}^{1-n} \text{L}^{1-n} \text{s}^{-1}$

Answer: C

Solution:

$$\text{Rate} = k[A]^n$$

comparing units

$$\frac{(\text{mol}/\text{l})}{\text{sec}} = k \left(\frac{\text{mol}}{\text{l}} \right)^n$$

$$\Rightarrow k = \text{mol}^{(1-n)} \text{l}^{(n-1)} \text{s}^{-1}$$

Question 125

For the reaction $A + B \rightleftharpoons 2C$

the value of equilibrium constant is 100 at 298K. If the initial concentration of all the three species is 1M each, then the equilibrium concentration of C is $x \times 10^{-1} \text{M}$. The value of x is _____ . (Nearest integer)

[25 Jul 2021 Shift 1]

Answer: 25

Solution:



$$\begin{array}{ccc} 1 & 1 & 1 \\ -x & -x & 2x \end{array}$$

$$K = \frac{[C]_{\text{eq}}^2}{[A]_{\text{eq}}[B]_{\text{eq}}} = \frac{(1+2x)^2}{(1-x)(1-x)}$$

$$100 = \left(\frac{1+2x}{1-x} \right)^2$$

$$\left(\frac{1+2x}{1-x} \right) = 10$$

$$x = \frac{3}{4}$$

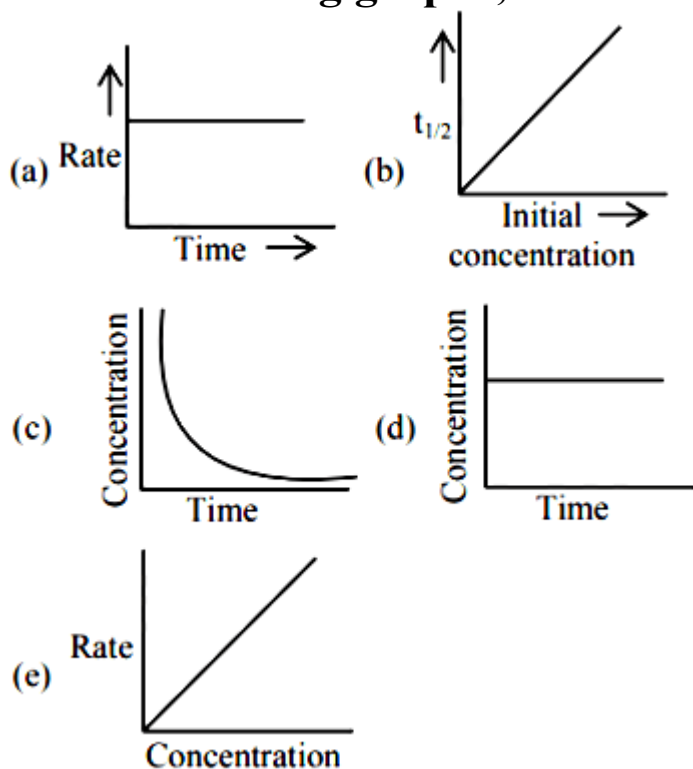
$$[C]_{\text{eq}} = 1 + 2x$$

$$= 1 + 2 \left(\frac{3}{4} \right)$$

$= 2.5\text{M}$
 $25 \times 10^{-1}\text{M}$

Question126

For the following graphs,



Choose from the options given below, the correct one regarding order of reaction is :

[25 Jul 2021 Shift 1]

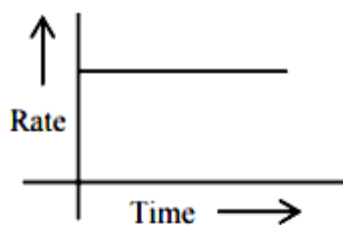
Options:

- A. (b) zero order (c) and (e) First order
- B. (a) and (b) Zero order (e) First order
- C. (b) and (d) Zero order (e) First order
- D. (a) and (b) Zero order (c) and (e) First order

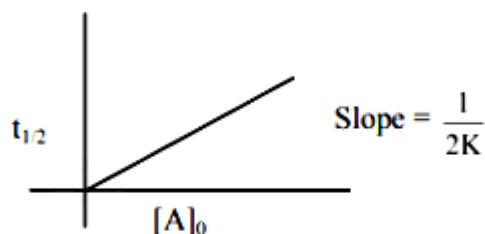
Answer: A

Solution:

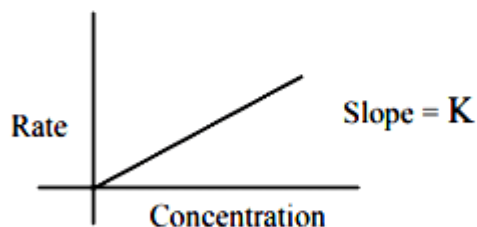
For zero order reaction's rate $= K [\text{Reactant}]^0$
 $\Rightarrow r = k$



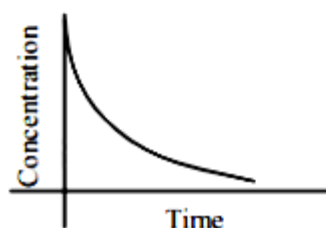
$$t_{1/2} = \frac{[A]_0}{2K} \text{ for zero order}$$



For first order reaction $\rightarrow$
 $r = K [\text{Concentration}]$



Reactant concentration after time $t \rightarrow C_t = C_0 e^{-kt}$



Reactant concentration after time $t \rightarrow C_t = C_0 e^{-kt}$

Question 127



In the above first order reaction the initial concentration of N_2O_5 is $2.40 \times 10^{-2} \text{ mol L}^{-1}$ at 318K. The concentration of N_2O_5 after 1 hour was $1.60 \times 10^{-2} \text{ mol L}^{-1}$. The rate constant of the reaction at 318K is $\underline{\hspace{2cm}} \times 10^{-3} \text{ min}^{-1}$. (Nearest integer)

[Given : $\log 3 = 0.477$, $\log 5 = 0.699$]

[22 Jul 2021 Shift 2]

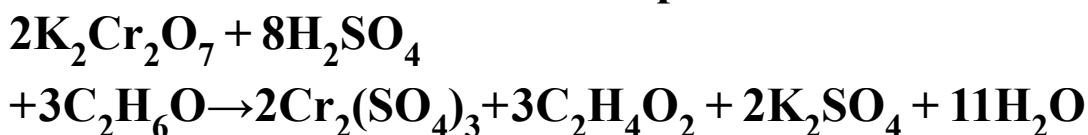
Answer: 7

Solution:

$$K = \frac{2.303}{t} \log \frac{[N_2O_5]_0}{[N_2O_5]_t}$$
$$= \frac{2.303}{60} \log \frac{2.4}{1.6} = 6.76 \times 10^{-3} \text{ min}^{-1} \approx 7 \times 10^{-3} \text{ min}^{-1}$$

Question 128

The reaction that occurs in a breath analyser, a device used to determine the alcohol level in a person's blood stream is



If the rate of appearance of $Cr_2(SO_4)_3$ is $2.67 \text{ mol min}^{-1}$ at a particular time, the rate of disappearance of C_2H_6O at the same time is mol min^{-1} . (Nearest integer)

[27 Aug 2021 Shift 1]

Answer: 4

Solution:

Rate of appearance of 2 moles of $Cr_2(SO_4)_3$,

$$\frac{d}{dt}[Cr_2(SO_4)_3] = 2.67 \text{ mol/min}$$

Rate of disappearance of 3 moles of C_2H_6O , $\frac{d[C_2H_6O]}{dt}$

$$= \frac{3}{2} \frac{d[Cr_2(SO_4)_3]}{dt} = \frac{2.67}{2} \times 3$$

$$= 4.005 = 4 \text{ mol/min}$$

Question129

The reaction rate for the reaction

$[\text{PtCl}_4]^{2-} + \text{H}_2\text{O} \rightleftharpoons [\text{Pt}(\text{H}_2\text{O})\text{Cl}_3]^- + \text{Cl}^-$ was measured as a function of concentrations of different species. It was observed that

$$\frac{-d[(\text{PtCl}_4)^{2-}]}{dt} = 4.8 \times 10^{-5}[(\text{PtCl}_4)^{2-}] = 2.4 \times 10^{-3}[\text{Pt}(\text{H}_2\text{O})\text{Cl}_3^-][\text{Cl}^-]$$

where, square brackets are used to denote molar concentrations.

The equilibrium constant, $K_c = \dots\dots\dots$ (Nearest integer)

[26 Aug 2021 Shift 2]

Answer: 50

Solution:

Consider, at equilibrium $\frac{d[(\text{PtCl}_4)^{2-}]}{dt} = 0$

$$\therefore 4.8 \times 10^{-5}[(\text{PtCl}_4)^{2-}] = 2.4 \times 10^{-3}[\text{Pt}(\text{H}_2\text{O})\text{Cl}_3^-][\text{Cl}^-]$$

Equilibrium constant,

$$\begin{aligned} K_c &= \frac{[\text{PtCl}_4]^{2-}}{[\text{Pt}(\text{H}_2\text{O})\text{Cl}_3^-][\text{Cl}^-]} \\ &= \frac{2.4 \times 10^{-3}}{4.8 \times 10^{-5}} = 0.50 \times 100 = 50 \end{aligned}$$

Question130

For a first order reaction, the ratio of the time for 75% completion of a reaction to the time for 50% completion is..... (Integer answer)

[31 Aug 2021 Shift 1]

Answer: 2

Solution:

For a first order reaction, rate constant is given as,

$$k = \frac{2.303}{t} \log \frac{a}{a-x}$$

where, a = initial concentration of reactant and
a - x = concentration of reactant after time t.

For 50% completion,

$$k = \frac{2.303}{t_{50\%}} \log \frac{100}{100-50} \dots(i)$$

For 75% completion,

$$k = \frac{2.303}{t_{75\%}} \log \frac{100}{100-75} \dots(ii)$$

From Eqs. (i) and (ii),

$$\frac{2.303}{t_{50\%}} \log \frac{100}{50} = \frac{2.303}{t_{75\%}} \log \frac{100}{25}$$

$$\frac{0.3010}{t_{50\%}} = \frac{0.6020}{t_{75\%}}$$

$$t_{75\%} = 2t_{50\%}$$

$$\therefore \frac{t_{75\%}}{t_{50\%}} = \frac{2}{1}$$

Question131

The first order rate constant for the decomposition of CaCO_3 at 700K is $6.36 \times 10^{-3} \text{ s}^{-1}$ and activation energy is 209 kJ mol^{-1} . Its rate constant (in s^{-1}) at 600K is $x \times 10^{-6}$. The value of x is (Nearest integer)

$$\begin{aligned} &[\text{Given, } R = 8.31 \text{ JK}^{-1} \text{ mol}^{-1}, \log 6.36 \times 10^{-3} \\ &= -2.19, 10^{-4.79} = 1.62 \times 10^{-5}] \end{aligned}$$

[27 Aug 2021 Shift 2]

Answer: 16

Solution:

According to Arrhenius theory,

$$k = Ae^{-E_a/RT}$$

(k is the rate constant, A is Arrhenius factor, E_a = Activation energy, T is absolute temperature and R is gas constant.)

$$\ln k = \ln A - \frac{E_a}{RT}$$

At temperature T_1 ,

$$\ln k_1 = \ln A - \frac{E_a}{RT_1} \dots (i)$$

At temperature T_2 ,

$$\ln k_2 = \ln A - \frac{E_a}{RT_2} \dots (ii)$$

Subtracting Eqs. (ii) from (i),

$$\ln k_1 - \ln k_2 = \frac{E_a}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

$$\ln \frac{k_1}{k_2} = \frac{E_a}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

$$2.303 \log \frac{k_1}{k_2} = \frac{E_a}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

$$\log \frac{k_1}{k_2} = \frac{E_a}{2.303R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

At, $T_1 = 700\text{K}$, $k_1 = 6.36 \times 10^{-3}\text{s}^{-1}$, $E_a = 209\text{kJmol}^{-1}$,

$R = 8.314\text{JK}^{-1}\text{mol}^{-1}$

At, $T_2 = 600\text{K}$, $k_2 = x \times 10^{-6}\text{s}^{-1}$

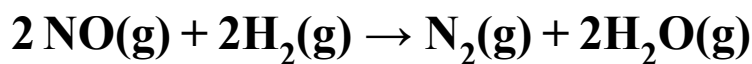
$$\log \frac{6.36 \times 10^{-3}}{x \times 10^{-6}} = \frac{209}{2.303 \times 8.314} \left(\frac{1}{600} - \frac{1}{700} \right)$$

$$= \frac{209}{2.303 \times 8.314} \left(\frac{100}{420000} \right)$$

$x = 16$.

Question132

The following data was obtained for chemical reaction given below at 975K.



	[NO] mol L ⁻¹	[H ₂] mol L ⁻¹	Rate mol L ⁻¹
A.	8×10^{-5}	8×10^{-5}	7×10^{-9}
B.	24×10^{-5}	8×10^{-5}	21×10^{-8}
C.	24×10^{-5}	32×10^{-5}	8.4×10^{-8}

The order of the reaction with respect to NO is [Integer answer]

[26 Aug 2021 Shift 1]

Answer: 1

Solution:

$$\text{Rate of the reaction} = K[\text{NO}]^x[\text{H}_2]^y$$

where K is rate constant and x, y are order of reaction w.r.t. NO and H₂ respectively.

From observations A and B

$$7 \times 10^{-9} = K(8 \times 10^{-5})^x(8 \times 10^{-5})^y \dots(i)$$

$$\text{and } 2.1 \times 10^{-8} = K(24 \times 10^{-5})^x(8 \times 10^{-5})^y \dots(ii)$$

Dividing Eqs. (i) by (ii)

$$\frac{7 \times 10^{-9}}{2.1 \times 10^{-8}} = \frac{K(8 \times 10^{-5})^x(8 \times 10^{-5})^y}{K(24 \times 10^{-5})^x(8 \times 10^{-5})^y}$$

$$\frac{1}{3} = \left(\frac{1}{3}\right)^x$$

$$x = 1$$

∴ Order of reaction w.r.t. NO is 1.

Question133

For the reaction $\text{A} \rightarrow \text{B}$, the rate constant k (in S⁻¹) is given by

$$\log_{10} k = 20.35 - \frac{(2.47 \times 10^3)}{T}. \text{The energy of activation in kJ mol}^{-1} \text{ is.....}$$

(Nearest integer)

[Given : $R = 8.314 \text{ JK}^{-1} \text{ mol}^{-1}$]
[31 Aug 2021 Shift 2]

Answer: 47

Solution:

According to Arrhenius theory,

$$\log k = \log A - \frac{E_a}{2.303 RT} \dots (i)$$

$$\log_{10} k = 20.35 - \frac{2.47 \times 10^3}{T} \dots (ii)$$

$$\frac{-2.47 \times 10^3}{T} = -\frac{E_a}{2.303 RT}$$

Comparing Eqs. (i) and (ii),

$$E_a = 2.47 \times 2.303 \times 8.314$$

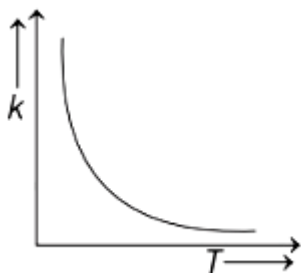
$$= 47.29 \text{ kJ/mol} \simeq 47 \text{ kJ/mol}$$

Question 134

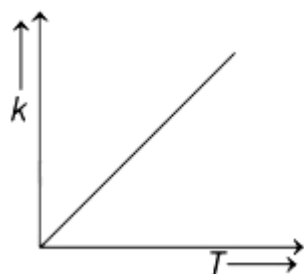
Which one of the following given graphs represents the variation of rate constant (k) with temperature (T) for an endothermic reaction?
[1 Sep 2021 Shift 2]

Options:

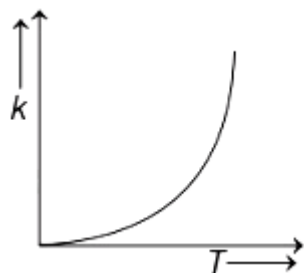
A.



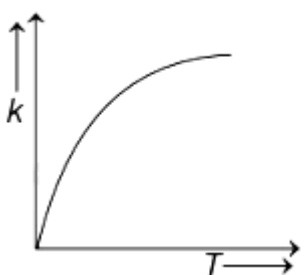
B.



C.



D.



Answer: C

Solution:

According to Arrhenius equation,

$$k = Ae^{-E_a/RT}$$

As the temperature increases, rate constant k also increases which will generally speed up the process (increase the rate).

So, graph will exponentially increase as in option (c).

Question 135

Gaseous cyclobutene isomerizes to butadiene in a first order process which has a ' k ' value of $3.3 \times 10^{-4} \text{ s}^{-1}$ at 153°C . The time in minutes it takes for the isomerization to proceed 40% to completion at this temperature is _____ (Rounded off to the nearest integer)

[2021]

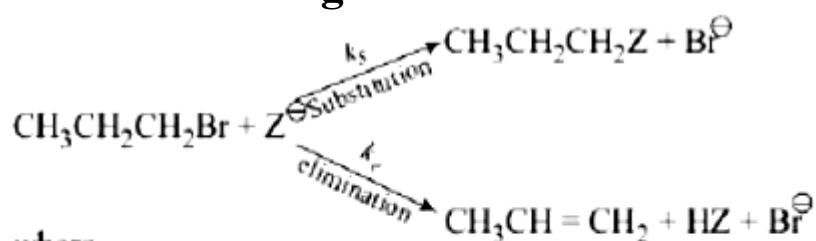
Answer: 26

Solution:

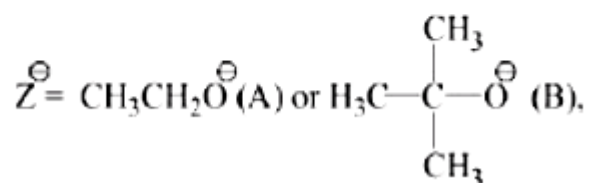
$$k = \frac{1}{t} \cdot \ln \frac{[A]_0}{[A]_t}$$
$$\Rightarrow t = \frac{1}{k} \ln \left(\frac{100}{60} \right) \Rightarrow t = \frac{1}{3.3 \times 10^{-4}}$$
$$\Rightarrow t = 1548.49 \text{ sec}$$
$$\Rightarrow t = 25.81 \text{ min} \approx 26 \text{ min.}$$

Question 136

For the following reactions



where



k_s and k_e , are, respectively, the rate constants for substitution and elimination, and $\mu = \frac{k_s}{k_e}$, the correct option is

[Jan. 07, 2020 (II)]

Options:

A. $\mu_B > \mu_A$ and $k_e(\text{A}) > k_e(\text{B})$

B. $\mu_A > \mu_B$ and $k_e(\text{B}) > k_e(\text{A})$

C. $\mu_B > \mu_A$ and $k_e(\text{B}) > k_e(\text{A})$

D. $\mu_A > \mu_B$ and $k_e(A) > k_e(B)$

Answer: B

Solution:

(b) Among the given bases (A) and (B), *t*-butoxide being bulky base favours elimination reaction and ethoxide favours substitution reaction.

$\therefore$ when $Z^\circ = \text{CH}_3\text{CH}_2\text{O}^-$, (substitution reaction favoured) and when $Z = \text{CH}_3 - \underset{\text{CH}_3}{\overset{\text{CH}_3}{\text{C}}} - \text{O}^-$,

(elimination reaction favoured)

Hence, $\mu_A > \mu_B$ and $k_e(B) > k_e(A)$

Question 137

For the reaction $2\text{H}_2(\text{g}) + 2\text{NO}(\text{g}) \rightarrow \text{N}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$ the observed rate expression is, $\text{rate} = k_e[\text{NO}]^2[\text{H}_2]$. The rate expression for the reverse reaction is:
[Jan. 07, 2020 (II)]

Options:

A. $k_b[\text{N}_2][\text{H}_2\text{O}]^2$

B. $k_b[\text{N}_2][\text{H}_2\text{O}]^2 / [\text{NO}]$

C. $k_b[\text{N}_2][\text{H}_2\text{O}]$

D. $k_b[\text{N}_2][\text{H}_2\text{O}]^2 / [\text{H}_2]$

Answer: D

Solution:

Rate of forward reaction $= k_f[\text{NO}]^2[\text{H}_2]$

Observed rate $= k_f[\text{NO}]^2[\text{H}_2]$

$$\text{Observed rate} = \frac{\text{Rate of forward reaction}}{[\text{H}_2]}$$

$$\therefore \text{Rate of backward reaction} = \frac{k_b[\text{N}_2][\text{H}_2\text{O}]^2}{[\text{H}_2]}$$

Question138

During the nuclear explosion, one of the products is ^{90}Sr with half life of 6.93 years. If $1\mu\text{g}$ of ^{90}Sr was absorbed in the bones of a newly born baby in place of Ca, how much time, in years, is required to reduce it by 90% if it is not lost metabolically.

[NV, Jan. 07, 2020 (I)]

Answer: 23.03

Solution:

$$t_{1/2} = 6.93 \text{ years,}$$

$$a = 10^{-6} \text{ g}$$

$$t_{1/2} = \frac{0.693}{K}$$

$$\Rightarrow K = \frac{0.693}{t_{1/2}} = \frac{0.693}{6.93} = 0.1$$

For Ist order reaction,

$$K = \frac{2.303}{t} \log \frac{a}{a-x}$$

$$t = \frac{2.303}{K} \log \frac{a}{a-x}$$

$$= \frac{2.303}{0.1} \log \frac{10^{-6}}{10^{-7}}$$

$$= \frac{2.303}{0.1} = 23.03 \text{ years}$$

Question139

For following reactions:



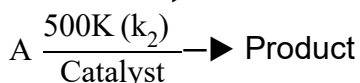
it was found that the E_a is decrease by 30kJ/mol in the presence of catalyst. If the rate remains unchanged, the activation energy for catalysed reaction is (Assume pre exponential factor is same):
[Jan. 09, 2020(I)]

Options:

- A. 75kJ/mol
- B. 105kJ/mol
- C. 135kJ/mol
- D. 198kJ/mol

Answer: A

Solution:



$$\text{Given: } k_1 = k_2$$

$$Ae^{-\frac{E_{a_1}}{RT_1}} = Ae^{-\frac{E_{a_2}}{RT_2}}$$

$$\frac{E_{a_1}}{T_1} = \frac{E_{a_2}}{T_2}$$

$$E_{a_2} = E_{a_1} - 30$$

$$\Rightarrow E_{a_1} = E_{a_2} + 30$$

$$\frac{E_{a_2} + 30}{T_1} = \frac{E_{a_2}}{T_2}$$

$$\frac{E_{a_2} + 30}{700} = \frac{E_{a_2}}{500}$$

$$\Rightarrow 150 = 2E_{a_2}$$

$$E_{a_2} = 75\text{kJ/mol}$$

Question140

A sample of milk splits after 60 min. at 300K and after 40 min. at 400K when the population of lactobacillus acidophilus in it doubles. The activation energy (in kJ/mol) for this process is closest to

(Given, $R = 8.3 \text{ J mol}^{-1} \text{ K}^{-1}$, $\ln\left(\frac{2}{3}\right) = 0.4$, $e^{-3} = 4.0$)

[NV, Jan. 09, 2020(II)]

Answer: -3.98

Solution:

For a first order reaction, $kt = \ln \frac{[A]}{[A_0]}$

At 300K, $k_1 \times 60 = \ln \frac{[A]}{[A_0]} \dots (i)$

At 400K, $k_2 \times 40 = \ln \frac{[A]}{[A_0]} \dots (ii)$

From equation (1) and (2),

$$\frac{k_2}{k_1} = \frac{60}{40}$$

$$\ln \frac{k_2}{k_1} = \frac{E_a}{R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$$

$$\ln \left(\frac{60}{40} \right) = \frac{E_a}{8.3} \times \frac{100}{400 \times 300}$$

$$\ln \left(\frac{3}{2} \right) \times 8.3 \times 1200 = E_a$$

$$\text{Given : } \ln \frac{2}{3} = 0.4$$

$$\text{Then, } \ln \frac{3}{2} = -0.4$$

$$\Rightarrow E_a = -0.4 \times 8.3 \times 1200$$

$$\Rightarrow E_a = -3984 \text{ J/mol}$$

$$\Rightarrow E_a = -3.984 \text{ kJ/mol}$$

Question141

**The rate of a certain biochemical reaction at physiological temperature (T) occurs 10^6 times faster with enzyme than without. The change in the activation energy upon adding enzyme is:
[Jan .08,2020(I)]**

Options:

A. $-6(2.303)RT$

B. $-6RT$

C. $+6(2.303)RT$

D. $+6RT$

Answer: A

Solution:

The rate constant of a reaction is given by

$$k = Ae^{-E_a/RT}$$

The rate constant in tpresence of catalyst is given by

$$k' = Ae^{-E'_a/RT}$$

$$\frac{k'}{k} = e^{-(E'_a - E_a)/RT}$$

$$10^6 = e^{-(E'_a - E_a)/RT}$$

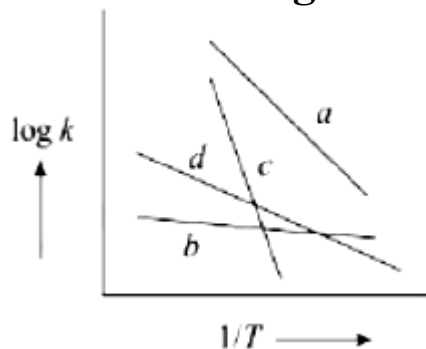
$$\ln 10^6 = -\frac{(E'_a - E_a)}{RT}$$

$$E'_a - E_a = -6(2.303)RT$$

Question142

Consider the following plots of rate constant versus $\frac{1}{T}$ for four different reactions. Which of the following orders is correct for the

activation energies of these reactions?



[Jan. 08, 2020 (II)]

Options:

A. $E_b > E_a > E_d > E_c$

B. $E_a > E_c > E_d > E_b$

C. $E_c > E_a > E_d > E_b$

D. $E_b > E_d > E_c > E_a$

Answer: C

Solution:

Arrhenius equation, $k = Ae^{-E_a/RT}$

$$\log k = \log A - \frac{E_a}{2.303RT}$$

$$\text{slope} = -\frac{E_a}{2.303R}$$

$\therefore$ More negative the slope greater will be the E_a .

So correct order is $E_c > E_a > E_d > E_b$

Question143

For the reaction $2A + 3B + \frac{3}{2}C \rightarrow 3P$, which statement is correct?

[Sep. 03,2020 (II)]

Options:

A. $\frac{dn_A}{dt} = \frac{3}{2} \frac{dn_B}{dt} = \frac{3}{4} \frac{dn_C}{dt}$

B. $\frac{dn_A}{dt} = \frac{dn_B}{dt} = \frac{dn_C}{dt}$

C. $\frac{dn_A}{dt} = \frac{2}{3} \frac{dn_B}{dt} = \frac{4}{3} \frac{dn_C}{dt}$

D. $\frac{dn_A}{dt} = \frac{2}{3} \frac{dn_B}{dt} = \frac{3}{4} \frac{dn_C}{dt}$

Answer: C

Solution:

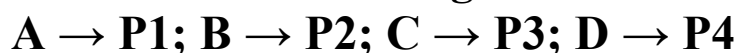
For a given reaction,

$$\text{rate} = -\frac{1}{2} \frac{dn_A}{dt} = -\frac{1}{3} \frac{dn_B}{dt} = -\frac{2}{3} \frac{dn_C}{dt}$$

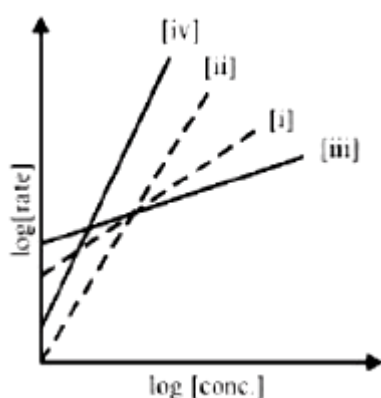
$$\text{rate} = \frac{dn_A}{dt} = \frac{2}{3} \frac{dn_B}{dt} = \frac{4}{3} \frac{dn_C}{dt}$$

Question144

Consider the following reactions



The order of the above reactions are (i), (ii), (iii), and (iv), respectively. The following graph is obtained when $\log [\text{rate}]$ vs. $\log [\text{conc.}]$ are plotted:



Among the following, the correct sequence for the order of the reactions is:

[Sep. 06, 2020 (I)]

Options:

A. (iv) > (i) > (ii) > (iii)

B. (i) > (ii) > (iii) > (iv)

C. (iii) > (i) > (ii) > (iv)

D. (iv) > (ii) > (i) > (iii)

Answer: D

Solution:

$$\text{Rate} = k[A]^n$$

$$\log[\text{Rate}] = \log k + n \log[A]$$

$$\text{Slope} = n \text{ [} n \text{ is order of the reaction]}$$

∴ Correct sequence for the order of the reaction is

$$(iv) > (ii) > (i) > (iii)$$

Question145

A flask contains a mixture of compounds A and B. Both compounds decompose by first-order kinetics. The half-lives for A and B are 300s and 180s, respectively. If the concentrations of A and B are equal initially, the time required for the concentration of A to be four times that of B (in s) is: (Use $\ln 2 = 0.693$)

[Sep. 05 , 2020 (I)]

Options:

A. 180

B. 900

C. 300

D. 120

Answer: B

Solution:

$$C_t = C_0 e^{-kt}; k = \frac{\ln 2}{t_{1/2}}$$

$$(C_t)_A = (C_0)_A e^{-k_A t}; k_A = \frac{\ln 2}{300}$$

$$(C_t)_B = (C_0)_B e^{-k_B t}; k_B = \frac{\ln 2}{180}$$

$$\frac{(C_t)_B}{(C_t)_A} = \frac{(C_0)_B}{(C_0)_A} \times e^{(k_B - k_A)t}$$

$$\Rightarrow 4 = e^{(k_B - k_A)t}$$

$$\Rightarrow 2 \ln 2 = \left[\frac{\ln 2}{180} - \frac{\ln 2}{300} \right] t$$

$$\Rightarrow 2 \ln 2 = \ln 2 \left[\frac{1}{180} - \frac{1}{300} \right] t$$

$$\Rightarrow 2 = \left(\frac{120}{180 \times 300} \right) t \Rightarrow t = \frac{2 \times 180 \times 300}{120} = 900 \text{ sec}$$

Question 146

If 75% of a first order reaction was completed in 90 minutes, 60% of the same reaction would be completed in approximately (in minutes)

_____ .
(Take : $\log 2 = 0.30$; $\log 2.5 = 0.40$)

[NV, Sep. 04, 2020 (I)]

Answer: 60

Solution:

$$t = \frac{2.303}{k} \log \left[\frac{100}{100 - x\%} \right]$$

$$t_{75\%} = \frac{2.303}{k} \log \left[\frac{100}{25} \right] = 90$$

$$t_{60\%} = \frac{2.303}{k} \log \left[\frac{100}{40} \right]$$

$$\frac{t_{75\%}}{t_{60\%}} = \frac{2 \log 2}{\log 2.5} \Rightarrow \frac{90}{t_{60\%}} = \frac{2 \times 0.3}{0.4}$$

$$\Rightarrow t_{60\%} = \frac{90 \times 4}{6} = 60 \text{ min.}$$

Question147

It is true that :
[Sep. 03, 2020 (I)]

Options:

- A. A second order reaction is always a multistep reaction
- B. A zero order reaction is a multistep reaction
- C. A first order reaction is always a single step reaction
- D. A zero order reaction is a single step reaction

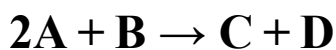
Answer: B

Solution:

zero order reaction is a multistep reaction .

Question148

The results given in the below table were obtained during kinetic studies of the following reaction:



Experiment	[A] mol L	[B] mol L ⁻¹	Initial rate V mol L ⁻¹ min ⁻¹
I	0.1	0.1	6.00×10^{-3}
II	0.1	0.2	2.40×10^{-2}
III	0.2	0.1	1.20×10^{-2}
IV	X	0.2	7.20×10^{-2}
V	0.3	Y	2.88×10^{-1}

X and Y in the given table are respectively:
[Sep. 02 , 2020(II)]

Options:

A. 0.4, 0.4

B. 0.4, 0.3

C. 0.3, 0.4

D. 0.3, 0.3

Answer: C

Solution:

$$\text{Rate (R)} = k[A]^a[B]^b$$

$$\text{Exp I} \Rightarrow 6.0 \times 10^{-3} = k[0.1]^a[0.1]^b$$

$$\text{Exp II} \Rightarrow 24.0 \times 10^{-3} = k[0.1]^a[0.2]^b$$

$$\text{Exp III} \Rightarrow 12.0 \times 10^{-3} = k[0.2]^a[0.1]^b$$

$$\text{Exp IV} \Rightarrow 72 \times 10^{-3} = k[X]^a[0.2]^b$$

$$\text{Exp V} \Rightarrow 288 \times 10^{-3} = k[0.3]^a[Y]^b$$

From Exp I & II,

$$\Rightarrow \frac{6.0 \times 10^{-3}}{24 \times 10^{-3}} = \frac{k[0.1]^a[0.1]^b}{k[0.1]^a[0.2]^b}$$

$$\Rightarrow \frac{1}{4} = \left(\frac{0.1}{0.2} \right)^b \Rightarrow \left(\frac{1}{2} \right)^2 = \left(\frac{1}{2} \right)^b$$

$$\therefore b = 2$$

Similarly, from exp I & III we get

$$\frac{1}{2} = \left(\frac{1}{2} \right)^a \Rightarrow a = 1$$

From Exp. II & IV,

$$\Rightarrow \frac{24 \times 10^{-3}}{72 \times 10^{-3}} = \frac{k[0.1]^a[0.2]^b}{k[X]^a[0.2]^b}$$

$$\Rightarrow \frac{1}{3} = \left(\frac{0.1}{X} \right)^a \Rightarrow \frac{1}{3} = \left(\frac{0.1}{X} \right)^1$$

$$\therefore X = 0.3$$

From Exp. I & V,

$$\Rightarrow \frac{6.0 \times 10^{-3}}{288 \times 10^{-3}} = \left(\frac{0.1}{0.3} \right)^a \left(\frac{0.1}{Y} \right)^b$$

$$\Rightarrow \frac{1}{48} = \left(\frac{1}{3} \right)^1 \left(\frac{0.1}{Y} \right)^2$$

$$\Rightarrow \frac{3}{48} = \left(\frac{0.1}{Y} \right)^2 \Rightarrow \frac{1}{16} = \left(\frac{0.1}{Y} \right)^2$$

$$\Rightarrow \left(\frac{1}{4}\right)^2 = \left(\frac{0.1}{Y}\right)^2 \Rightarrow \frac{1}{4} = \frac{0.1}{Y}$$

$$\therefore Y = 0.4$$

Question149

The rate of a reaction decreased by 3.555 times when the temperature was changed from 40°C to 30°C. The activation energy (in kJ mol⁻¹) of the reaction is _____. Take;

$$R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1} \ln 3.555 = 1.268$$

[NV, Sep. 06, 2020 (II)]

Answer: 100

Solution:

The Arrhenius equation is

$$k = A e^{\frac{E_a}{RT}}$$

Assuming A and E_a to be independent of temperature

$$\ln \frac{k_2}{k_1} = \frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

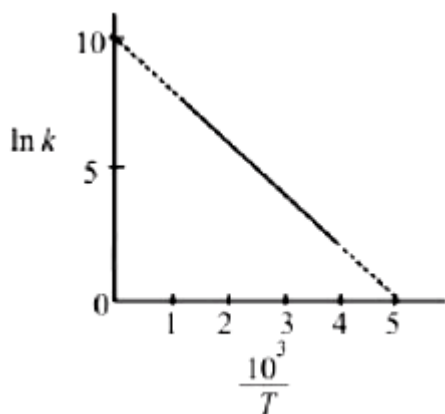
$$\ln 3.555 = \frac{E_a}{8.314} \left(\frac{1}{303} - \frac{1}{313} \right)$$

$$\Rightarrow E_a = \frac{1.268 \times 8.314 \times 303 \times 313}{10}$$

$$= 99980.7 = 99.98 \text{ kJ/mol}$$

Question150

The rate constant (k) of a reaction is measured at different temperatures (T), and the data are plotted in the given figure. The activation energy of the reaction in kJ mol⁻¹ is:
(R is gas constant)



[Sep. 05, 2020 (II)]

Options:

A. $2/R$

B. $1/R$

C. R

D. $2R$

Answer: D

Solution:

Arrhenius equation:

$$k = Ae^{-E_a/RT}$$

$$\ln k = \ln A - \left(\frac{E_a}{R} \right) \frac{1}{T}$$

$$\ln k = \ln A - \left(\frac{E_a}{R \times 10^3} \right) \times \frac{10^3}{T}$$

$$\text{Slope of graph} = \frac{-E_a}{R \times 10^3} = \frac{-10}{5}$$

$$E_a = 2R \times 10^3 \text{ J} = 2R \text{ kJ}$$

Question 151

The number of molecules with energy greater than the threshold energy for a reaction increases five fold by a rise of temperature from 27°C to 42°C . Its energy of activation in J/mol is _____ .

(Take $\ln 5 = 1.6094$; $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$)
[NV, Sep. 04, 2020 (II)]

Answer: 84297.48

Solution:

$$\begin{aligned} \because k &= A e^{-E_a/RT} \\ \ln \frac{k_2}{k_1} &= \frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right) \Rightarrow \ln 5 = \frac{E_a}{R} \left(\frac{1}{300} - \frac{1}{315} \right) \\ \Rightarrow E_a &= \frac{1.6094 \times 8.314 \times 300 \times 315}{15} = 84297.48 \text{ J/mol} . \end{aligned}$$

Question152

the expression for $\frac{d[A]}{dt}$ is:
[Jan. 10,2019 (II)]

Options:

- A. $k_1[A_2] - k_{-1}[A]^2$
- B. $2k_1[A_2] - k_{-1}[A]^2$
- C. $k_1[A_2] + k_{-1}[A]^2$
- D. $2k_1[A_2] - 2k_{-1}[A]^2$

Answer: D

Solution:

Given: $A_2 \xrightleftharpoons[k_{-1}]{k_1} 2A$

Now, $-\frac{1}{2} \frac{d[A]}{dt} = k_{-1}[A]^2 - k_1[A_2]$

$$\frac{d[A]}{dt} = -2k_{-1}[A]^2 + 2k_1[A_2]$$

$$\Rightarrow \frac{d[A]}{dt} = 2k_1[A_2] - 2k_{-1}[A]^2$$

Question153

Decomposition of X exhibits a rate constant of $0.05\mu\text{g}/\text{year}$. How many years are required for the decomposition of $5\mu\text{g}$ of X into $2.5\mu\text{g}$?

[Jan. 12, 2019 (I)]

Options:

- A. 50
- B. 25
- C. 20
- D. 40

Answer: A

Solution:

Rate constant of decomposition of X = $0.05\mu\text{g}/\text{year}$. Unit of rate constant confirms that the decomposition of X is a zero order reaction.

For zero order kinetics,

$$[X] = [X]_0 - kt$$

$$kt = [X]_0 - [X]$$

$$t = \frac{[X]_0 - [X]}{k}$$

$$t = \frac{5 - 2.5}{0.05}$$

$$= \frac{2.5}{0.05} = 50 \text{ years}$$

Question154

The reaction $2X \rightarrow B$ is a zeroth order reaction. If the initial concentration of X is 0.2M , the half-life is 6h. When the initial

concentration of X is 0.5M , the time required to reach its final concentration of 0.2M will be :

[Jan. 11, 2019 (II)]

Options:

A. 9.0h

B. 12.0h

C. 18.0h

D. 7.2h

Answer: C

Solution:

For the reaction $2X \rightarrow B$, follow zeroth order Rate equation is

$$K t = [A]_0 - [A]$$

For the half-life; $t = t_{1/2}$ and $[A] = 0.1$

$$K t_{1/2} = 0.2 - 0.1$$

$$\frac{0.2 - 0.1}{6} = \frac{0.1}{6} \text{ M hr}^{-1}$$

$\therefore$ Time required to reach from 0.5M to 0.2M

$$K t = [A]_0 - [A]$$

$$\frac{0.1}{6} \times t = (0.5 - 0.2); t = 18 \text{ hour}$$

Question 155

The following results were obtained during kinetic studies of the reaction;

$2A + B \rightarrow \text{Products}$

Experiment	[A] (in mol L^{-1})	(in mol L^{-1})	Initial Rate of reaction (in $\text{mol L}^{-1} \text{min}^{-1}$)
I	0.10	0.20	6.93×10^{-3}
II	0.10	0.25	6.93×10^{-3}
III	0.20	0.30	1.386×10^{-2}

**The time (in minutes) required to consume half of A is:
[Jan. 9, 2019 (I)]**

Options:

A. 5

B. 10

C. 1

D. 100

Answer: A

Solution:

From experiment I and II, it is observed that order of reaction w.r.t. (B) is zero.

From experiment II and III, x can be calculated as:

$$\frac{6.93 \times 10^{-3}}{13.86 \times 10^{-3}} = \left(\frac{0.1}{0.2} \right)^x \left(\frac{0.25}{0.3} \right)^y$$

∴ Order of reaction w.r.t. B = 0,

so $y = 0$

$$\therefore \frac{6.93 \times 10^{-3}}{13.86 \times 10^{-3}} = \left(\frac{0.1}{0.2} \right)^x$$

$$\left(\frac{1}{2} \right)^1 = \left(\frac{1}{2} \right)^x$$

$x = 1$

Now, Rate = $K [A]^1 [B]^0$

or, $6.93 \times 10^{-3} = K [0.1]^1 [0.2]^0$

$$K = 6.93 \times 10^{-2}$$

For the reaction, $2A + B \rightarrow \text{Products}$

$$2K t = \ln \frac{[A]_0}{[A]}$$

$$\therefore t_{1/2} = \frac{0.693}{2K} = \frac{0.693}{0.693 \times 10^{-2} \times 2}$$

Question156

For the reaction, $2A + B \rightarrow \text{products}$, when the concentrations of A and B both were doubled, the rate of the reaction increased from $0.3 \text{ mol L}^{-1} \text{ s}^{-1}$ to $2.4 \text{ mol L}^{-1} \text{ s}^{-1}$. When the concentration of A alone is doubled, the rate increased from $0.3 \text{ mol L}^{-1} \text{ s}^{-1}$ to $0.6 \text{ mol L}^{-1} \text{ s}^{-1}$.

Which one of the following statements is correct?

[Jan. 9, 2019 (II)]

Options:

- A. Total order of the reaction is 4
- B. Order of the reaction with respect to B is 2
- C. Order of the reaction with respect to B is 1
- D. Order of the reaction with respect to A is 2

Answer: B

Solution:

$$r = K [A]^x [B]^y$$

$$\frac{r_2}{r_1} = 2^x \cdot 2^y = 8 \Rightarrow x + y = 3$$

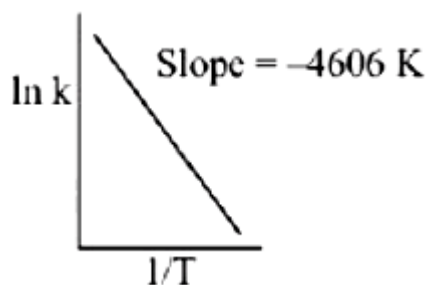
$$\frac{r_3}{r_1} = 2^x = 2 \Rightarrow x = 1$$

$$\therefore y = 2$$

$$t_{1/2} = 5$$

Question157

For a reaction, consider the plot of $\ln k$ versus $1/T$ given in the figure. If the rate constant of this reaction at 400K is 10^{-5} s^{-1} , then the rate constant at 500K is :



[Jan. 12, 2019 (II)]

Options:

- A. 10^{-6} s^{-1}

B. $2 \times 10^{-4} \text{s}^{-1}$

C. 10^{-4}s^{-1}

D. $4 \times 10^{-4} \text{s}^{-2}$

Answer: C

Solution:

From Arrhenius equation,

$$\ln K = \ln A - \frac{E_a}{RT}$$

$$\text{Slope} = \frac{-E_a}{R} = -4606 \text{K}$$

$$\log \left(\frac{K_2}{K_1} \right) = \frac{E_a}{2.303R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

$$\log \left(\frac{K_2}{10^{-5}} \right) = \frac{1}{2.303} \times 4606 \left(\frac{1}{400} - \frac{1}{500} \right)$$

$$= \frac{4606 \times (100)}{2.303 \times 400 \times 500} = 1$$

$$\Rightarrow \log \left(\frac{K_2}{10^{-5}} \right) = 1; \frac{K_2}{10^{-5}} = \text{Antilog}(1)$$

$$\frac{K_2}{10^{-5}} = 10$$

$$\Rightarrow K_2 = 10^{-5} \times 10 = 10^{-4} \text{s}^{-1}$$

Question158

If a reaction follows the Arrhenius equation, the plot $\ln k$ vs $\frac{1}{(RT)}$ gives straight line with a gradient $(-y)$ unit.

**The energy required to activate the reactant is:
[Jan. 11,2019(I)]**

Options:

A. y/R unit

B. y unit

C. yR unit

D. $-y$ unit

Answer: B

Solution:

From Arrhenius equation,

$$k = Ae^{-E_a/RT}$$

$$\ln k = \ln A - E_a/RT$$

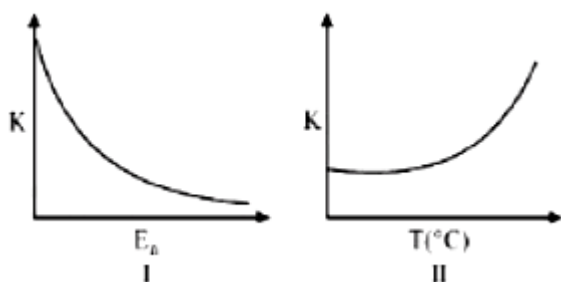
$$\text{slope} = -y \text{ (given)}$$

$$-y = -E_a$$

$$\Rightarrow E_a = y$$

Question159

Consider the given plots for a reaction obeying Arrhenius equation ($0^\circ\text{C} < T < 300^\circ\text{C}$) : (k and E_a are rate constant and activation energy, respectively)



Choose the correct option:
[Jan. 10,2019 (I)]

Options:

- A. I is right but II is wrong
- B. Both I and II are correct
- C. I is wrong but II is right
- D. Both I and II are wrong

Answer: B

Solution:

From Arrhenius equation,

$$k = Ae^{-E_a/RT}$$

So, as E_a increases, $e^{-E_a/RT}$ decreases, k decreases

and as T increases, $\frac{E_a}{RT}$ decreases, $e^{-E_a/RT}$ increases.

Question 160

In the following reaction :

$$x\text{A} \rightarrow y\text{B} \quad \log_{10} \left[-\frac{d[\text{A}]}{dt} \right] = \log_{10} \left[\frac{d[\text{B}]}{dt} \right] + 0.3010$$

'A' and 'B' respectively can be:

[April 12, 2019 (I)]

Options:

A. n-Butane and Iso-butane

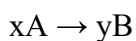
B. C_2H_2 and C_6H_6

C. C_2H_4 and C_4H_8

D. N_2O_4 and NO_2

Answer: C

Solution:



$$\therefore \frac{-d\text{A}}{xdt} = \frac{1}{y} \frac{dB}{dt}$$

$$\frac{-d\text{A}}{dt} = \frac{dB}{dt} \times \frac{x}{y}$$

$$\log \left[\frac{-d\text{A}}{dt} \right] = \log \left[\frac{dB}{dt} \right] + \log \left(\frac{x}{y} \right)$$

Comparing this equation with the equation given in question. We get,

$$\log \frac{x}{y} = 0.3010 \text{ or } \log \frac{x}{y} = \log 2$$

$$\therefore \frac{x}{y} = 2$$

$\therefore$ The reaction is of type $2\text{A} \rightarrow \text{B}$.

Hence, option (3) is correct.

Question161

N O₂ required for a reaction is produced by the decomposition of N₂O₅ in CCl₄ as per the equation, $2\text{N}_2\text{O}_5(\text{g}) \rightarrow 4\text{N O}_2(\text{g}) + \text{O}_2(\text{g})$
The initial concentration of N₂O₅ is 3.00mol L⁻¹ and it is 2.75mol L⁻¹ after 30 minutes. The rate of formation of N O₂ is:
[April 12,2019 (II)]

Options:

- A. $4.167 \times 10^{-3} \text{mol L}^{-1} \text{min}^{-1}$
- B. $1.667 \times 10^{-2} \text{mol L}^{-1} \text{min}^{-1}$
- C. $8.333 \times 10^{-3} \text{mol L}^{-1} \text{min}^{-1}$
- D. $2.083 \times 10^{-3} \text{mol L}^{-1} \text{min}^{-1}$

Answer: B

Solution:



$$\text{rate of reaction} = \frac{-1}{2} \frac{d[\text{N}_2\text{O}_5]}{dt} = \frac{1}{4} \frac{d[\text{N O}_2]}{dt} = \frac{d[\text{O}_2]}{dt}$$

According to the question

$$\frac{-d[\text{N}_2\text{O}_5]}{dt} = -\frac{(2.75 - 3)}{30} = \frac{0.25}{30} \text{M min}^{-1}$$

$$\begin{aligned} \therefore \frac{d[\text{N O}_2]}{dt} &= 2 \times \frac{-d[\text{N}_2\text{O}_5]}{dt} = 2 \times \frac{0.25}{30} \\ &= 1.67 \times 10^{-2} \text{M min}^{-1} \end{aligned}$$

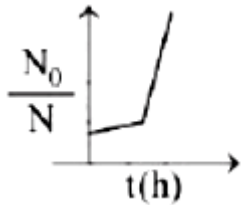
Question162

A bacterial infection in an internal wound grows as $N'(t) = N_0 \exp(t)$, where the time t is in hours. A dose of antibiotic, taken orally, needs 1 hour to reach the wound. Once it reaches there, the

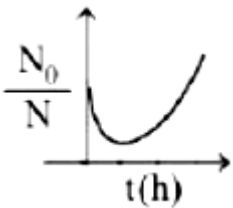
bacterial population goes down as $\frac{dN}{dt} = -5N^2$. What will be the plot of $\frac{N_0}{N}$ vs .t after 1 hour?
[April 10,2019 (I)]

Options:

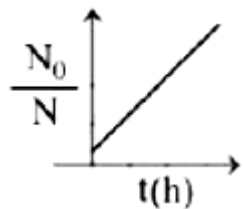
A.



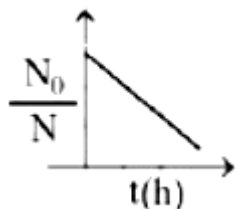
B.



C.



D.



Answer: C

Solution:

From 0 to 1h, $N' = N_0 e^t$

When drug is administered bacterial growth is given by

$$\frac{dN}{dt} = -5N^2$$

So, at $t = 1h$, $N' = eN_0$

Integrating the following equation,

$$\frac{dN}{dt} = -5N^2$$

$$\int_{eN_0}^N \frac{dN}{N^2} = -5 \int_1^t dt$$

$$\frac{1}{N} - \frac{1}{eN_0} = 5(t - 1)$$

$$\frac{N_0}{N} - \frac{1}{e} = 5N_0(t - 1)$$

$$\frac{N_0}{N} = 5N_0(t - 1) + \frac{1}{e}$$

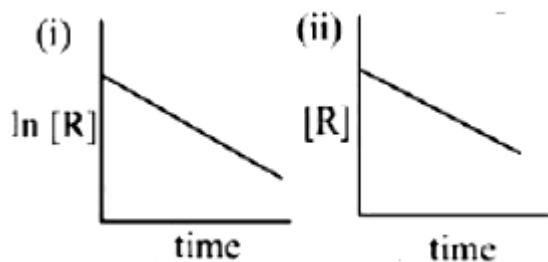
$$\frac{N_0}{N} = 5N_0t + \left(\frac{1}{e} - 5N_0 \right)$$

The above equation is similar to straight line equation with positive slope.

Thus $\frac{N_0}{N}$ increases linearly with t .

Question163

The given plots represents the variation of the concentration of a reactant R with time for two different reactions (i) and (ii). The respective orders of the reactions are:



[April 9, 2019 (I)]

Options:

- A. 1,0
- B. 1,1
- C. 0,1
- D. 0,2

Answer: A

Solution:

In graph (i), $\ln[\text{Reactant}]$ vs time is linear with positive intercept and negative slope. Hence it is 1st order. In graph (ii), $[\text{Reactant}]$ vs time is linear with positive intercept and negative slope. Hence, it is zero order.

Question 164

For the reaction $2A + B \rightarrow C$, the values of initial rate at different reactant concentrations are given in the table below. The rate law for the reaction is:

$[A](\text{mol L}^{-1})$	$[B](\text{mol L}^{-1})$	Initial Rate($\text{mol L}^{-1}\text{s}^{-1}$)
0.05	0.05	0.045
0.10	0.05	0.090
0.20	0.10	0.72

[April 8, 2019 (I)]

Options:

A. Rate = $k[A][B]^2$

B. Rate = $k[A]^2[B]^2$

C. Rate = $k[A][B]$

D. Rate = $k[A]^2[B]$

Answer: A

Solution:



$$\text{Rate} = k[A]^x[B]^y$$

Exp-1, $0.045 = k[0.05]^x[0.05]^y \dots (i)$

Exp-2, $0.090 = k[0.1]^x[0.05]^y \dots (ii)$

Exp-3, $0.72 = k[0.2]^x[0.1]^y \dots (iii)$

Divide equation (i) by equation (ii)

$$\frac{0.045}{0.090} = \left(\frac{1}{2}\right)^x \Rightarrow x = 1$$

Divide equation (i) by equation (iii)

$$\frac{0.045}{0.72} = \left(\frac{0.05}{0.1}\right)^y \left(\frac{0.05}{0.2}\right)^1$$

$$\frac{0.045}{0.72} \times \frac{0.2}{0.05} = \left(\frac{0.05}{0.1}\right)^y$$

$$\left(\frac{1}{2}\right)^2 = \left(\frac{1}{2}\right)^y \Rightarrow y = 2$$

$$\text{Rate law} = k[A]^1[B]^2.$$

Question165

For a reaction scheme $A \xrightarrow{k_1} B \xrightarrow{k_2} C$, if the rate of formation of B is set to be zero then the concentration of B is given by:

[April 8, 2019 (II)]

Options:

A. $(k_1 - k_2)[A]$

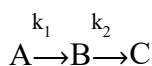
B. $k_1 k_2 [A]$

C. $(k_1 + k_2)[A]$

D. $\left(\frac{k_1}{k_2}\right)[A]$

Answer: D

Solution:



$$\frac{d[B]}{dt} = k_1[A] - k_2[B] = 0$$

$$[B] = \frac{k_1[A]}{k_2}$$

Question166

For the reaction of H_2 with I_2 , the rate constant is $2.5 \times 10^{-4} \text{ dm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ at 327°C and $1.0 \text{ dm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ at 527°C .

The activation energy for the reaction, in kJ mol^{-1} is:

($R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$)

[April 10, 2019 (II)]

Options:

A. 166

B. 150

C. 72

D. 59

Answer: A

Solution:

$$K = e^{-\frac{E_a}{RT}} \text{ or } \log K = \frac{-E_a}{2.303RT}$$

$$\text{So, } \log \frac{K_2}{K_1} = \frac{E_a}{2.303R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

$$\log \frac{1}{2.5 \times 10^{-4}} = \frac{E_a}{8.314 \times 2.303} \left(\frac{1}{600} - \frac{1}{800} \right)$$

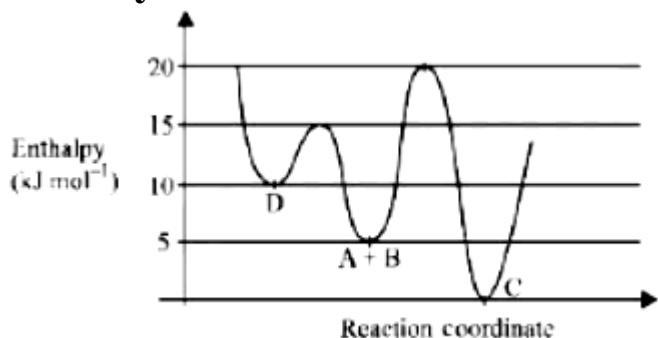
$$3.6 = \frac{E_a}{8.314 \times 2.303} \times \frac{200}{600 \times 800}$$

$$E_a = 165.4 \text{ kJ / mol} \approx 166 \text{ kJ / mol}$$

Question 167

Consider the given plot of enthalpy of the following reaction between A and B, $\text{A} + \text{B} \rightarrow \text{C} + \text{D}$.

Identify the incorrect statement.



[April 9, 2019 (II)]

Options:

- A. Activation enthalpy to form C is 5 kJ mol^{-1} less than that to form D.
- B. C is the thermodynamically stable product.
- C. D is kinetically stable product.
- D. Formation of A and B from C has highest enthalpy of activation.

Answer: A

Solution:

As we can see from the graph that activation enthalpy to form D from A + B is $15 - 5 = 10\text{ kJ mol}^{-1}$, whereas, to form C from A + B is $20 - 5 = 15\text{ kJ mol}^{-1}$. Therefore, activation enthalpy to form C is 5 kJ more than that to form D.

Question 168

At 518°C , the rate of decomposition of a sample of gaseous acetaldehyde, initially at a pressure of 363 Torr, was 1.00 Torr s^{-1} when 5% had reacted and 0.5 Torr s^{-1} when 33% had reacted. The order of the reaction is:

[2018]

Options:

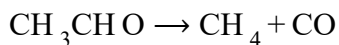
- A. 2
- B. 3

C. 1

D. 0

Answer: A

Solution:



$$r \propto (a-x)^m$$

Generally

m = order of reaction

a - x = unreacted

$r_1 = 1 \text{ torr s}^{-1}$, when 5% reacted

$r_2 = 0.5 \text{ torr s}^{-1}$, when 33% reacted

$(a - x_1) = 0.95$ (unreacted)

$(a - x_2) = 0.67$ (unreacted)

$$\frac{r_1}{r_2} = \left[\frac{(a - x_1)}{(a - x_2)} \right]^m; \frac{1}{0.5} = \left(\frac{0.95}{0.67} \right)^m$$

$$2 = (1.41)^m \Rightarrow 2 = (\sqrt{2})^m$$

$$\Rightarrow m = 2$$

Question169

**If 50% of a reaction occurs in 100 seconds and 75% of the reaction occurs in 200 seconds, the order of this reaction is:
[Online April 16, 2018]**

Options:

A. 2

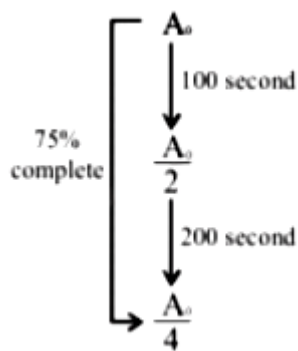
B. 3

C. Zero

D. 1

Answer: D

Solution:



First order reaction as half life is constant.

Question170

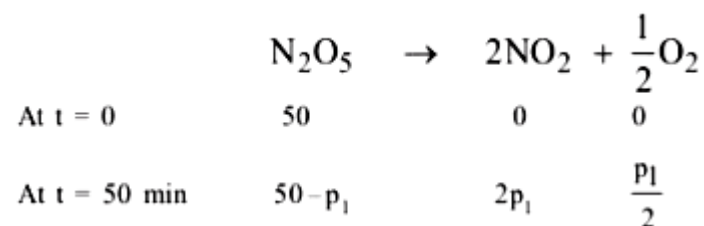
N_2O_5 decomposes to N O_2 and O_2 and follows first order kinetics. After 50 minutes, the pressure inside the vessel increases from 50mmH g to 87.5mmH g. The pressure of the gaseous mixture after 100 minutes at constant temperature will be _____.
[Online April 15, 2018 (I)]

Options:

- A. 136.25mmH g
- B. 106.25mmH g
- C. 175.0mmH g
- D. 116.25mmH g

Answer: B

Solution:



Total pressure at 50 minutes

$$= 50 - p_1 + 2p_1 + \frac{p_1}{2} = 87.5$$

$$50 + \frac{3p_1}{2} = 87.5$$

$$\frac{3p_1}{2} = 37.5$$

$$\therefore p_1 = \frac{37.5 \times 2}{3} = 25$$

At, $t = 100\text{min}$

$$50 - p_2 \quad 2p_2 \quad \frac{p_2}{2}$$

50 minutes is half life period

For 100 minutes i.e. for 2 half lives $50 - p_2 = 12.5$

$$\therefore p_2 = 37.5\text{mm of Hg}$$

Total pressure at 100 minutes

$$= 50 - p_2 + 2p_2 + \frac{p_2}{2}$$

$$= 50 + \frac{3p_2}{2} = 50 + \frac{3}{2} \times 37.5$$

$$= 50 + 56.25$$

$$= 106.25\text{mm of Hg}$$

Question171

For a first order reaction, $A \rightarrow P$, $t_{1/2}$ (half-life) is 10 days.

The time required for $\frac{1}{4}$ conversion of A(in days) is:

($\ln 2 = 0.693$, $\ln 3 = 1.1$).

[Online April 15,2018(II)]

Options:

A. 3.2

B. 2.5

C. 4.1

D. 5

Answer: C

Solution:

The half life $t_{1/2} = 10$ days

The decay constant

$$k = \frac{0.693}{t_{1/2}} = \frac{0.693}{10\text{ days}} = 0.0693\text{ days}^{-1}$$

The time required for one fourth conversion

$$t = \frac{2.303}{k} \log_{10} \frac{a}{a-x}$$

$$= \frac{2.303}{0.0693 \text{ day}^{-1}} \log_{10} \frac{1}{1-(1/4)} = 4.1 \text{ days}$$

Question 172

Two reactions R_1 and R_2 have identical pre-exponential factors.

Activation energy of R_1 exceeds that of R_2 by 10 kJ mol^{-1} . If k_1 and k_2 are rate constants for reactions R_1 and R_2 respectively at 300 K , then $\ln(k_2/k_1)$ is equal to:

($R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$)
[2017]

Options:

- A. 8
- B. 12
- C. 6
- D. 4

Answer: D

Solution:

From Arrhenius equation,

$$k = A \cdot e^{\frac{-E_a}{RT}}$$

$$\text{so, } k_1 = A \cdot e^{\frac{-E_{a_1}}{RT}} \dots (i)$$

$$k_2 = A \cdot e^{\frac{-E_{a_2}}{RT}} \dots (ii)$$

On dividing equation (ii) by (i)

$$\Rightarrow \frac{k_2}{k_1} = e^{\frac{(E_{a_1} - E_{a_2})}{RT}}$$

$$\ln\left(\frac{k_2}{k_1}\right) = \frac{E_{a_1} - E_{a_2}}{RT} = \frac{10,000}{8.314 \times 300} = 4$$

Question173

The rate of a reaction A doubles on increasing the temperature from 300 to 310K . By how much, the temperature of reaction B should be increased from 300K so that rate doubles if activation energy of the reaction B is twice to that of reaction A.

[Online April 8, 2017]

Options:

A. 9.84K

B. 4.92K

C. 2.45K

D. 19.67K

Answer: B

Solution:

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left[\frac{T_2 - T_1}{T_1 T_2} \right]$$

For reaction A -

Given, $\frac{k_2}{k_1} = 2$, $T_1 = 300\text{K}$, $T_2 = 310\text{K}$

$$\log 2 = \frac{E_a}{2.303R} \left[\frac{1}{300} - \frac{1}{310} \right] \dots (i)$$

For reaction B -

Given, $\frac{k_2}{k_1} = 2$, $E_a = 2E_a$, $T_1 = 300\text{K}$, $T_2 = ?$

$$\log 2 = \frac{E_a}{2.303R} \left[\frac{1}{300} - \frac{1}{T_2} \right] \dots (ii)$$

From equation (i) and (ii),

$$\frac{2E_a}{2.303R} \left[\frac{1}{300} - \frac{1}{T_2} \right] = \frac{E_a}{2.303R} \left[\frac{1}{300} - \frac{1}{310} \right]$$

$$\Rightarrow 2 \left[\frac{1}{300} - \frac{1}{T_2} \right] = \frac{310 - 300}{300 \times 310}$$

$$\Rightarrow T_2 = 304.92\text{K}$$

$$T_1 = 300\text{K}, T_2 = 304.92\text{K}$$

$$\Delta T = T_2 - T_1 = 4.92\text{K}$$

Question174

The rate of a reaction quadruples when the temperature changes from 300 to 310K . The activation energy of this reaction is :
(Assume activation energy and pre-exponential factor are independent of temperature; $\ln 2 = 0.693$; $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$)
[Online April 9, 2017]

Options:

A. $107.2 \text{ kJ mol}^{-1}$

B. 53.6 kJ mol^{-1}

C. 26.8 kJ mol^{-1}

D. $214.4 \text{ kJ mol}^{-1}$

Answer: A

Solution:

$$\ln \frac{k_1}{k_2} = \frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

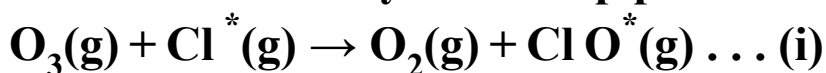
$$\ln 4 = \frac{E_a}{8.314} \left(\frac{310 - 300}{310 \times 300} \right)$$

$$2 \ln 2 = \frac{E_a}{8.314} \left(\frac{310 - 300}{310 \times 300} \right)$$

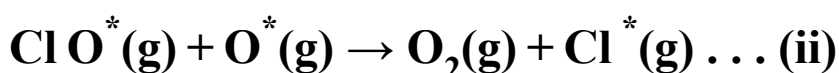
$$E_a = \frac{0.693 \times 2 \times 8.314 \times 300 \times 310}{10} = 107.2 \text{ kJ /mol}$$

Question175

The reaction of ozone with oxygen atoms in the presence of chlorine atoms can occur by a two step process shown below:

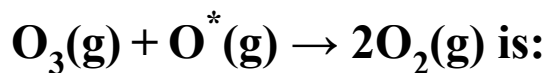


$$[K_i = 5.2 \times 10^9 \text{ L mol}^{-1} \text{ s}^{-1}]$$



$$[K_{ai} = 2.6 \times 10^{10} \text{Lmol}^{-1} \text{s}^{-1}]$$

The closest rate constant for the overall reaction



[Online April 9, 2016]

Options:

A. $1.4 \times 10^{20} \text{Lmol}^{-1} \text{s}^{-1}$

B. $3.1 \times 10^{10} \text{Lmol}^{-1} \text{s}^{-1}$

C. $5.2 \times 10^9 \text{Lmol}^{-1} \text{s}^{-1}$

D. $2.6 \times 10^{10} \text{Lmol}^{-1} \text{s}^{-1}$

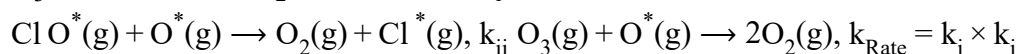
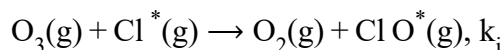
Answer: A

Solution:

On adding eq. (i) and eq. (ii) we get $\text{O}_3(\text{g}) + \text{O}(\text{g}) \rightarrow 2\text{O}_2(\text{g})$

Hence overall rate constant $= k_i \times k_{ii}$

$$= 5.2 \times 10^9 \times 2.6 \times 10^{10} \approx 1.4 \times 10^{20} \text{mol}^{-1} \text{Ls}^{-1}$$



Question176

The rate law for the reaction below is given by the expression

$$k[\text{A}][\text{B}]$$



If the concentration of B is increased from 0.1 to 0.3mol e, keeping the value of A at 0.1mol e, the rate constant will be:

[Online April 10, 2016]

Options:

A. 3k

B. 9k

C. $k/3$

D. k

Answer: D

Solution:

Rate constant is independent of concentration.

Question 177

Decomposition of H_2O_2 follows a first order reaction. In fifty minutes the concentration of H_2O_2 decreases from 0.5 to 0.125M in one such decomposition. When the concentration of H_2O_2 reaches 0.05M, the rate of formation of O_2 will be:
[2016]

Options:

A. 2.66Lmin^{-1} at ST P

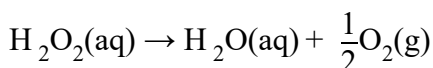
B. $1.34 \times 10^{-2}\text{mol min}^{-1}$

C. $6.96 \times 10^{-2}\text{mol min}^{-1}$

D. $6.93 \times 10^{-4}\text{mol min}^{-1}$

Answer: D

Solution:



For a first order reaction

$$k = \frac{2.303}{t} \log \frac{a}{(a-x)}$$

Given $a = 0.5$, $(a-x) = 0.125$, $t = 50\text{min}$

$$\therefore k = \frac{2.303}{50} \log \frac{0.5}{0.125}$$

$$r = k[\text{H}_2\text{O}_2] = 2.78 \times 10^{-2} \text{min}^{-1}$$

$$\text{Now } -\frac{d[\text{H}_2\text{O}_2]}{dt} = \frac{d[\text{H}_2\text{O}]}{dt} = \frac{2d[\text{O}_2]}{dt}$$

$$\therefore \frac{2d[\text{O}_2]}{dt} = -\frac{d[\text{H}_2\text{O}_2]}{dt}$$

$$\therefore \frac{d[\text{O}_2]}{dt} = -\frac{1}{2} \times \frac{d[\text{H}_2\text{O}_2]}{dt}$$

$$= \frac{1.386 \times 10^{-3}}{2} = 6.93 \times 10^{-4} \text{mol min}^{-1}$$

Question178

A + 2B → C, the rate equation for this reaction is given as Rate = k[A][B].

If the concentration of A is kept the same but that of B is doubled what will happen to the rate itself?

[Online April 11,2015]

Options:

A. halved

B. the same

C. doubled

D. quadrupled

Answer: C

Solution:

$$\text{Rate} = k[\text{A}][\text{B}] = R$$

$$R' = k[\text{A}][2\text{B}]$$

$$\frac{R}{R'} = \frac{k[\text{A}][\text{B}]}{k[\text{A}][2\text{B}]} = \frac{k[\text{A}][\text{B}]}{2k[\text{A}][\text{B}]}$$

$$\Rightarrow 2R = R' \text{ i.e., rate become doubles.}$$

Question179

**Higher order (>3) reactions are rare due to :
[2015]**

Options:

- A. shifting of equilibrium towards reactants due to elastic collisions
- B. loss of active species on collision
- C. low probability of simultaneous collision of all the reacting species
- D. increase in entropy and activation energy as more molecules are involved

Answer: C

Solution:

Reactions of higher order (>3) are very rare due to very less chances of many molecules to undergo effective collisions.

Question180

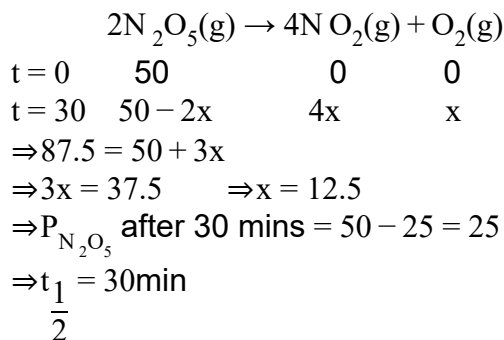
**The reaction $2\text{N}_2\text{O}_5(\text{g}) \rightarrow 4\text{N}_2\text{O}(\text{g}) + \text{O}_2(\text{g})$ follows first order kinetics. The pressure of a vessel containing only N_2O_5 was found to increase from 50mm Hg to 87.5mmHg in 30min. The pressure exerted by the gases after 60min. will be (assume temperature remains constant):
[Online April 10,2015]**

Options:

- A. 106.25mmHg
- B. 150mmHg
- C. 125mmHg
- D. 116.25mmHg

Answer: A

Solution:



Hence after 60 min, (two half lives), $P_{\text{N}_2\text{O}_5} \text{ remaining} = \frac{50}{4} = 12.5\text{torr}$

$\Rightarrow$ Hence decrease in $P_{\text{N}_2\text{O}_5} = 50 - 12.5 = 37.5 \text{ torr}$

$\Rightarrow P_{\text{N}_2\text{O}} = 2 \times 37.5 = 75 \text{ torr}$

$P_{\text{O}_2} = \frac{37.5}{2} = 18.75 \text{ torr}$

$\Rightarrow P_{\text{total}} = 12.5 + 75 + 18.75$
 $= 106.25 \text{ torr}$

Question181

For the equilibrium, $\text{A}(\text{g}) \rightleftharpoons \text{B}(\text{g})$, ΔH is $-40\text{kJ} / \text{mol}$. If the ratio of the activation energies of the forward (E_f) and reverse (E_b)

reactions is $\frac{2}{3}$ then :

[Online April 11, 2015]

Options:

A. $E_f = 80\text{kJ} / \text{mol}$; $E_b = 120\text{kJ} / \text{mol}$

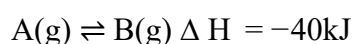
B. $E_f = 60\text{kJ} / \text{mol}$; $E_b = 100\text{kJ} / \text{mol}$

C. $E_f = 30\text{kJ} / \text{mol}$; $E_b = 70\text{kJ} / \text{mol}$

D. $E_f = 70\text{kJ} / \text{mol}$; $E_b = 30\text{kJ} / \text{mol}$

Answer: A

Solution:



If activation energy of the reaction = x

since, $\frac{E_f}{E_b} = \frac{2}{3}$, therefore, $E_f = \frac{2x}{5}$ and $E_b = \frac{3x}{5}$

$$E_b - E_f = +40$$

$$= \frac{3x}{5} - \frac{2x}{5} = 40; \quad \frac{x}{5} = 40; \quad x = 200$$

Therefore

$$E_b = \frac{3x}{5} = \frac{3 \times 200}{5} = 120 \text{ kJ mol}^{-1}$$

$$E_f = \frac{2x}{5} = \frac{2 \times 200}{5} = 80 \text{ kJ mol}^{-1}$$

Question182

For the non – stoichiometric reaction $2A + B \rightarrow C + D$, the following kinetic data were obtained in three separate experiments, all at 298K .

InitialConcentration(A)	InitialConcentration(B)	Initial rate of formation of C ($\text{mol L}^{-1} \text{s}^{-1}$)
0.1M	0.1M	1.2×10^{-3}
0.1M	0.2M	1.2×10^{-3}
0.2M	0.1M	2.4×10^{-3}

The rate law for the formation of C is:
[2014]

Options:

A. $\frac{dc}{dt} = k[A][B]$

B. $\frac{dc}{dt} = k[A]^2[B]$

C. $\frac{dc}{dt} = k[A][B]^2$

D. $\frac{dc}{dt} = k[A]$

Answer: D

Solution:

Let rate of reaction = $\frac{d[C]}{dt} = k[A]^x[B]^y$

Now from the given data

$$1.2 \times 10^{-3} = k[0.1]^x[0.1]^y \dots (i)$$

$$1.2 \times 10^{-3} = k[0.1]^x[0.2]^y \dots (ii)$$

$$2.4 \times 10^{-3} = k[0.2]^x[0.1]^y \dots (iii)$$

Dividing equation (i) by (ii)

$$\Rightarrow \frac{1.2 \times 10^{-3}}{1.2 \times 10^{-3}} = \frac{k[0.1]^x[0.1]^y}{k[0.1]^x[0.2]^y}$$

We find, $y = 0$

Now dividing equation (i) by (iii)

$$\Rightarrow \frac{1.2 \times 10^{-3}}{2.4 \times 10^{-3}} = \frac{k[0.1]^x[0.1]^y}{k[0.2]^x[0.1]^y}$$

We find, $x = 1$

Hence, $\frac{d[C]}{dt} = k[A]^1[B]^0$

Question183

In the reaction of formation of sulphur trioxide by contact process

$2SO_2 + O_2 \rightleftharpoons 2SO_3$ the rate of reaction was measured as

$\frac{d[O_2]}{dt} = -2.5 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$. The rate of reaction in terms of $[SO_2]$ in $\text{mol L}^{-1} \text{ s}^{-1}$ will be:

[Online April 11, 2014]

Options:

A. -1.25×10^{-4}

B. -2.50×10^{-4}

C. -3.75×10^{-4}

D. -5.00×10^{-4}

Answer: D

Solution:

From rate law

$$-\frac{1}{2} \frac{d[SO_2]}{dt} = -\frac{d[O_2]}{dt} = \frac{1}{2} \frac{d[SO_3]}{dt}$$

$$\begin{aligned}\therefore -\frac{d[\text{SO}_2]}{dt} &= -2 \times \frac{d[\text{O}_2]}{dt} \\ &= -2 \times 2.5 \times 10^{-4} \\ &= -5 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}\end{aligned}$$

Question184

For the reaction, $2\text{N}_2\text{O}_5 \rightarrow 4\text{N O}_2 + \text{O}_2$, the rate equation can be expressed in two ways – $\frac{d[\text{N}_2\text{O}_5]}{dt} = k[\text{N}_2\text{O}_5]$ and $+\frac{d[\text{N O}_2]}{dt} = k'[\text{N}_2\text{O}_5]$ k and k' are related as:
[Online April 11, 2014]

Options:

- A. $k = k'$
- B. $2k = k'$
- C. $k = 2k'$
- D. $k = 4k'$

Answer: B

Solution:

Rate of disappearance of reactant = Rate of appearance of products

$$-\frac{1}{2} \frac{d[\text{N}_2\text{O}_5]}{dt} = \frac{1}{4} \frac{d[\text{N O}_2]}{dt}$$

$$\frac{1}{2}k[\text{N}_2\text{O}_5] = \frac{1}{4}k'[\text{N}_2\text{O}_5]$$

$$\frac{k}{2} = \frac{k'}{4}$$

$$\therefore k' = 2k$$

Question185

For the reaction, $3\text{A} + 2\text{B} \rightarrow \text{C} + \text{D}$ the differential rate law can be written as:

[Online April 19, 2014]

Options:

A. $\frac{1}{3} \frac{d[A]}{dt} = \frac{d[C]}{dt} = k[A]^n[B]^m$

B. $-\frac{d[A]}{dt} = \frac{d[C]}{dt} = k[A]^n[B]^m$

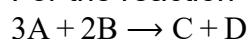
C. $+\frac{1}{3} \frac{d[A]}{dt} = -\frac{d[C]}{dt} = k[A]^n[B]^m$

D. $-\frac{1}{3} \frac{d[A]}{dt} = \frac{d[C]}{dt} = k[A]^n[B]^m$

Answer: D

Solution:

For the reaction



Rate of disappearance of A = Rate of appearance of C reaction

$$= -\frac{1}{3} \frac{d[A]}{dt} = \frac{d[C]}{dt} = k[A]^n[B]^m$$

Question186

The half-life period of a first order reaction is 15 minutes. The amount of substance left after one hour will be:

[Online April 9, 2014]

Options:

A. $\frac{1}{4}$ of the original amount

B. $\frac{1}{8}$ of the original amount

C. $\frac{1}{16}$ of the original amount

D. $\frac{1}{32}$ of the original amount

Answer: C

Solution:

Given $t_{1/2} = 15$ minutes

Total time (T) = 1hr = 60min

From $T = n \times t_{1/2}$

$$n = \frac{60}{15} = 4$$

Now from the formula $\frac{N}{N_0} = \left(\frac{1}{2}\right)^n = \left(\frac{1}{2}\right)^4 = \frac{1}{16}$

Where N_0 = initial amount

N = amount left after time t

Hence the amount of substance left after 1 hour will be $\frac{1}{16}$

Question187

The rate coefficient (k) for a particular reactions is $1.3 \times 10^{-4} \text{M}^{-1} \text{s}^{-1}$ at 100°C , and $1.3 \times 10^{-3} \text{M}^{-1} \text{s}^{-1}$ at 150°C . What is the energy of activation (E_0)(in kJ) for this reaction? (R = molargas constant = $8.314 \text{J K}^{-1} \text{mol}^{-1}$)

[Online April 12, 2014]

Options:

A. 16

B. 60

C. 99

D. 132

Answer: B

Solution:

According to Arrhenius equation $\log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$

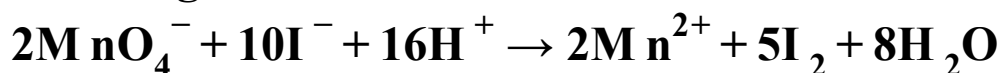
$$\log \frac{1.3 \times 10^{-3}}{1.3 \times 10^{-4}} = \frac{E_a}{2.303 \times 8.314} \left[\frac{1}{373} - \frac{1}{423} \right]$$

$$1 = \frac{E_a}{2.303 \times 8.314} \left[\frac{1}{373} - \frac{1}{423} \right]$$

$$E_a = 60 \text{ kJ/mol}$$

Question 188

The instantaneous rate of disappearance of MnO_4^- ion in the following reaction is $4.56 \times 10^{-3} \text{ M s}^{-1}$



The rate of appearance I_2 is:

[Online April 9, 2013]

Options:

A. $4.56 \times 10^{-4} \text{ M s}^{-1}$

B. $1.14 \times 10^{-2} \text{ M s}^{-1}$

C. $1.14 \times 10^{-3} \text{ M s}^{-1}$

D. $5.7 \times 10^{-3} \text{ M s}^{-1}$

Answer: B

Solution:

Given $-\frac{d[\text{MnO}_4^-]}{dt} = 4.56 \times 10^{-3} \text{ M s}^{-1}$

From the reaction given,

$$-\frac{1}{2} \frac{d[\text{MnO}_4^-]}{dt} = \frac{4.56 \times 10^{-3}}{2} \text{ M s}^{-1}$$

$$-\frac{1}{2} \frac{d[\text{MnO}_4^-]}{dt} = \frac{1}{5} \frac{d\text{I}_2}{dt}$$

$$\therefore -\frac{5}{2} \frac{d[\text{MnO}_4^-]}{dt} = \frac{d\text{I}_2}{dt}$$

On substituting the given value

$$\therefore \frac{d\text{I}_2}{dt} = \frac{4.56 \times 10^{-3} \times 5}{2} = 1.14 \times 10^{-2} \text{ M/s}$$

Question189

The rate constant of a zero order reaction is $2.0 \times 10^{-2} \text{ mol L}^{-1} \text{ s}^{-1}$. If the concentration of the reactant after 25 seconds is 0.5M . What is the initial concentration?

[Online April 23, 2013]

Options:

- A. 0.5M
- B. 1.25M
- C. 12.5M
- D. 1.0M

Answer: D

Solution:

For a zero order reaction Rate constant

$$= k = \frac{a - x}{t}$$

$$2 \times 10^{-2} = \frac{a - 0.5}{25}$$

$$a - 0.5 = 0.5$$

$$a = 1.0\text{M}$$

Question190

A radioactive isotope having a half - life period of 3 days was received after 12 days. If 3g of the isotope is left in the container, what would be the initial mass of the isotope?

[Online April 25, 2013]

Options:

- A. 12g
- B. 36g

C. 48g

D. 24g

Answer: C

Solution:

Given $t_{1/2} = 3$

Total time $T = 12$

No. of half lives $(n) = \frac{12}{3} = 4$

$$\left(\frac{1}{2}\right)^n = \frac{N}{N_0}$$

$$\therefore \left(\frac{1}{2}\right)^4 = \frac{3}{N}$$

$$\frac{3}{N} = \frac{1}{16}$$

$$N = 48\text{g}$$

Question191

The rate of a reaction doubles when its temperature changes from 300K to 310K . Activation energy of the reaction will be :

($R = 8.314\text{J K}^{-1}\text{mol}^{-1}$ and $\log 2 = 0.301$)

[2013]

Options:

A. 53.6kJ mol^{-1}

B. 48.6kJ mol^{-1}

C. 58.5kJ mol^{-1}

D. 60.5kJ mol^{-1}

Answer: A

Solution:

Activation energy can be calculated from the equation

$$\frac{\log k_2}{\log k_1} = \frac{E_a}{2.303R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

Given $\frac{k_2}{k_1} = 2$; $T_2 = 310\text{K}$; $T_1 = 300\text{K}$

$$\therefore \log 2 = \frac{E_a}{2.303 \times 8.314} \left(\frac{1}{300} - \frac{1}{310} \right)$$

$$E_a = 53598.6\text{J/mol} = 53.6\text{kJ/mol}$$

Question192

The reaction $X \rightarrow Y$ is an exothermic reaction. Activation energy of the reaction for X into Y is 150 kJ mol^{-1} . Enthalpy of reaction is 135kJ mol^{-1} . The activation energy for the reverse reaction, $Y \rightarrow X$ will be:

[Online April 22, 2013]

Options:

A. 280kJ mol^{-1}

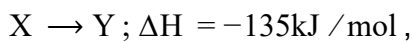
B. 285kJ mol^{-1}

C. 270kJ mol^{-1}

D. 15kJ mol^{-1}

Answer: B

Solution:



$$E_a = 150\text{kJ/mol}$$

For an exothermic reaction

$$E_{a(\text{F.R.})} = \Delta H + E_{a(\text{B.R.})}$$

$$150 = -135 + E_{a(\text{B.R.})}$$

$$E_{a(\text{B.R.})} = 285\text{kJ/mol}$$

Question193

**For a first order reaction (A) → products, the concentration of A changes from 0.1M to 0.025M in 40 minutes. The rate of reaction when the concentration of A is 0.01M is :
[2012]**

Options:

A. $1.73 \times 10^{-5} \text{M} / \text{min}$

B. $3.47 \times 10^{-4} \text{M} / \text{min}$

C. $3.47 \times 10^{-5} \text{M} / \text{min}$

D. $1.73 \times 10^{-4} \text{M} / \text{min}$

Answer: B

Solution:

For a first order reaction

$$\begin{aligned} k &= \frac{2.303}{t} \log \frac{a}{a-x} = \frac{2.303}{40} \log \frac{0.1}{0.025} \\ &= \frac{2.303}{40} \log 4 = \frac{2.303 \times 0.6020}{40} \\ &= 3.47 \times 10^{-2} \end{aligned}$$

$$\begin{aligned} R &= k(A)^1 = 3.47 \times 10^{-2} \times 0.01 \\ &= 3.47 \times 10^{-4} \end{aligned}$$

Question194

**In a chemical reaction A is converted into B. The rates of reaction, starting with initial concentrations of A as $2 \times 10^{-3} \text{M}$ and $1 \times 10^{-3} \text{M}$, are equal to $2.40 \times 10^{-4} \text{M s}^{-1}$ and $0.60 \times 10^{-4} \text{M s}^{-1}$ respectively. The order of reaction with respect to reactant A will be
[Online May 12,2012]**

Options:

A. 0

B. 1.5

C. 1

D. 2

Answer: D

Solution:

$A \rightarrow B$

Initial concentration

$$2 \times 10^{-3} \text{M}$$

$$1 \times 10^{-3} \text{M}$$

Rate of reaction

$$2.40 \times 10^{-4} \text{M s}^{-1}$$

$$0.60 \times 10^{-4} \text{M s}^{-1}$$

rate of reaction

$$r = k[A]^x$$

where x = order of reaction

hence

$$2.40 \times 10^{-4} = k[2 \times 10^{-3}]^x \dots\dots (i)$$

$$0.60 \times 10^{-4} = k[1 \times 10^{-3}]^x \dots\dots (ii)$$

On dividing eqn.(i) from eqn. (ii) we get

$$4 = (2)^x$$

$$\therefore x = 2$$

i.e. order of reaction = 2

Question195

For a reaction $A \rightarrow \text{Products}$, a plot of $\log t_{1/2}$ versus $\log a_0$ is shown in the figure. If the initial concentration of A is represented by a_0 , the order of the reaction is
[Online May 19, 2012]

Options:

A. one

B. zero

C. two

D. three

Answer: B

Solution:

Plot given is for zero order reaction.

Question196

The activation energy for a reaction which doubles the rate when the temperature is raised from 298K to 308K is
[Online May 26, 2012]

Options:

A. 59.2kJ mol^{-1}

B. 39.2kJ mol

C. 52.9kJ mol^{-1}

D. 29.5kJ mol^{-1}

Answer: C

Solution:

Activation energy can be calculated from the equation

$$\frac{\log k_2}{\log k_1} = \frac{E_a}{2.303R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

Given, $\frac{\log k_2}{\log k_1} = 2$ $T_2 = 308$; $T_1 = 298$

$$\therefore \log 2 = \frac{E_a}{2.303 \times 8.314} \left(\frac{1}{298} - \frac{1}{308} \right)$$

$$E_a = 52.9\text{kJ mol}^{-1}$$

Question197

Reaction rate between two substance A and B is expressed as following:

$$\text{rate} = k[A]^n[B]^m$$

If the concentration of A is doubled and concentration of B is made half of initial concentration, the ratio of the new rate to the earlier rate will be:

[Online May 7, 2012, Offline 2003]

Options:

A. $m + n$

B. $n - m$

C. $\frac{1}{2^{(m+n)}}$

D. $2^{(n-m)}$

Answer: D

Solution:

$$\text{Rate}_1 = k[A]^n[B]^m$$

$$\text{Rate}_2 = k[2A]^n\left[\frac{1}{2}B\right]^m$$

$$\begin{aligned}\therefore \frac{\text{Rate}_2}{\text{Rate}_1} &= \frac{k[2A]^n\left[\frac{1}{2}B\right]^m}{k[A]^n[B]^m} = (2)^n \left(\frac{1}{2}\right)^m \\ &= 2^n \cdot (2)^{-m} = 2^{n-m}\end{aligned}$$

Question 198

A reactant (A) from two products :

$A \xrightarrow{k_1} B$, Activation energy E_{a_1}

$A \xrightarrow{k_2} C$, Activation energy E_{a_2}

If $E_{a_2} = 2E_{a_1}$, then k_1 and k_2 are related as :

[2011RS]

Options:

A. $k_2 = k_1 e^{E_{a_1}/RT}$

$$\text{B. } k_2 = k_1 e^{E_{a_2}/RT}$$

$$\text{C. } k_1 = A k_2 e^{E_{a_1}/RT}$$

$$\text{D. } k_1 = 2 k_2 e^{E_{a_2}/RT}$$

Answer: C

Solution:

$$k_1 = A_1 e^{-E_{a_1}/RT}$$

$$k_2 = A_2 e^{-E_{a_2}/RT} \dots\dots(\text{ii})$$

On dividing eqn. (i) by eqn.(ii)

$$\frac{k_1}{k_2} = \frac{A_1}{A_2} (E_{a_2} - E_{a_1})/RT \dots\dots(\text{iii})$$

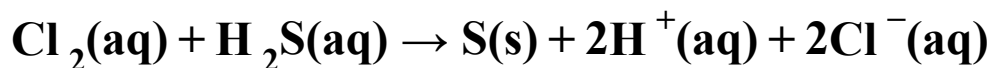
$$\text{Given } E_{a_2} = 2E_{a_1}$$

On substituting this value in eqn. (iii)

$$k_1 = k_2 A \times e^{E_{a_1}/RT}$$

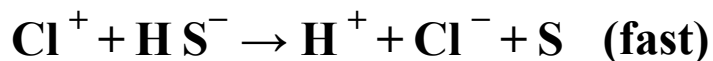
Question199

Consider the reaction:



The rate equation for this reaction is rate = $k[\text{Cl}_2][\text{H}_2\text{S}]$

Which of these mechanisms is/are consistent with this rate equation?



[2010]

Options:

A. B only

B. Both A and B

C. Neither A nor B

D. A only

Answer: D

Solution:

Since the slow step is the rate determining step, hence if we consider mechanism A. We find
 $\text{Rate} = k[\text{Cl}_2][\text{H}_2\text{S}]$

Now, if we consider mechanism B. We find

$$\text{Rate} = k[\text{Cl}_2][\text{HS}^-] \dots (i)$$

From step 1 (fast equilibrium) of mechanism B.

$$k = \frac{[\text{H}^+][\text{HS}^-]}{\text{H}_2\text{S}}$$

$$\text{or } [\text{HS}^-] = \frac{k[\text{H}_2\text{S}]}{\text{H}^+}$$

Substituting this value in equation (i) we find

$$\text{Rate} = k[\text{Cl}_2]k \frac{[\text{H}_2\text{S}]}{\text{H}^+} = k' \frac{[\text{Cl}_2][\text{H}_2\text{S}]}{[\text{H}^+]}$$

Hence, only mechanism A is consistent with the given rate equation.

Question200

The time for half life period of a certain reaction $\text{A} \rightarrow \text{Products}$ is 1 hour. When the initial concentration of the reactant 'A' is 2.0 mol L^{-1} , how much time does it take for its concentration to come from 0.50 to 0.25 mol L^{-1} if it is a zero order reaction?
[2010]

Options:

A. 4h

B. 0.5h

C. 0.25h

D. 1h

Answer: C

Solution:

For the reaction

$A \rightarrow \text{Product}$

Given $t_{1/2} = 1$ hour

For a zero order reaction

$$t_{\text{completion}} = \frac{[A_0]}{k} = \frac{\text{Initial conc.}}{\text{Rate constant}}$$

$$\therefore t_{1/2} = \frac{[A_0]}{2k}$$

$$\text{or } k = \frac{[A_0]}{2t_{1/2}} = \frac{2}{2 \times 1} = 1 \text{ mol lit}^{-1} \text{ hr}^{-1}$$

Further for a zero order reaction

$$k = \frac{dx}{dt} = \frac{\text{change in concentration}}{\text{time}}$$

$$1 = \frac{0.50 - 0.25}{\text{time}}$$

$$\therefore \text{time} = 0.25 \text{ hr.}$$

Question201

The half life period of a first order chemical reaction is 6.93 minutes. The time required for the completion of 99% of the chemical reaction will be ($\log 2 = 0.301$)

[2009]

Options:

A. 23.03 minutes

B. 46.06 minutes

C. 460.6 minutes

D. 230.03 minutes

Answer: B

Solution:

For first order reaction,

$$k = \frac{2.303}{t} \log \frac{100}{100 - 99}$$

$$\frac{0.693}{6.93} = \frac{2.303}{t} \log \frac{100}{1}$$

$$\frac{0.693}{6.93} = \frac{2.303 \times 2}{t} 6.93$$

$$t = 46.06 \text{ min}$$

Question202

For a reaction $\frac{1}{2}\text{A} \rightarrow 2\text{B}$, rate of disappearance of 'A' is related to the rate of appearance of 'B' by the expression [2008]

Options:

A. $-\frac{d[\text{A}]}{dt} = \frac{1}{2} \frac{d[\text{B}]}{dt}$

B. $-\frac{d[\text{A}]}{dt} = \frac{1}{4} \frac{d[\text{B}]}{dt}$

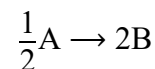
C. $-\frac{d[\text{A}]}{dt} = \frac{d[\text{B}]}{dt}$

D. $-\frac{d[\text{A}]}{dt} = 4 \frac{d[\text{B}]}{dt}$

Answer: B

Solution:

The rate of reaction for the reaction



can be written either as

$$-2 \frac{d}{dt}[\text{A}] \text{ with respect to 'A' or } \frac{1}{2} \frac{d}{dt}[\text{B}] \text{ with respect to 'B'}$$

From the above, we have

$$-2 \frac{d}{dt}[\text{A}] = \frac{1}{2} \frac{d}{dt}[\text{B}]$$

$$\text{or } -\frac{d}{dt}[\text{A}] = \frac{1}{4} \frac{d}{dt}[\text{B}]$$

Question203

Consider the reaction, $2A + B \rightarrow$ products. When concentration of B alone was doubled, the half-life did not change. When the concentration of A alone was doubled, the rate increased by two times. The unit of rate constant for this reaction is [2007]

Options:

- A. s^{-1}
- B. $Lmol^{-1}s^{-1}$
- C. no unit
- D. $mol\ L^{-1}s^{-1}$.

Answer: B

Solution:

For a first order reaction, $t_{1/2} = \frac{0.693}{K}$ i.e. for a first order reaction $t_{1/2}$ does not depend up on the concentration. From the given data, we can say that order of reaction with respect to B = 1 because change in concentration of B does not change half life. Order of reaction with respect to A = 1 because rate of reaction doubles when concentration of B is doubled keeping concentration of A constant.

$\therefore$ Order of reaction = $1 + 1 = 2$ and units of second order reaction are $Lmol^{-1}sec^{-1}$.

Question 204

A radioactive element gets spilled over the floor of a room. Its half-life period is 30 days. If the initial velocity is ten times the permissible value, after how many days will it be safe to enter the room? [2007]

Options:

- A. 100 days
- B. 1000 days

C. 300 days

D. 10 days.

Answer: A

Solution:

Suppose activity of safe working = A

Given $A_0 = 10A$

$$\lambda = \frac{0.693}{t_{1/2}} = \frac{0.693}{30}$$

$$t_{1/2} = \frac{2.303}{\lambda} \log \frac{A_0}{A} = \frac{2.303}{0.693/30} \log \frac{10A}{A}$$
$$= \frac{2.303 \times 30}{0.693} \times \log 10 = 100 \text{ days.}$$

Question205

The energies of activation for forward and reverse reactions for $A_2 + B_2 \rightleftharpoons 2AB$ are 180kJ mol^{-1} and 200kJ mol^{-1} respectively. The presence of a catalyst lowers the activation energy of both (forward and reverse) reactions by 100kJ mol^{-1} . The enthalpy change of the reaction ($A_2 + B_2 \rightarrow 2AB$) in the presence of a catalyst will be (in kJ mol^{-1})

[2007]

Options:

A. 20

B. 300

C. 120

D. 280

Answer: A

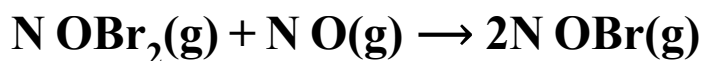
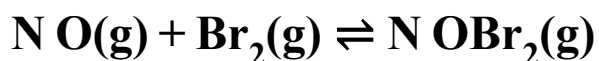
Solution:

$$\Delta H_R = E_f - E_b = 180 - 200 = -20 \text{ kJ/mol}$$

The nearest correct answer given in choices may be obtained by neglecting sign.

Question 206

The following mechanism has been proposed for the reaction of NO with Br₂ to form N OBr :



If the second step is the rate determining step, the order of the reaction with respect to N O(g) is

[2006]

Options:

A. 3

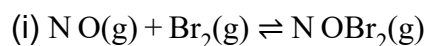
B. 2

C. 1

D. 0

Answer: B

Solution:



$$\text{Rate law equation} = k[\text{N OBr}_2][\text{N O}]$$

But N OBr₂ is intermediate and must not appear in the rate law equation.

From 1st step,

$$K_c = \frac{[\text{N OBr}_2]}{[\text{N O}][\text{Br}_2]}$$

$$\therefore [\text{N OBr}_2] = K_c [\text{N O}][\text{Br}_2]$$

$$\therefore \text{Rate law equation} = k \cdot K_c [\text{N O}]^2 [\text{Br}_2]$$

Hence order of reaction w.r.t. NO is 2 .

Question207

A reaction was found to be second order with respect to the concentration of carbon monoxide. If the concentration of carbon monoxide is doubled, with everything else kept the same, the rate of reaction will

[2006]

Options:

A. increase by a factor of 4

B. double

C. remain unchanged

D. triple

Answer: A

Solution:

Since the reaction is 2nd order w.r.t CO. Thus, rate law is given as.

$$r = k[\text{CO}]^2$$

Let initial concentration of CO is a

$$\text{i.e. } [\text{CO}] = a$$

$$\therefore r_1 = k(a)^2 = ka^2$$

When concentration becomes doubled, i.e. $[\text{CO}] = 2a$

$$\therefore r_2 = k(2a)^2 = 4ka^2 \quad \therefore r_2 = 4r_1$$

So, the rate of reaction becomes 4 times.

Question208

Rate of a reaction can be expressed by Arrhenius equation as:

$$k = Ae^{-E_a/RT}$$

In this equation, E_a represents

[2006]

Options:

- A. the total energy of the reacting molecules at a temperature, I
- B. the fraction of molecules with energy greater than the activation energy of the reaction
- C. the energy above which all the colliding molecules will react
- D. the energy below which all the colliding molecules will react

Answer: C

Solution:

In Arrhenius equation, $k = Ae^{-E_a/RT}$, E_a is the energy of activation, which is required by the colliding molecules to react resulting in the formation of products.

Question 209

A reaction involving two different reactants can never be [2005]

Options:

- A. bimolecular reaction
- B. second order reaction
- C. first order reaction
- D. unimolecular reaction

Answer: D

Solution:

The molecularity of a reaction is the number of reactant molecules taking part in a single step of the reaction. Note: The reaction involving two different reactants can never be unimolecular.

Question 210

$t_{\frac{1}{4}}$ can be taken as the time taken for the concentration of a reactant to drop to $\frac{3}{4}$ of its initial value. If the rate constant for a first order reaction is k , the $t_{\frac{1}{4}}$ can be written as
[2005]

Options:

A. $0.75/k$

B. $0.69/k$

C. $0.29/k$

D. $0.10/k$

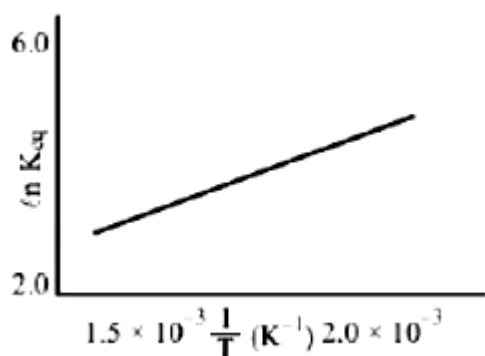
Answer: C

Solution:

$$\begin{aligned} t_{1/4} &= \frac{2.303}{K} \log \frac{1}{3/4} = \frac{2.303}{K} \log \frac{4}{3} \\ &= \frac{2.303}{K} (\log 4 - \log 3) \\ &= \frac{2.303}{K} (2 \log 2 - \log 3) \\ &= \frac{2.303}{K} (2 \times 0.301 - 0.4771) = \frac{0.29}{K} \end{aligned}$$

Question211

A schematic plot of $\ln K_{eq}$ versus inverse of temperature for a reaction is shown below



The reaction must be
[2005]

Options:

- A. highly spontaneous at ordinary temperature
- B. one with negligible enthalpy change
- C. endothermic
- D. exothermic

Answer: D

Solution:

The graph shows that reaction is exothermic.

$$\log k = \frac{-\Delta H}{RT} + 1$$

For exothermic reaction $\Delta H < 0$

$\therefore \log k$ vs $\frac{1}{T}$ would be negative straight line with positive slope.

Question212

Consider an endothermic reaction $X \rightarrow Y$ with the activation energies E_b and E_f for the backward and forward reactions, respectively. In general
[2005]

Options:

- A. there is no definite relation between E_b and E_f
- B. $E_b = E_f$
- C. $E_b > E_f$
- D. $E_b < E_f$

Answer: D

Solution:

Enthalpy of reaction (ΔH) = $E_f - E_b$

For an endothermic reaction, $\Delta H = +ve$ hence for ΔH to be negative

$$E_b < E_f$$

Question213

The rate equation for the reaction $2A + B \rightarrow C$ is found to be: rate = $k[A][B]$. The correct statement in relation to this reaction is that the
[2004]

Options:

- A. rate of formation of C is twice the rate of disappearance of A
- B. $t_{1/2}$ is a constant
- C. unit of k must be s^{-1}
- D. value of k is independent of the initial concentrations of A and B

Answer: D

Solution:

The velocity constant depends on temperature only. It is independent of concentration of reactants.

Question214

In a first order reaction, the concentration of the reactant, decreases from 0.8M to 0.4M in 15 minutes. The time taken for the concentration to change from 0.1M to 0.025 M is
[2004]

Options:

- A. 7.5 minutes
- B. 15 minutes
- C. 30 minutes
- D. 60 minutes

Answer: C

Solution:

As the concentration of reactant decreases from 0.8 to 0.4 in 15 minutes hence the $t_{1/2}$ is 15 minutes. To fall the concentration from 0.1 to 0.025 we need two half-lives i.e., 30 minutes.

Question 215

The half-life of a radioisotope is four hours. If the initial mass of the isotope was 200g, the mass remaining after 24 hours undecayed is [2004]

Options:

- A. 3.125g
- B. 2.084g
- C. 1.042g
- D. 4.167g

Answer: A

Solution:

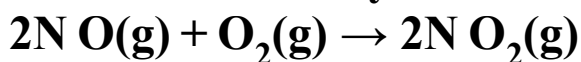
$N_t = N_0 \left(\frac{1}{2} \right)^n$ where n is number of half life periods.

$$n = \frac{\text{Total time}}{\text{half life}} = \frac{24}{4} = 6$$

$$\therefore N_t = 200 \left(\frac{1}{2} \right)^6 = 3.125\text{g}$$

Question216

For the reaction system:



volume is suddenly reduced to half its value by increasing the pressure on it. If the reaction is of first order with respect to O_2 and second order with respect to N O , the rate of reaction will [2003]

Options:

- A. diminish to one-eighth of its initial value
- B. increase to eight times of its initial value
- C. increase to four times of its initial value
- D. diminish to one-fourth of its initial value

Answer: B

Solution:

$r = k[\text{O}_2][\text{N O}]^2$. When the volume is reduced to $1/2$, the conc. will double

$\therefore \text{New rate} = k[2\text{O}_2][2\text{N O}]^2 = 8k[\text{O}_2][\text{N O}]^2$

The new rate increases to eight times of its initial.

Question217

The half-life of a radioactive isotope is three hours. If the initial mass of the isotope were 256g, the mass of it remaining undecayed after 18 hours would be [2003]

Options:

- A. 8.0g
- B. 12.0g

C. 16.0g

D. 4.0g

Answer: D

Solution:

$$t_{1/2} = 3\text{h}, T = 18\text{h}$$

$$\because T = n \times t_{1/2}$$

$$\therefore n = \frac{18}{3} = 6$$

$$\text{Initial mass } (C_0) = 256\text{g}$$

$$\therefore C_n = \frac{C_0}{2^n} = \frac{256}{(2)^6} = \frac{256}{64} = 4\text{g}$$

Question218

**In respect of the equation $k = Ae^{-E_a/RT}$ in chemical kinetics, which one of the following statements is correct?
[2003]**

Options:

A. A is adsorption factor

B. E_a is energy of activation

C. R is Rydberg's constant

D. k is equilibrium constant

Answer: B

Solution:

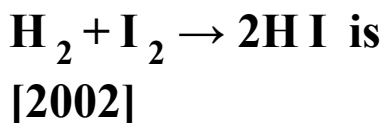
In equation $k = Ae^{-E_a/RT}$;

A = Frequency factor

k = velocity constant, R = gas constant and E_a = energy of activation

Question219

The differential rate law for the reaction



Options:

A. $-\frac{d[\text{H}_2]}{dt} = -\frac{d[\text{I}_2]}{dt} = -\frac{d[\text{HI}]}{dt}$

B. $\frac{d[\text{H}_2]}{dt} = \frac{d[\text{I}_2]}{dt} = \frac{1}{2} \frac{d[\text{HI}]}{dt}$

C. $\frac{1}{2} \frac{d[\text{H}_2]}{dt} = \frac{1}{2} \frac{d[\text{I}_2]}{dt} = -\frac{d[\text{HI}]}{dt}$

D. $-2 \frac{d[\text{H}_2]}{dt} = -2 \frac{d[\text{I}_2]}{dt} = \frac{d[\text{HI}]}{dt}$

Answer: D

Solution:

$$\text{Rate of appearance of HI} = \frac{1}{2} \frac{d[\text{HI}]}{dt}$$

$$\text{Rate of disappearance of H}_2 = \frac{-d[\text{H}_2]}{dt}$$

$$\text{Rate of disappearance of I}_2 = \frac{-d[\text{I}_2]}{dt}$$

$$\text{hence } -\frac{d[\text{H}_2]}{dt} = -\frac{d[\text{I}_2]}{dt} = \frac{1}{2} \frac{d[\text{HI}]}{dt}$$

$$\text{or } -2 \frac{d[\text{H}_2]}{dt} = -2 \frac{d[\text{I}_2]}{dt} = \frac{d[\text{HI}]}{dt}$$

Question220

The integrated rate equation is

$$Rt = \log C_0 - \log C_i$$

The straight line graph is obtained by plotting
[2002]

Options:

A. time vs $\log C_t$,

B. $\frac{1}{\text{time}}$ vs C_t

C. time vs C_t

D. $\frac{1}{\text{time}}$ vs $\frac{1}{C_t}$

Answer: A

Solution:

$$Rt = \log C_0 - \log C_t$$

It is clear from the equation that if we plot a graph between $\log C_t$ and time, a straight line with a slope equal to $-\frac{k}{2.303}$ and intercept equal to $\log[C_0]$ will be obtained.

Question221

Units of rate constant of first and zero order reactions in terms of molarity M unit are respectively [2002]

Options:

A. sec^{-1} , M sec^{-1}

B. sec^{-1} , M

C. M sec^{-1} , sec^{-1}

D. M, sec^{-1} .

Answer: A

Solution:

For a zero order reaction.

rate = $k[A]^0$ i.e. rate = k

hence unit of $k = M \cdot \text{sec}^{-1}$

For a first order reaction. rate = $k[A]$

$\therefore k = M \cdot \text{sec}^{-1} / M = \text{sec}^{-1}$

Question222

For the reaction $A + 2B \rightarrow C$, rate is given by $R = [A][B]^2$ then the order of the reaction is
[2002]

Options:

A. 3

B. 6

C. 5

D. 7

Answer: A

Solution:

Note : Order is the sum of the power of the concentration terms in rate law expression. Hence the order of reaction is $= 1 + 2 = 3$

Question223

If half-life of a substance is 5 yrs, then the total amount of substance left after 15 years, when initial amount is 64 grams is
[2002]

Options:

A. 16 grams

B. 2 grams

C. 32 grams

D. 8 grams

Answer: D

Solution:

$t_{1/2} = 5$ years, $T = 15$ years.

Hence total number of half life periods $= \frac{15}{5} = 3$

$\therefore$ Amount left $= \frac{64}{(2)^3} = 8\text{g}$
